SUPERSEDING
MIL-T-81517A(AS)
17 April 1970

MILITARY SPECIFICATION

TEST SET, MISSILE CONTROL SYSTEM AN/AWM-21

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment listed herein shall be an integrated assembly of Contractor Furnished and Government Furnished Equipment to facilitate the detailed maintenance of the AN/AWG-10 Missile Control System (MCS) at the Intermediate Level of Maintenance where a center line bench is suitable. Facilities where a sidebench is suitable should refer to MIL-T-81527(AS) Test Set, Missile Control System AN/AWG-10.

1.2 Classification - The complete AN/AWM-21 MCS Test Set shall be comprised from the following items:

Item	Type Designation
Table-Rack, Radar Maintenance	FN-199C/AWM-21
Fixture, Antenna Maintenance	MX-7340/AWM-21
Waveguide Extension	MX-8491/AWG-10
Cable and Adapter Group	OA-8124B/AWM-21
Tool Kit, Test Set	TK 217/AWM-21
Tool Kit Maintenance	TK 216/AWM-21
Fixture, Antenna Transporting	MX-7339B/AWM-21
Test Set, Reference Rate Gyro	TS-2512/AWM-21
Generator, Signal	SG-696B/AWM-21

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Item	Type Designation
Compressor - Dehydrator Electronic Equipment	HD-755/AWM-21
Counter, Electronic Digital	CP-870/AWM-21
Oscilloscope	OS-180/AWM-21
Test Set, Missile Control System	TS-2511C/AWG-10*
Analyzer, Radio Frequency Noise	TS-2573E/AWG-10
Cabinet, Electrical Equipment	CY-6667A/AWM-21*
Multimeter	TS-505/U
Multimeter	AN/PSM-4
Radar Set	AN/APG-59
Voltmeter Digital	ME-371/AWM-21
Generator Signal	SG-844/AWM-21

1.2.1 Government Furnished Equipment - The Radar Set AN/APG-59, Multimeter AN/PSM-4 and Multimeter, Electronic, TS-505/U shall be Government Furnished Equipment (GFE).

2. APPLICABLE DOCUMENTS

2.1 General - The following documents form a part of this specification to the extent specified and goes into effect on the date of invitation for bids.

SPECIFICATIONS

Military

MIL-C-17	Cables, Radio Frequency; Coaxial, Dual Twin Conductor and Twin Lead.
MIL-D-3716	Dessicant, Activated for dynamic dehumidification.
MIL-C-3767	Connector, Plug, Electrical (Power, Three Wire, Polarized Spring-loaded pivoted grounding type).
MIL-T-5422	Testing, Environmental, Aircraft
MIL-I-6181	Interference Control Requirements, Aircraft Equipment

* Pending Navy Approval

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MIL-M-15134	Multimeter AN/PSM-4
MIL-H-15362	Harnesses; Test Bench Avionic Equipment
MIL-W-16878	Wire, Electrical, Insulated High Temperature
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Deliver Of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment Format For
MIL-T-21200	Test Equipment for Use With Electronic and Fire Control Systems, General Specification For
MIL-T-22910	Tool, Crimping, Hand, For Crimp Type Electronic Terminal and Shield Ferrule
MIL-T-23336	Technical Directives (Letter Type) Preparation Of
MIL-H-25579	Hose Assembly, Tetrafluoroethylene, High Temperature, Power Plant, Aircraft
MIL-P-26514	Polyurethane Foam, Rigid or Elastic, For Packaging Radar Set AN/APG-59
MIL-H-27267	Hose Assembly, Tetrafluoroethylene, High Temperature, Medium Pressure
MIL-F-27272	Fitting, Tetrafluoroethylene, High Temperature, Medium Pressure, General Requirements For
MIL-C-22750	Coating, Epoxy-Polyamide
MIL-M-55088	Multimeter, Electronic TS-505/U
MIL-R-81114	Radar Set AN/APG-59
MIL-T-81527	Test Set, Missile Control System, AN/AWG-10
Federal	
GGG-K-00450	Knife, Craftsman S, Multipurpose
GGG-S-121	Screwdriver and Screw Starter, Hand
GGG-T-558	Tool Box, Mechanics
GGG-W-641	Wrench, Socket
GGG-W-680	Wrenches, Tap and Reamer Adjustable
GGG-W-686	Wrench, Torque
STANDARDS	
Military	
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

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MIL-STD-189	Racks Electrical Equipment 19-inch and Associated Panels
MIL-STD-701	Preferred and Guidance List of Semiconductor Devices
MIL-STD-781	Reliability Test, Exponential Distribution
MIL-STD-803	Human Engineering Design Criteria for Aerospace Systems and Equipment

Federal

FED-STD-595	Colors

DRAWINGS

Military

BuShips S9101-212410 - Shipboard Workbench for Aviation Electronic Shops

NAMC 610415	Guidance Plan, CVA 61, Avionics Shop No. 1 Main Deck FRS 29-39
MS 3100	Connector, Electrical
MS 3119	Connector, Receptical, Electric, Thru-Bulkhead Mounting
MS 3120	Connectors, Receptacle Electric, Crimp Type
MS 3126	Connector, Plug, Electrical
MS 3191	Tool, Hand, Crimp Type
MS 3196	Gage, Inspection For Content Crimping Tool
MS 16108	Jacks, Tip, Low Voltage
MS 17321	Meter, Time Totalizing, Miniature
MS 17322	Meter, Time Totalizing
MS 18137	Tool, Content, Crimp Type Electrical, Insertion and Removal
MS 28889	Valve, Air, High Pressure Charging
MS 33514	Fitting End, Standard Dimensions For Flareless Tube Connection and Gasket Seal
MS 33656	Fitting End Standard Dimensions for Flared Tube
MS 35168	Connector, Plug UG-88E/U
MS 35173	Adapter, Connector UG-274B/U
MS 35175	Connector, Receptacle, Electrical UG-291C/U

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2.1.1 Availability of Documents - Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office. (See paragraph 6.4.)

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

a. Parts and Materials as approved by MIL-T-21200 shall be given first consideration.

b. Guidelines for the selection of nonstandard parts shall be per MIL-T-21200, Selection of Parts and Materials.

c. In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass electrical and environmental test.

d. Paint for transit cases for test equipment used at shop level maintenance shall be epoxy polyamide per MIL-C-22750 except that the color shall be aircraft gray, color number 16473, per FED-STD-595.

3.1.1 Nonstandard Parts - Nonstandard parts and materials shall be as outlined in MIL-T-21200, Materials, Parts and Processes, except that mechanical parts shall be considered as nonstandard parts only when the material requirements of MIL-T-21200, as modified herein, are not met.

3.1.2 Tube Types and Semiconductor Types - The choice and application of electron tubes and semiconductor types used in the equipment shall be in accordance with MIL-T-21200 and MIL-STD-701.

3.1.3 Noncorrosive Hardware - External Military Standard (MS) hardware which is used during servicing or installing test equipments shall be of corrosive-resistant type and shall not be further finish-treated.

3.2 Design and Construction - Unless otherwise specified herein or in the detail equipment specification, Contractor-Furnished Equipment (CFE) shall conform with all the requirements of MIL-H-15362 or MIL-T-21200 as applicable for design, construction, and workmanship. Government Furnished Equipment (GFE) shall conform to the applicable detail equipment specification. The total weight of the AN/AWM-21 CFE shall not exceed 6000 pounds.

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3.2.1 Operational Stability - Unless otherwise specified in the individual equipment specification, the equipment shall operate with satisfactory performance, continuously or intermittently, for a period of 24 hours without necessity for readjustment of any operator controls which are inaccessible to the operator during use, except where adjustments associated with parts failures are required following replacement of those parts.

3.2.2 Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts.

3.2.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200 and MIL-H-15362, as applicable.

3.2.4 Interchangeability - Unless otherwise specified herein or in the detail equipment specification, the equipment shall meet the interchangeability requirements as defined in MIL-T-21200.

3.2.5 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance. The color of front panel indicator lights shall be in accordance with MIL-STD-803.

3.2.6 Calibration - Test and measuring equipment shall be calibrated at periodic intervals established on the basis of stability, purpose, and degree of usage. Calibration intervals shall be adjusted as required to assure continued accuracy as evidenced by previous calibration results.

3.2.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient: 25° +15° -5° Centigrade (C) 77° +27° -9° Fahrenheit (F)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90 percent relative humidity
Input Voltage	3-phase, 115 ± 1 Volts (V) line to neutral, 60 Hz ±2 percent 3-phase, 115 ± 1 Volts (V) line to neutral, 400 Hz ±2 percent 27.5 ±0.5 Volts Direct Current (Vdc)

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3.2.8 Warmup Time - Unless otherwise specified herein or in the detail equipment specification, the time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions, 20 minutes at extreme service conditions.

3.2.9 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-I-6181.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein, or in the detail equipment specification.

3.3.2 Function - The AN/AWM-21 shall provide an integrated maintenance facility. This shall include simulated aircraft controls and equipment test bench harnesses; equipment support and handling fixtures; test equipment both special and standard; special tools; cooling equipment; and a hydraulic power supply.

3.3.3 Form Factor - The AN/AWM-21 shall be compatible with avionics shop layouts. Guidance Plan CVA61 Avionics Shop No. 1, Main Deck, FRS 29-39; Drawing 610415, bench space 119 through 126 shall be used as a guide to shop layout and bench design.

3.4 Detail Requirements

3.4.1 Table-Rack, Radar Maintenance FN-199C/AWM-21

3.4.1.1 Function - The Radar Maintenance Table-Rack shall be for shop use and shall provide monitoring, mounting, interconnection, hydraulic power and cooling facilities for the MCS. The equipment shall electrically simulate MCS required aircraft flight functions and shall contain provisions for fault isolation, adjustment, and repair or replacement of defective pallets, circuits, plugin, assemblies, and components of the MCS.

3.4.1.2 Form Factor - The Table-Rack shall be compatible with the Shipboard work bench for Aviation Electronic shops per S9101-212410 and shall require centerline space 19 by 11 by 7-1/2 feet. The total weight of the equipment, including cables, shall not exceed 4200 pounds.

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3.4.1.3 Mean Time Between Failures (MTBF) - When maintained in accordance with applicable preventive maintenance procedures, the equipment shall demonstrate at least 400-hours MTBF. Accept-reject criteria shall be defined by paragraph 4.2.8.1 of MIL-STD-781.

3.4.1.4 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H-15362 except that the panels of the Test and Control Rack shall comply with MIL-T-21200, Class 3.

3.4.1.5 Primary Input Power Requirements - The equipment shall require power of the following type and shall not exceed the specified amounts.

AC Power: 3-phase, 115/200V ±5 percent, 380 to 420 Hz, 48 Amperes (A) phase
AC Power: 3-phase, 115/200V ±5 percent, 50 to 70 Hz, 35 A/phase
DC Power: 27V ±2V dc, 65 Amperes

3.4.1.6 Contents of Unit - The Table-Rack shall consist of the following major units:

a. MCS Support Structure Assembly
b. Test and Control Rack
c. Bench Assembly
d. Hydraulic Supply
e. Liquid Cooling System
f. Blower Assembly
g. Remote Safety Unit
h. Test Accessories

3.4.1.6.1 MCS Support Structure Assembly

3.4.1.6.1.1 Function - This structure supports the AN/APG-59 Radar on its slide rail, provides a work area beneath the MCS, and provides space and mechanical support for the Liquid Cooling System, Hydraulic Supply and the Remote Safety Unit.

3.4.1.6.1.2 Form Factor - The maximum overall dimensions of this unit shall be 44-3/4 by 82-1/2 by 89-5/8 inches with the Antenna Guard Ring extended. The weight of the structure shall not exceed 1150 pounds.

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3.4.1.6.1.3 Contents of Unit - The MCS Support Structure Assembly shall contain the following assemblies:

a. MCS Support Structure

b. MCS Support Bench

3.4.1.6.1.3.1 MCS Support Structure

3.4.1.6.1.3.1.1 Function - This structure shall support the AN/APG-59 system on its Slide Rail. It shall have forward support arms which can be removed for increased accessibility when required. A surfaced work area shall be provided below the AN/APG-59 System. An extendable antenna ring shall furnish protection to personnel from the hydraulic antenna of the AN/APG-59. Mounting shall also be provided on the left rear of the MCS Support Structure (when looking to the rear) for the Remote Safety Unit.

3.4.1.6.1.3.1.2 Form Factor - The MCS Support Structure shall be a cantilever beam mounted on a rear 'A' frame structure with two secondary removable forward support arms. Eye bolt attachments shall be provided on the top of the Support Structure for handling of the Structure. The MCS Support Structure shall be similar to the System Support Structure of the shipping container. The overall dimensions of the structure shall not exceed 44-3/4 by 49-1/8 by 64 inches. An antenna guard ring shall be attached to the Structure and can be extended 33-1/2 inches in front of the Structure. The weight of the Structure shall not exceed 550 pounds. A nonmetallic, hard, smooth work surface shall be provided for attachment to the top of the base of the MCS Support Structure.

3.4.1.6.1.3.1.3 Service Conditions - With the forward support arms in place, the MCS Support Structure shall meet the service conditions specified in paragraph 3.4.1.4. With the arms removed the unloaded mechanical support structure shall withstand a maximum continuous load of 1600 pounds.

3.4.1.6.1.3.2 MCS Support Bench

3.4.1.6.1.3.2.1 Function - The MCS Support Bench shall support the MCS Support Structure at a height convenient for work, and provide mounting provisions for the Liquid Cooling System and the Hydraulic Supply.

3.4.1.6.1.3.2.2 Form Factor - The MCS Support Bench shall not exceed 45 by 40 by 26 inches and the weight shall not exceed 375 pounds. The Base Structure shall provide mounting attachments for the MCS Support Structure so that the work surface will be located approximately 30 inches from the deck. The under structure shall be capable of bolting to the deck

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and provide attachments for the Liquid Cooling System and the Hydraulic Supply. Clamps shall be provided to hold the Grounding Bar while not in use.

3.4.1.6.2 Test and Control Rack

3.4.1.6.2.1 Function - The Test and Control Rack shall provide in one cabinet the necessary controls for primary power, hydraulics, liquid coolant system, and other bench functions; provide master MCS control, aircraft simulation and safety interlocks. It shall also contain the interconnecting test harness, air distribution ducts and test equipment mounting provisions.

3.4.1.6.2.2 Form Factor - This shall be a standard 19-inch equipment rack containing the items specified herein. The maximum dimensions of this unit shall be 35 by 24 by 75 inches. The weight of the Test and Control Rack shall not exceed 700 pounds, including the interconnecting Cabling, Shock Mounts, and Bottom Frame.

3.4.1.6.2.3 Contents of Unit - The Test and Control Rack shall contain the following assemblies:

a. Rack Assembly
b. Power Control Panel
c. Aircraft Simulator Panel
d. MCS Control Panel
e. Manual BIT Panel
f. Umbilical Plug and Tuning Drive Test Panel
g. Voltage Regulator
h. Shelf
i. Grounding Bar
j. IFF Simulator Panel
k. Test Point Bracket

3.4.1.6.2.3.1 Rack Assembly

3.4.1.6.2.3.1.1 Function - The Rack Assembly shall provide mounting provisions for the panels, shelf and equipment listed in (3.4.1.6.2.3) and the Antenna Control of the serial number 1527 configuration of the MCS. It shall provide all electrical interconnections between the Table-Rack and

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and the MCS and Convenience Outlets (60 Hz or 400 Hz). Ducts for cooling air to the MCS shall be located in the Rack Assembly.

3.4.1.6.2.3.1.2 Form Factor - The Rack Assembly shall be a standard 19-inch rack conforming to MIL-STD-189. Shock mounts shall be located between the rack and the base frame of the assembly. A second set of shock mounts shall be located between the Rack and the MCS Support Structure. The Base Frame shall be bolted or welded to the deck or floor. The weight of the unit shall not exceed 425 pounds.

3.4.1.6.2.3.1.3 Harness - All electrical interconnections shall be accomplished by the harness. The AN/APG-59 cable harness shall connect directly to the Test and Control Rack.

3.4.1.6.2.3.1.4 Cables - The interconnecting cables to the Radar Set Control and the Selector - Test Programmer (mounted in the MCS Control Panel) shall be of sufficient length to remove these units from their mountings and place them on the Shelf for maintenance while still electrically connected. Provisions shall be made for connecting in the Test Set, Missile Control System, TS-2511C/AWG-10 on either side of the Rack.

3.4.1.6.2.3.1.5 Air Distribution - The Rack Assembly shall provide air distribution from the Blower Assembly to the AN/APG-59 System Assembly. Air ducts shall carry air from the Blower Assembly hose entry in the right side of the Rack Structure to a connection in the left side which shall mate directly with the AN/APG-59 air fitting.

3.4.1.6.2.3.1.6 Connectors - Connections to the AN/APG-59 Radar Set shall be made with the following connectors:

Reference Designation		Function
7J1	*3, *4, *5	Radar Main
7J2	*3, *5	Radar Main
7J3	*5	BIT Main
7J4		BIT Main
7J5		Coaxial Attenuator
7J6		Simulated Doppler Main
8P1	*5	Unregulated Power Supply
8P2	*3	Unregulated Power Supply
9P1		Command Indicator
9P2		Command Indicator

*3 See paragraph 6.5.3
*4 See paragraph 6.5.4
*5 See paragraph 6.5.5

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Reference Designation	Function
10P1 *1	Radar Indicator
10P2 *3	Radar Indicator
11P1 *3	Control Indicator
11P2	Control Indicator
11P3	Control Indicator
11P4 *3	Control Indicator
12P1 *3, *5	Radar Set Control
12P2 *2, *3, *5	Radar Set Control
12P3 *3	Radar Set Control
13P1	Antenna Control
14P1	Selector-Test Programmer
14P2	Selector-Test Programmer
15P1	Velocity Signal Computer
15P2	Velocity Signal Computer
15P3	Velocity Signal Computer
16P1	Launch Signal Computer
16P2 *3	Launch Signal Computer
16P3 *5	Launch Signal Computer
17P1	Pilot Command Signal Computer
17P2	Pilot Command Signal Computer
17P3	Pilot Command Signal Computer

3.4.1.6.2.3.1.7 Shipping List - A durable list shall be affixed to the side of the Test and Control Rack and shall contain the numbers of the major assemblies and the numerous small accessories forming a complete Radar Maintenance Table-Rack. The list shall be for inventory control especially whenever the bench is transported or shipped.

3.4.1.6.2.3.1.8 Test Point Bracket

3.4.1.6.2.3.1.8.1 Function - The bracket shall contain test points associated with the transmitter and capable of continuous monitoring. The test points shall back up the BIT and FIM comparators.

3.4.1.6.2.3.1.8.2 Form Factor - The size of the bracket shall not exceed 8 inches by 2 inches by 2 inches. The bracket shall be designed to attach to the side of the Test and Control Rack and its weight shall not exceed 1 pound.

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*1 See paragraph 6.5.1
*2 See paragraph 6.5.2
*3 See paragraph 6.5.3
*4 See paragraph 6.5.4
*5 See paragraph 6.5.5

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3.4.1.6.2.3.1.8.3 Test Points - The following test point BNC-type female connectors for the signals specified shall be contained in the bracket.

a. PD Video

b. PD Sync

c. Pulse Video (B-Gun Video)

d. Pulse Sync (T_o)

e. Line Frequency (400 Hz)

f. Pretrigger Pulse

3.4.1.6.2.3.2 Power Control Panel

3.4.1.6.2.3.2.1 Function - This panel shall provide the power control and monitoring for the Table-Rack.

3.4.1.6.2.3.2.2 Form Factor - This unit shall not exceed 19 by 7 by 14 inches. The weight of the unit shall not exceed 27 pounds.

3.4.1.6.2.3.2.3 Controls - The following controls shall be provided on the front panel. System response to the operation of the controls listed in (a) through (h) below shall be indicated by the illumination of the control itself.

a. PWR ON (Power On) - A green pushbutton switch shall transfer +28 Vdc to power control relays thus enabling power to be supplied throughout the Table-Rack.

b. PWR OFF (Power Off) - A red pushbutton switch shall remove +28 Vdc from the power control relays thus enabling all power to be removed from Table-Rack.

c. RDR PWR (Radar Power) - The Radar Power Control shall be a green pushbutton switch which shall be used to control the 115/200V, 3-phase 400 Hz and the 28 Vdc power to the MCS.

d. PNL LTS (Panel Lights)- The Panels Lights Control shall be a white pushbutton switch which shall be used to turn the MCS cockpit panel lights on or off.

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Preparing Activity:

e. LIQ COOL (Liquid Cooling) - The Liquid Cooling Control shall be a green pushbutton switch which shall be used to turn on the Liquid Cooling System when the MCS is not on. The button shall also be lit when the RSC FUNCTION switch is turned to STANDBY (STBY) or OPERATE (OPR), and energizing cooler.

f. AIR COOL (Air Cooling) - The Air Cooling Control shall be a green pushbutton switch which shall be used to turn on the Blower Assembly when the MCS is not on. The button is also lit when the RSC FUNCTION switch is turned to STBY or OPR, and energizing blower.

g. HYD ON (Hydraulics On) - The Hydraulics On Control shall be a green, momentary pushbutton switch which shall be used to turn on the Hydraulic Supply. The Hydraulics On Control shall be interlocked with the RSC FUNCTION switch and must be operated each time the RSC FUNCTION switch is turned from OFF to any other position. The Hydraulics On control shall also be interlocked with the Hydraulic Panel of the Antenna Maintenance Fixture, when the Maintenance Fixture is used with the Table-Rack. The control shall be operable only when the Hydraulic Panel is in the BENCH position.

h. HYD OFF (Hydraulics Off) - The Hydraulics Off control shall be a red, momentary pushbutton switch which shall be used to turn off the hydraulics while the MCS is on.

i. PHASE SELECTOR - The Phase Selector control shall be a 3-position rotary switch which shall select the phase of the 3 phase, 400 Hz to be measured on the 400-Hz meter. Positions shall be marked A, B, and C.

3.4.1.6.2.3.2.4 Indication - The following monitors and indicating lights shall be provided on the front panel.

a. RADAR PHASE INDICATOR (400-Hz Phase Selector) - The Radar Phase Indicator shall be a neon indicator light with a translucent white lens which shall show when the 3-phase, 400-Hz line voltage is connected in the proper phase sequence.

b. 400-HZ VOLTAGE (AC Voltage Monitor) - A voltmeter of 3 percent full scale accuracy shall measure the MCS primary 400-Hz line voltage between 104 and 125 V.

c. DC VOLTAGE (DC Voltage Monitor) - A voltmeter of 3 percent full scale accuracy shall measure the primary dc line voltage between 22 and 31 Vdc.

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d. ELAPSED TIME (Elapsed Time Indicator) - A meter which conforms to MS17322-6A shall indicate total "on time" of the Bench Mounted Radar.

e. LIQ FLOW (Liquid Flow Indicator) - The Liquid Flow Indicator shall be a red indicator light which shall show the state of the coolant flow interlock. When the flow through the Liquid Cooling System is less than 2.5 + 0.1, -0.0 gallons per minute (gpm), the indicator light shall be lit.

f. HIGH TEMP (High Temperature Indicator) - The High Temperature Indicator shall be a red indicator light which shall show the state of the Temperature High interlock switch. When the return temperature to the Liquid Cooling System rises above 107 ±3°C (225 ±5°F) the indicator light shall be lit.

g. CMPTR AIR (Computer Air Indicator) - The Computer Air Indicator shall be a red indicator light which shall show the state of the Sensing Element installed in the plenum of the Computer Support Structure and the state of the Sensing Element installed in the plenum of the Blower Assembly. When the pressure is below nominal 5-1/2 inches of water, in either plenum, the indicator light shall be illuminated.

h. MAG TRAP (Magnetic Trap Indicator) - The Magnetic Trip Indicator shall be a red indicator light which shall show the state of the Magnetic Trap filter of the Liquid Cooling System. When the trap accumulates sufficient magnetic particles, the indicator light shall be lit to indicate that trap cleaning is necessary.

i. LOW TEMP (Low Temperature Indicator) - The Low Temperature Indicator shall be a red indicator light which shall show the state of the Temperature Low interlock switch of the Liquid Cooling System. The indicator light shall be lit until the system coolant temperature exceeds 43 ±3°C (110 ±5°F) in a rising direction. When this temperature is exceeded, the indicator light shall be off to indicate that the coolant temperature is near the regulated temperature range.

3.4.1.6.2.3.2.5 Protective Devices - The following protective devices shall be provided:

a. Circuit Breakers - There shall be four pushbutton circuit breakers:

(1) 400 HZ - A 3-phase, 50-A pushbutton breaker shall be provided for the 115/200V, 400-Hz power.

(2) +28 VDC - A 70-A pushbutton breaker shall be provided for 28 Vdc power.

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(3) 60 HZ - A 3-phase, 35-A bushbutton breaker shall be provided for 115/200V, 60-Hz power.

(4) +28 VDC STBY (Standby) - 15-A pushbutton breaker shall be provided for the +28 Vdc Stdby Supply.

b. Fuse - A 1-A fuse shall be provided for protection of the MCS panel light circuitry.

3.4.1.6.2.3.3 Aircraft Simulator Panel

3.4.1.6.2.3.3.1 Function - This panel shall simulate those signals that are normally supplied to the MCS by the aircraft and provide loads where needed or readout lights on those signals from the MCS that normally go to the aircraft.

3.4.1.6.2.3.3.2 Form Factor - This unit shall be 19 by 7 by 16-1/4 inches. The weight of the unit shall not exceed 12 pounds.

3.4.1.6.2.3.3.3 Controls - The following controls shall be provided on the panel face. System response to the operation of the controls listed in (c) through (h), (n), and (r) below shall be indicated by illumination of the control itself.

a. STATION SELECT - The Station Select Control shall be a two-position toggle switch which shall be used to test the missile station select wiring. The positions shall be marked BIT and ACFT (Aircraft).

b. MSL SEL (Missile Select) - The Missile Select Control shall be a three-position toggle switch which shall be used to energize the respective missile circuits in the radar. The positions shall be marked RADAR GUNS and HEAT.

c. CW STBY (Continuous Wave Standby) - The CW Standby Control shall be a yellow pushbutton switch which shall be used to place the CW Illuminator of the MCS in the standby mode.

d. CW ON - The CW On Control shall be a green pushbutton switch which shall be used to turn the CW Illuminator on.

e. CW OFF - The CW Off Control shall be a red pushbutton switch which shall be used to turn the CW Illuminator off.

f. XMTR ON (Transmitter On) - The Transmitter On Control shall be a green pushbutton switch which shall be used to turn the MCS transmitter on when the RSC FUNCTION switch is in the OPR position.

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g. ACCEL (Acceleration) - The Acceleration Control shall be a three-position toggle switch shall be used to select simulated acceleration signals representative of vertical acceleration of +3g, level flight, or -3g. Positions shall be marked +3, 0, and -3.

h. ALT (Altitude) - The Altitude Control shall be a two-position toggle switch which shall be used to select simulated altitude (see paragraph 3.4.1.6.2.3.3.6 (m)) signals representative of the aircraft altitudes to the computer. Positions shall be marked HIGH and LOW.

i. MBC/GSO INPUTS (Main Beam Clutter/Ground Speed Oscillator Inputs) - A two-position toggle switch shall be provided to switch from normal operation to variable simulation to Main Beam Clutter and Ground Speed. Positions shall be marked NORM (Normal) and MAN (Manual).

j. AIM-7E (AN/AIM-7E) - A white pushbutton switch which shall be used to activate circuits which generate a simulated missile identity signal (+28 Vdc) to the Pilot Command Signal Computer.

k. MAN MBC (Manual Main Beam Clutter) - A 10-turn potentiometer shall be provided for controlling variable simulation of Main Beam Clutter.

l. MAN GRD SPEED (Manual Ground Speed) - A 10-turn potentiometer shall be provided for controlling variable simulation of Ground Speed.

m. LAT OFFSET CURSOR (Lateral Offset Cursor Signal) - A three-position toggle switch shall select simulated Lateral Offset Cursor (3.4.1.6.2.3.3.6(i)) signals of 6.25-miles right (lying on the 30-degrees right line on the indicators), 0 miles (line in center of the indicators), or 6.25-miles left (lying on the 30-degree left line). Positions shall be marked 30 degrees-RIGHT, ZERO, and 30 degrees-LEFT.

n. DATA LINK - The Data Link switch shall be a white pushbutton switch which applies a +28V signal to the Indicator Control Unit of the radar to energize the circuits which enable the display of data link signals.

o. PLM ACQ (Pilot's Lockon Mode-Acquisition) - A five-position three-pole rotary switch to provide simulation of the half-action, full-action switch on the pilot's control stick. The switch position shall be fixed and shall not simulate the spring return feature of the pilot's switch. Two modes shall be provided, a normal mode which shall simulate electrically the pilot's switch and the antenna test mode which interrupts the boresight signal to the Scan Pattern Generator in the half-action position so that the Antenna will remain in the elevated position for checking.

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p. DF (Dogfight) - A single-pole single-throw toggle switch shall simulate the pilot's switch and apply a +28 Vdc control signal to the Tracker.

q. MSL TRIG (Missile Trigger) - A single-pole single-throw toggle switch shall simulate the pilot's missile firing button and apply a +28 Vdc firing signal to the computer firing relay contacts.

r. SEAM (Sidewinder Extended Acquisition Mode) - A three-position toggle switch shall simulate inputs to the AWG-10 indicators from the AN/ASA-63 programmer. The inputs shall be a +28 Vdc SEAM lock signal to activate the SEAM displays and the azimuth and elevation signals for the aim dot. Two aim dot positions shall be simulated, one at 0 V, 0 V negative, and the other in the lower right quadrant +6 V, -6 V.

s. ANN LAMP (Annunicator Lamp) - A single-pole single-throw toggle switch shall energize all lamps in the pilot's indicator (Command Indicator) via a set of steering diodes as a lamp check.

t. GEAR INTLK (Gear Interlock) - A single-pole double-throw toggle switch shall simulate the interlock switches activated by weight on the aircraft landing gear and the nose wheel retraction and associated relays. The switch shall be connected to the waveguide switch logic circuits in the radar.

3.4.1.6.2.3.3.4 Indication - The following indicating lights shall be provided on the front panel.

a. FIRE INTLK (Fire Interlock) - A white indicator light shall show the presence of the Fire Interlock signal from the Launch Signal Computer.

b. SPEED GATE - A white indicator light shall show the presence of the Speed Gate signal from the Velocity Signal Computer.

c. RDR WIND (Radar Wind) - A white indicator light shall show the presence of the Radar Wind Override signal from the Velocity Signal Computer.

d. ALT #1 (Altitude No. 1) - A white indicator light shall show the presence of the Altitude Command No. 1 signal from the Launch Signal Computer.

e. ALT #2 (Altitude No. 2) - A white indicator light shall show the presence of the Altitude Command No. 2 signal from the Launch Signal Computer.

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f. BOOST SP III (AN/AIM-7E Sparrow III Sensing and Band Switch Down) - A white indicator light shall show the presence of the AN/AIM-7E Sensing and Band Switch Down signal from the Launch Signal Computer.

g. CW RDY (Continuous Wave Ready) - A green indicator light shall indicate that the CW Illuminator's warmup delays have time out.

h. CW DLY (Continuous Wave Delay) - A white indicator light shall show the presence of the Delayed signal from the CW Illuminator.

i. XMTR RDY (Transmitter Ready) - A green indicator light shall indicate that the transmitter's warmup delays have timed out.

j. RAD ANT (Radiating from the Antenna) - A white indicator light shall show when the system is programmed to radiate out of the antenna.

k. STOP SCAN - A white indicator light shall show that the antenna is not scanning.

l. BIT 7E & 9D (AN/AIM-7E/9D BIT Command) - A white indicator light shall show the presence of the AN/AIM-7E/9D BIT command from the Selector-Test Programmer.

3.4.1.6.2.3.3.5 Test Points - Test points shall be provided on the front panel for the following signals from the AN/APG-59.

Test Point	Test Point Source Impedance
a. ENG BIAS (English Bias)	
AZ (Azimuth)	100 kilohms (k) in parallel with 240 microfarads (μF)
EL (Elevation)	100 K in parallel with 240 μF
b. HEAD AIM	
ψ_H (Roll)	180K in parallel with 0.60 μF
θ_H (Pitch)	180K in parallel with 0.60 μF
c. SWEEP CONT (Simulated Doppler DC Sweep Control)	Not applicable
d. SIM DOPP (Simulated Doppler AC)	Not applicable
e. MAN MBC	Not applicable

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Test Point	Test Point Source Impedance
f. VERT DIS POINTER (Vertical Display Pointer) OFF BIAS GRD (Ground) PLUS MINUS } Connected by 330 Ω load	Not applicable
g. MAN GRD SPEED	Not applicable
h. DIM LAMP SUP (Dimmer Lamp Supply)	10 ohms, 5 watts
i. CMD IND LAMP TEST VOLTS (Command Indicator Lamp Test Volts)	10 ohms, 5 watts
j. RDR IND LAMP TEST VOLTS (Radar Indicator Lamp Test Volts)	10 ohms, 5 watts
k. WARNING FLAG	1K, 1/8 watt
l. SIGNAL GRD (Signal Ground)	Not applicable
m. SCAN END (Scan Turn Around)	Not applicable
n. ANT POS (Antenna Position) AZ (Azimuth) EL (Elevation) RET (Return)	Not applicable

3.4.1.6.2.3.3.6 Simulation - The following signals normally provided by the aircraft systems shall be provided.

a. Navigational Computer Simulation - A 10 ± IV, 400 Hz, phase A signal shall be supplied to simulate the North (Vwn) and East (Vwe) Wind Velocity excitation. Load resistors of 360 ohms ± 5 percent, 1/2 watt, shall be provided to simulate the wind signal loads.

b. Vertical Reference - A three-wire synchro signal for both the pitch and roll inputs to the radar at an angle of 0 degree shall be simulated.

c. Compass Heading - A three-wire synchro signal input to the radar which simulates at 0-degree heading shall be provided.

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d. Drift Angle - A three-wire synchro signal input to the radar which simulates a 0-degree drift angle shall be provided.

e. Pressure Ratio (P/PSL) - The panel shall accept an ac input from the AN/APG-59 and provide a simulated pressure ratio by attenuating the input signal with a transfer ratio of 0.672 ± 0.007 to 1.

f. True Air Speed (TAS) (AC) - The panel shall accept the TAS (ac) excitation voltage from the Velocity Signal Computer and simulate a 1200-knot velocity by attenuating the excitation voltage with a transfer ratio 0.80 ± 0.01 to 1.

g. TAS (DC) - The panel shall accept the TAS (dc) excitation voltage from the Launch Signal Computer and simulate a 1200-knot velocity by attenuating the excitation voltage with a transfer ratio of 0.80 ± 0.01 to 1.

h. Angle of Attack (α t) - The panel shall accept an ac input from the launch signal computer and simulate a 4-degree angle of attack by attenuating the ac input with a transfer ratio of 0.080±0.002 to 1.

i. Lateral Offset Cursor - Simulated Lateral Offset signals of 6.25-miles right, 6.25 miles left, and 0 mile shall be provided for a scale factor of 1 mile/volt (dc) for selection by the LAT OFFSET CURSOR Switch. The source impedance shall be 5000 ohms ± 10 percent.

j. Main Beam Clutter - A variable MBC signal having a magnitude of 0 to -16 Vdc shall be provided by means of a 10-turn potentiometer MAN MBC. This signal shall be fed through a low-pass filter having a corner frequency of 210 Hz and shall be selected by the MBC/GSO INPUTS switch to replace fixed signals which originate in the velocity signal computer.

k. Ground Speed - A variable Ground Speed signal having a magnitude of 0 to -16 Vdc shall be provided by means of a 10-turn potentiometer MAN GRD SPEED. This signal shall be fed through a low-pass filter having a corner frequency of 210 Hz and shall be selected by the MBC/GSO INPUTS switch to replace the fixed signals which originate in the velocity signal computer.

l. Acceleration - Simulated vertical acceleration signals, having a scale factor as specified in MIL-R-81114 shall be provided representing +3 g's, -3 g's, and level flight. Selection of these signals shall be by the ACCEL switch.

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m. Altitude - Two altitude signals of 4.08 Vdc ± 2 percent and 5.26 Vdc ± 2 percent with scale factors as specified in paragraph 3.3.12.1(2) of MIL-R-81114 shall be simulated. Selection of these signals shall be by the ALT switch.

n. Climb Angle Simulation - The Panel shall accept a 25 V 400-Hz ac input and shall provide a simulated climb angle signal through a voltage divider ratio of 0.400 ± 0.004 to 1.

o. Data Link - The following test voltages shall be provided for checking the display of signals which normally come from the One-Way Data Link

(1) H	±3.50 Vdc ±3 percent
(2) Heading Error	-8.73 Vdc ±3 percent
(3) Target Relative Bearing	-3.50 Vdc ±3 percent
(4) Range to target	+8.45 Vdc ±3 percent when loaded with look.

3.4.1.6.2.3.4 MCS Control Panel

3.4.1.6.2.3.4.1 Function - This panel shall provide the structure and mounting facilities for holding the RSC and the Selector-Test Programmer of the MCS.

3.4.1.6.2.3.4.2 Form Factor - This unit shall not exceed 19 by 14 by 11-7/8 inches. The weight of the unit shall not exceed 30 pounds.

3.4.1.6.2.3.4.3 Controls - The controls provided shall be those controls contained on the AN/APG-59 system units installed in the panel. The function of the following control of the Radar Set Control (RSC) will be modified as indicated.

a. When the FUNCTION switch is moved from OFF to STBY, the Liquid Cooling System and Blower Assembly of the Bench shall be energized immediately. When interlocks on these units are satisfied, the system will be on Standby.

b. WES - A lighted pushbutton switch shall control power to the Waveguide Extension (WE) (3.4.3). The switch shall be illuminated only if the WE actually comes on.

3.4.1.6.2.3.5 Manual Built-In-Test (BIT) Panel

3.4.1.6.2.3.5.1 Function - This panel shall provide the following two functions:

a. Visual indication shall be made of the Selector-Test Programmer output control signals.

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b. Means shall be provided to initiate commands manually when the Selector-Test Programmer is not being used, and readout test points shall be furnished for monitoring purposes.

3.4.1.6.2.3.5.2 Form Factor - This unit shall not exceed 19 by 8-3/4 by 14 inches. The weight of the unit shall not exceed 20 pounds.

3.4.1.6.2.3.5.3 Controls and Interface - There shall be the following on the panel:

a. BIT SELECT - This shall be a three-position rotary switch to select either the NORM, MAN, or LAMP TEST Mode. When in the MAN Mode, the Selector-Test Programmer shall be disabled and this panel shall be activated. The LAMP TEST position shall test the lamps specified in (b). When in the NORMAL Mode, the BIT circuits will be activated by BIT or FIM in response to tape commands. In response to a tape command, a relay in the Manual BIT Panel shall permit the Test Set, Missile Control Group, OQ-50A/AWM-21 to use pushbutton commands of the Manual BIT Panel.

b. Lamps A1 Through E14 - Seventy illuminated white pushbuttons shall be provided to in initiate commands when the BIT SELECT control is set to the MAN Mode. The lights shall show the presence of a command signal when the BIT SELECT control is set to either the NORM or MAN Mode.

c. MAN BIT TGT (Manual BIT Targets) - A 10-turn potentiometer shall provide variable BIT targets when the BIT TGT control is set to the MAN position. An output from 0 to -16 Vdc shall be fed through a low-pass filter with a 3-decibel (dB) corner frequency of 21 Hz. Manual targets shall be selected by BIT TGT switch to replace the Digital Analog outputs of the Selector-Test Programmer. The output signal shall be monitored by means of test point MAN BIT TGT (3.4.1.6.2.3.5.3 (e)(6)).

d. BIT TGT (BIT Targets Select) - A two-position toggle switch shall be provided to select either Manual BIT Targets or the normal outputs of the Selector-Test Programmer. Positions shall be marked MAN and NORM.

e. TEST POINTS - The following test points shall be provided on the front panel for monitoring purposes.

(1) BIT READOUT RADAR 1 - It shall permit test probe access to signals from AN/APG-59 Radar Set Receiver and transmitter subassemblies that are selected by and routed to the Selector-Test Programmer.

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(2) BIR 2 - It shall also permit test probe access to signals ladar Set Receiver and transmitter subassemblies, buse at BIT Readout Radar 1, which are selected by and ro-Test Programmer.

(3) BIR 1 (BIT Readout Computer 1) - It shall permit test probe om the computers that are selected by and routed to the Sammer.

(4) BIR 2 (BIT Readout Computer 2) - It shall also permit test pals from the computers but different from those at BIT 1, that are selected and routed to the Selector-Test Pro;

(5) GR- It shall provide the signal ground reference for the surement.

(6) M/ble BIT Targets) - It shall permit test probe access to mestablished by the potentiometer.

(7) CCSelector-Test Programmer Ground) - It shall provide sigce to the Selector-Test Programmer when making BIT Iments.

3.4.1.6: - Self-test shall be incorporated to check the A1 throue power supplies to the Selector-Test Programmer.

a. Self-test ough E14 shall be accomplished by energizing all the lhe BIT SELECT switch.

b. Self-test ver supplies shall be accomplished by means of a green ied on the front panel for each of the power supply voltage is above 85 percent -0 percent +10 percent of nominalht shall light.

(1) -6

(2) +6

(3) +12

(4) +25

(5) +28

(6) 28

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3.4.1.6.2.3.6 Umbilical Plug and Tuning Drive Test Panel

3.4.1.6.2.3.6.1 Function - This panel, in conjunction with the Test Set Missile Control System, TS-2511C/AWG-10(FIM), shall provide the means to test the Umbilical Test Plugs of the Test Set, Umbilical Plug TS-2515D/AWM-22. The panel shall also test the four-channel (TG-75/APA-128 or TG-185/AWG-10) and single-channel (TG-76/APA-128 or TG-184/AWG-10) Tuning Drives.

3.4.1.6.2.3.6.2 Form Factor - The unit dimensions shall not exceed 19 by 7 by 15-1/2 inches. The weight of the unit shall not exceed 19 pounds.

3.4.1.6.2.3.6.3 Controls - The following controls shall be provided on the panel:

a. CHANNEL SELECT - A four-position rotary switch shall be provided to select any one of four channels of the TG-75/APA-128 or TG-185/AWG-10 Drive, Tuning. The positions of the switch shall be marked 1, 2, 3, and 4.

b. CIRCUIT BREAKERS - There shall be three pushbutton circuit breakers.

(1) 400 Hz - 115/200 V, 400 Hz, 3-phase, 10 A

(2) 28 VDC - 10 A

(3) LOAD TEST - 28 Vdc, 5 A

c. TUN DR SEL (Tuning Drive Select) - This switch shall be a two-position toggle switch to select between signals from the TG-75 and TG-76 or TG-185 and TG-184 Tuning Drive Motors and Firing interlocks without one loading the other. The two positions of the switch shall be marked SGL CHAN (Single Channel) and FOUR CHAN (Four Channel).

d. 28 VDC RESET - This normally closed switch is provided to reset certain relays which lock-up when running tests by interrupting the +28 Vdc line.

e. MNL TEST PRGM (Manual Test Program) - This switch shall be a 30-position thumb-wheel switch with two thumb wheels and three modules. The switch shall be capable of setting up, in proper sequence, the various test conditions to check for proper operation of any of the four Tuning Drives.

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f. AIM-7F - This single-pole double-throw toggle switch shall apply a 10.2 V signal to the limit output in the ON position, and other voltages (3.4.1.6.2.3.6.6.1) shall be applied in the OFF position or selected by other switching.

g. PD ILL (Pulse Doppler Illuminate) - This single-pole single-throw toggle switch shall apply +28 Vdc to the +28 Vdc delayed output line in the ON position, and the OFF position shall be an open circuit.

h. NO GO - RECYCLE - This single-pole double-throw toggle switch shall select the 3.2 Vdc signal at the recycle line in the ON position, and other voltages (3.4.1.6.2.3.6.6.3) shall be applied in the OFF position as selected by other switching.

i. NO GO - AFC/LIM (NO GO-AFC/Limit) - This double-pole double-throw toggle switch, in conjunction with other switching, shall apply the 1.4 V NO GO signals to the AFC line and the 7.9 V NO GO signal to the limit line in the ON position. In the OFF position, other switching shall select the signals on this line (3.4.1.6.2.3.6.6.1 and 3.4.1.6.2.3.6.6.2).

3.4.1.6.2.3.6.4 Indicator Lights - The following indicator lights shall be provided on the front panel.

a. TUNE-UP - A white TUNE-UP indicator light shall show that the selected Klystron Tuning Motor signal is +28 Vdc for a TUNE-UP command.

b. TUNE-DOWN - A white TUNE-DOWN indicator light shall show that the selected Klystron Tuning Motor Signal is ground for a TUNE-DOWN command.

c. FIRE INTLK (Fire Interlock) - A white Fire Interlock indicator light shall show that the tuner has been tuned and has switched to Automatic Frequency Control (AFC).

d. GO - A green GO indicator shall indicate when continuity exists during the check of the five ground tests on the Tuning Drives. It shall also light during the checks of the roll commands when the inputs are written within the proper voltage ranges.

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3.4.1.6.2.3.6.5 Electrical Connectors - Connectors to the units under test shall be as follows:

Reference Designation	Function
J1	Input Power
J2	Motor Fire, Wind
J3	Motor Fire, Fuselage
J4	AN/AIM-7D/7E, Umbilical
J5	AN/AIM-9B/9D, Umbilical
J6	Test Set Missile Control System, TS-2511C/AWG-10
J7	Drive, Tuning, TG-75/APA-128 or TG-185/AWG-10
J8	Drive, Tuning, TG-75/APA-128 or TG-185/AWG-10
J9	Drive, Tuning, TG-76/APA-128 or TG-184/AWG-10
J10	Cartridge Firing Input
J11	AN/AIM-9B/9D, Motor Fire
J12	Test Point: HOLD
J13	Test Point: LIM (Limit)
J14	Test Point: AFC
J15	Test Point: RECYCLE
J16	Test Point: FIRE INTLK (Fire Interlock)
J17	Test Point: TUN MOT DR (Tuning Motor Drive)
J18	Test Point: STBD-ROLL CMD (Starboard Roll Command)
J19	Test Point: PORT ROLL CMD (Port Roll Command)
J20	Test Point: 28 V DELAYED
J21	Test Point: SIG GND (Signal Ground)

3.4.1.6.2.3.6.6 Test Outputs - The Test Panel shall provide the output simulations listed below.

3.4.1.6.2.3.6.6.1 Limit Voltage - The following voltages shall be simulated from a 12 kΩ ±10 percent source impedance. The MNL TEST PRGM switch, and other switches, shall select the voltage to be supplied.

a. +4.1 ±0.8 Vdc for AIM-7D/7E series of missiles

b. -4.1 ±0.8 Vdc for AIM-7D/7E series of missiles

c. +7.9 ±0.16 Vdc for upper limit NO GO test

d. -7.9 ±0.16 Vdc for upper limit NO GO test

e. +10.2 ±0.2 Vdc for AIM-7F series of missiles

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3.4.1.6.2.3.6.6.2 AFC Voltages - The following voltages shall be simulated from a 75 kΩ ±10 percent source impedance. The MNL TEST PRGM and AFC/LIMIT NO GO switches shall select the one to be applied.

a. +1.4 + 0.06, - 0 Vdc (increasing from 0)

b. -(1.4 + 0.06, - 0) Vdc (increasing from 0)

c. +2.2 ±0.04 Vdc (increasing from 0)

d. -2.2 ±0.04 Vdc (increasing from 0)

e. +1.0 + 0.04, - 0 Vdc (decreasing toward 0)

f. -(1.0 + 0.04, - 0) Vdc (decreasing toward 0)

3.4.1.6.2.3.6.6.3 Recycle - The voltages shall be simulated from a source impedance of 150 kΩ ±10 percent. The MNL TEST PRGM and Recycle NO GO switches shall select which is to be applied.

a. +3.2 ±0.06 Vdc (increasing from 0)

b. +3.8 + 0, - 0.19 Vdc (increasing from 0)

c. +2.7 + 0.05, - 0 Vdc (decreasing toward 0)

3.4.1.6.2.3.6.6.4 28 Vdc Delayed - A +28 Vdc signal shall be present at this output when the PD ILL switch is ON.

3.4.1.6.2.3.6.6.5 28 Vdc Hold - When the MNL TEST PRGM switch reaches the appropriate step in the test sequence, +28 Vdc shall be applied at this output. A latching circuit shall be provided which will keep the signal on for subsequent steps until the 28 Vdc RESET switch is momentarily depressed.

3.4.1.6.2.3.6.7 Test Inputs - The Test Panel shall provide for monitoring the following inputs coming back from the Tuning Drives in response to the switching and outputs described in 3.4.1.6.2.3.6.3 and 3.4.1.6.2.3.6.6.

3.4.1.6.2.3.6.7.1 Roll Channels - The Test Panel shall verify that the Right Roll command is +18.8 Vdc ±10 percent and that the Left Roll command is -18.8 Vdc ±10 percent when checking four-channel tuners.

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3.4.1.6.2.3.6.7.2 Tuning - Provisions for indicating which of the following five conditions exists on each of the tuning inputs shall be made.

a. TUNE-UP: +28 Vdc

b. TUNE-UP with Recycle: +28 Vdc through 180 ohms

c. TUNE-DOWN: ground

d. TUNE-DOWN with Recycle: ground through 180 ohms

e. Open Circuit

3.4.1.6.2.3.6.7.3 Grounds - Provisions shall be made for checking all the separate ground circuits for the single- and four-channel tuners.

3.4.1.6.2.3.6.7.4 Firing Interlocks - Presence/absence of the +28 Vdc Firing Interlock voltage shall be monitored for the selected missile channel.

3.4.1.6.2.3.6.8 Cable - The J7, J8, and J9 connectors shall be part of a cable wired directly to the test panel. This cable shall enable any of the four Tuning Drives to be connected to the panel while resting on the drop leaf of the Receiver Section Bench Assembly. The cable shall retract into the panel when not in use.

3.4.1.6.2.3.7 Voltage Regulator

3.4.1.6.2.3.7.1 Function - An ac Voltage Regulator shall be provided for regulating the 115V, 400 Hz, Phase A which is used by Tuning Drives TG-75/APA-128 and TG-76/APA-128 or TG-185/AWG-10 and TG-184/AWG-10. This regulator shall maintain the 400 Hz voltage at 115V ±1 percent when the line voltage varies between 107 and 120V. The current carrying capability of the regulator shall be 1A minimum.

3.4.1.6.2.3.7.2 Form Factor - The Voltage Regulator shall be hermetically sealed and shall not exceed 6-1/2 by 5 by 3-1/2 inches. It shall be bracket-mounted on the inside right of the Test and Control Rack behind the MCS Control Panel. The weight of this unit shall be approximately 6 pounds.

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3.4.1.6.2.3.8 Shelf

3.4.1.6.2.3.8.1 Function - A pullout Shelf shall be provided to hold an Umbilical Test Plug or a Tuning Drive while it is being tested by the Umbilical Plug and Tuning Drive Test Panel.

3.4.1.6.2.3.8.2 Form Factor - This shall be a standard rack shelf 19 by 1-3/4 by 14 inches. The weight of the Shelf shall not exceed 9 pounds.

3.4.1.6.2.3.9 Grounding Bar - A portable, hand-held, high-voltage, insulated handle Grounding Bar shall be provided for personnel safety. A cable at least 10-feet long shall connect to the Grounding Bar and shall be terminated with a lug suitable for attachment to the main ground bus of the bench.

3.4.1.6.2.3.10 IFF Simulator Panel*2

3.4.1.6.2.3.10.1 Function - The IFF Simulator Panel shall provide test points, circuits, and indicator lights to verify proper operation of the SEA circuits.

3.4.1.6.2.3.10.2 Form Factor - This unit shall be 10 by 4 by 6 inches and designed to mount against the upper right side of the Test and Control Rack. It shall not weight more than 5 pounds.

3.4.1.6.2.3.10.3 Controls - The following controls shall be on the unit:

a. IFF SIMULATION - This switch shall be a DPDT toggle switch for selecting whether the SEA tests shall be controlled from the Table Rack or the Selector Test Programmer and together with associated relays permit exercising the operation enable, video No. 1, and IFF display video lines.

b. INTERROGATE TEST - This switch shall be a momentary pushbutton switch to apply +28 Vdc to the interrogate No. 1 circuit when this is required.

3.4.1.6.2.3.10.4 Indicator Lights - The following white indicator Lights shall be mounted on the front panel to indicate when the respective circuits in the radar are energized with +28 Vdc:

a. INTEROG #1

b. INTEROG #2

c. T C CMD

d. LONG RANGE DISPLAY ENABLE

*2 See paragraph 6.5.2

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3.4.1.6.2.3.10.5 Test Points - The following front panel test points shall provide a monitoring point for the respective signals within the radar:

a. PRETRIGGER #1 - The cable for this signal shall be terminated in 91 ohms and 100 pF.

b. RDR TRIGGER (Radar Trigger) - The cable for this signal shall be terminated in 91 ohms.

c. PULSE RDR VIDEO (Pulse Radar Video) - The cable for this signal shall be terminated in 91 ohms and 100 pF.

d. t_o PULSE - The cable for this signal shall be terminated in 91 ohms and 100 pF.

e. WARNING PULSE - The cable for this signal shall be terminated in 91 ohms and 100 pF.

f. EARLY R GATE (Early Range Gate) - The cable for this signal shall be either terminated in 91 ohms and 100 pF or returned on the video lines as they are exercised per 3.4.1.6.2.3.11.3 (a).

g. SIGNAL GRD

3.4.1.6.2.3.10.6 Test Signal - A test signal, EL STROBE JITTER shall be provided from a -6 Vdc input through a 15 kohm ±1 percent, 1/8W resistor.

3.4.1.6.3 Bench Assembly

3.4.1.6.3.1 Function - The Bench Assembly shall hold the sections of the MCS as listed below, provide limited working space for necessary testing and repair and provide power convenience outlets.

3.4.1.6.3.2 Form Factor - The Bench Assembly shall not exceed 72 by 48 by 71 inches. The weight of the Bench Assembly shall not exceed 750 pounds and shall consist of the following items:

a. Bench

b. Radar Indicator Support Structure

c. Command Indicator Support Structure

d. Unregulated Power Supply Support Structure

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e. Computer Support Structure

f. Control Indicator Support Structure

3.4.1.6.3.2.1 Bench

3.4.1.6.3.2.1.1 Function - This Bench shall support the items specified in 3.4.1.6.3.2 and the Blower Assembly and shall provide work and storage space. Three 115 Vac, single-phase, 60 Hz, duplex power convenience outlets shall be provided on both sides of the structure.

3.4.1.6.3.2.1.2 Form Factor - The Bench shall be nominal 48 by 72 by 32 inches bench. A shelf, 30 by 72 inches, shall be installed 38 inches above the bench. This shelf shall be detachable. A second shelf 72 by 48 inches, shall be installed beneath the Bench and shall be raised from deck level by 6 inches. The weight of this structure shall not exceed 500 pounds.

3.4.1.6.3.2.2 Radar Indicator Support Structure

3.4.1.6.3.2.2.1 Function - This structure shall support the Radar Indicator so that it can be viewed easily from several angles and can be rotated for easy maintenance.

3.4.1.6.3.2.2.2 Form Factor - The normal mounting points of the Radar Indicator shall be used, and the structure shall be capable of rotating the Radar Indicator 45 ±5 degree in the horizontal plane and of lifting and locking at 30 ± 5 degree in the vertical plane. The weight of this structure shall not exceed 38 pounds.

3.4.1.6.3.2.3 Command Indicator Support Structure

3.4.1.6.3.2.3.1 Function - This structure shall support the Command Indicator and shall make it accessible for repair.

3.4.1.6.3.2.3.2 Form Factor - The normal mounting points of Command Indicator shall be utilized. The structure shall be capable of lifting and locking at 30 ± 5 degree and 90 ±5 degree in the vertical plane so that the underside of the Indicator can be reached. The weight of this structure shall not exceed 14 pounds.

3.4.1.6.3.2.4 Unregulated Power Supply Support Structure

3.4.1.6.3.2.4.1 Function - This structure shall support the Unregulated Power Supply, in its normal vertical position and shall be positioned on the Bench top or the lower storage shelf.

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3.4.1.6.3.2.4.2 Form Factor - The structure shall support the unit from its normal mounting points. Maximum dimensions shall be 13 by 10 by 21 inches. The weight of this structure shall not exceed 8 pounds.

3.4.1.6.3.2.5 Computer Support Structure.

3.4.1.6.3.2.5.1 Function - The Computer Support Structure shall provide mounting for the AN/APG-59 Radar Set Velocity, Launch, and Pilot Command Signal Computers and shall distribute cooling air to them so that they can be operated and maintained.

3.4.1.6.3.2.5.2 Form Factor - The structure shall consist of a four-sided vertical plenum chamber mounted on a swivel base capable of being rotated 85 ±5 degrees in azimuth, and shall receive air at the bottom from the Blower Assembly. The three signal computers shall mount to three sides of the plenum chamber, each computer being assigned to a given side of the chamber. The computers shall be attached to the plenum by normal mounting points, and shall receive cooling air distributed through ports in the side of the plenum. Three removable plenum covers shall also be furnished, each to cover one set of computer air ports, to prevent the loss of cooling air pressure when the computers are not in place. Nominal dimensions shall be 18 by 26 by 23 inches. The weight of this unit, including covers, shall not exceed 97 pounds.

3.4.1.6.3.2.5.3 Sensing Element - A differential pressure switch in the plenum shall sense static pressure difference between the plenum and ambient air. The switch shall activate at air pressures of 5.50, -0, +0.25 inches of water and shall be electrically connected to the CMPTR AIR (3.4.1.6.2.3.2.4(g)) indicator.

3.4.1.6.3.2.6 Indicator Control Support Structure

3.4.1.6.3.2.6.1 Function - This structure shall support the Indicator Control and shall be attached to the lower surface of the overhead shelf.

3.4.1.6.3.2.6.2 Form Factor - This unit shall fasten to the underside of the shelf and support the Indicator Control from its normal mounting fixtures. Nominal dimensions shall be 17 by 8 by 12 inches. The weight of this unit shall not exceed 8 pounds.

3.4.1.6.4 Hydraulic Supply

3.4.1.6.4.1 Function - The Hydraulic Supply shall provide hydraulic power to the Table-Rack for operation of the antenna. This unit shall be

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mounted in the MCS Support Bench. It shall be capable of operating an antenna on the Antenna Maintenance Fixture located up to 40 feet from the Table Rack when suitable controls and hoses (furnished with the Maintenance Fixture) are installed.

3.4.1.6.4.2 Contents - The supply shall be a Hydraulic Unit, HD413/APM-92. Electrical cabling and hydraulic hoses provide for connection to the Table Rack and the radar antenna.

3.4.1.6.5 Liquid Cooling System

3.4.1.6.5.1 Function - The Liquid Cooling System shall provide OS25 liquid coolant to the AN/APG-59. The system shall pump, cool, and regulate coolant fluid and shall operate as either a liquid-to-air or liquid-to-seawater heat exchanger.

3.4.1.6.5.2 Form Factor - The Liquid Cooling System shall be housed in an enclosure 20 by 20 by 43 inches. Four flush mounted carrying handles, two on each side, shall be provided. A recessed control panel on the top surface of the unit shall contain all necessary connections and controls for operation. Weight, excluding hoses and coolant, shall not exceed 400 pounds. The Liquid Cooling System shall be mounted in the MCS Support Bench or at a remote station located no more than 40 feet distant from the AN/APG-59 Radar Set coolant connections.

3.4.1.6.5.3 Primary Input Power Requirements - The equipment power requirements shall be a 3-phase, 115/200 V, 60 Hz, 1600 VA.

3.4.1.6.5.4 Seawater Requirements - The liquid-to-seawater heat exchanger shall be capable of the heat transfer specified under Performance (3.4.1.6.5.6) with supplied seawater having the following characteristics.

a. Pressure - Supply pressure shall be 175 psig maximum.

b. Temperature - Maximum input temperature shall be 32°C (90°F)

c. Flow Rate - Flow Rate shall be 5 ± 0.5 gpm

3.4.1.6.5.5 Noise Level - The acoustical noise level produced by this equipment shall not exceed 80 dB (0.0002 microbar reference) when measured 6 feet from the unit.

3.4.1.6.5.6 Performance - When operating as a complete system the Liquid Cooling System shall be capable of removing 10,700 watts of heat from the coolant fluid and shall provide the following:

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a. Flow Rate - Liquid coolant flow rate shall be 2.7 gpm minimum.

b. Pressure - The pumping system shall be capable of producing the the rated flow with an external pressure drop of 135 psi maximum from supply to return connections with a pressure at the return connector which shall not exceed 35 psig. The supply pressure shall not exceed 225 psig under all flow conditions.

c. Temperature - The output temperature shall be maintained to 52 ± 5.5°C (125 ± 10°F) except when the inlet fluid temperature shall fall below 49°C (120°F).

3.4.1.6.5.7 Contents of Unit - The Liquid Cooling System shall consist of the following:

3.4.1.6.5.7.1 Pump Motor Assembly - There shall be a motor-pump assembly capable of circulating the coolant as specified in 3.4.1.6.5.6. There shall be a fan either as part of this Assembly or driven by a separate motor to circulate ambient air through the unit when the Liquid-to-Air Heat Exchanger is in use. The air flow shall be sufficient to meet performance specified in 3.4.1.6.5.6 under conditions specified in 3.4.1.6.5.7.2.

3.4.1.6.5.7.2 Liquid-to-Air Heat Exchanger - This Exchanger shall be capable of heat transfer as specified in 3.4.1.6.5.6 with ambient air temperature and altitude profile as follows:

a. 40°C (104°F) from sea level to 8,000 feet.

b. Linear decrease from 40°C (104°F) at 8,000 feet to 94°F at 10,000 feet.

3.4.1.6.5.7.3 Liquid-to-Seawater Heat Exchanger - This Exchanger shall be capable of heat transfer specified in 3.4.1.6.5.6 with seawater available as specified in 3.4.1.6.5.4. A replaceable zinc anode plug shall be provided on the seawater side.

3.4.1.6.5.7.4 Reservoir - A 2.5-gallon minimum capacity Reservoir shall be provided with a sight glass and relief valve. The relief valve shall be in the supply pressure line and shall limit the liquid coolant delivered to the AN/APG-59 Radar Set to a maximum pressure of 225 psig. The sight glass shall be graduated with at least 1/4, 1/2, 3/4, and FULL markings.

3.4.1.6.5.7.5 Control Panel - A recessed Control Panel shall be provided on the top surface of the enclosure. The panel shall mount all necessary external connections, consisting of coolant quick disconnect couplings, reservoir filler with cap, seawater fittings, and an electrical

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power receptacle. A selector valve and an ELAPSED TIME indicator conforming to MS 17321-2 shall also be provided on this panel.

3.4.1.6.5.7.5.1 Connections - The following connections shall be provided on the Control Panel:

a. Coolant - A SUPPLY and a RETURN quick disconnect connector shall be provided to mate with coolant lines specified in 3.4.1.6.5.7.6.

b. Seawater - An OUT and an IN water connector shall each be supplied. Connectors shall be 1/2-inch outside diameter tube, stainless steel fittings with 3/4-16 UNF-2A standard thread.

c. Electrical - An electrical receptacle shall be provided to mate with MS3126-F14-12S.

3.4.1.6.5.7.6 Coolant Lines - Two 5/8-inch diameter lines 40 ± 1-feet long with quick disconnect fittings at each end shall be provided. Connectors shall be per MS 33656/8.

3.4.1.6.5.8 Controls - The selection of air or seawater as the heat exchange medium shall be made by a manual selector valve located on the Control Panel (3.4.1.6.5.7.5). Positions shall be marked WATER and AIR. Electrical power shall be applied through the LIQ COOL (3.4.1.6.2.3.2.3(e)) pushbutton control.

3.4.1.6.5.9 Protective Devices

3.4.1.6.5.9.1 Safety Interlocks - The following Safety Interlocks shall be provided:

a. Temperature High - A temperature switch to allow interlocking of the MCS 28 Vdc Standby and Operate signal with the fluid temperature shall be interrupted when the temperature of the coolant at the RETURN Coolant conntector exceeds 107 ± 3°C (225 ±5°V) on rising temperature. It shall close at 99°C (210°F) minimum on a falling temperature. The contacts shall be rated for 2 A at 28 Vdc. The HIGH TEMP indicator shall be illuminated when the switch is open.

b. Temperature Low - A temperature switch shall allow +28 Vdc to be applied to the LOW TEMP indicator to show when the coolant temperature is below the regulated temperature (3.4.1.6.5.6(c)) range. The switch shall be interrupted when the temperature of the coolant at the SUPPLY Coolant connector reaches 43 +3°C (110 ±5°F) in a rising direction.

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It shall close at 32°C (+90°F) minimum on a falling temperature. The contacts shall be rated for 2A at 28 Vdc.

c. Flow - A flow switch shall allow interlocking of the MCS 28 Vdc Standby and Operate signal with the fluid flow. It shall be closed when the fluid flow exceeds 2.75 gpm and shall be interrupted when the flow falls below 2.5 gpm. The contacts shall be rated for 2A at 28 Vdc. The LIQ FLOW indicator shall be illuminated when the switch is open.

3.4.1.6.5.9.2 Filters

a. Magnetic Trap - A magnetic trap shall be provided in the coolant fluid supply line which shall accumulate magnetic particles from the fluid. When the trap collects magnetic impurities sufficient to close an electrical circuit, the MAG TRAP indicator shall be illuminated.

b. Coolant Fluid - A nominal 30-micron filter shall be provided in the supply line.

c. Seawater - A bronze seawater strainer of 50 to 125 micron size shall be provided in the inlet line.

3.4.1.6.6 Blower Assembly

3.4.1.6.6.1 Function - The Blower Assembly shall provide filtered cooling air to the AN/APG-59 Radar Set, the Computer Support Structure, and the Unregulated Power Supply mounted on the Unregulated Power Supply Support Structure.

3.4.1.6.6.2 Form Factor - The Blower Assembly shall be 20 by 25 by 28 inches exclusive of the ducting and shall be mounted beneath the Bench (3.4.1.6.3.2.1). The weight of the assembly shall not exceed 300 pounds.

3.4.1.6.6.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in section 3.4.1.4.

3.4.1.6.6.4 Primary Input Power Requirements - The Blower Assembly shall operate with input power of 3-phase, 115/200 Vac, 60 Hz, 2250 Va maximum.

3.4.1.6.6.5 Noise Level - Acoustical noise levels shall not be produced from this Assembly in excess of 85 dB (0.0002 microbar reference) when measured 6 feet from the unit and 5 feet above the deck.

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3.4.1 6.6.6 Performance - The Blower Assembly shall provide a minimum of 10.5 pounds per minute (ppm) of air at a pressure head of 12.8 inches of water. The thermal rise across the unit shall not exceed 9°C (16°F), and energy dissipated in the motor shall not be carried into the output air stream.

3.4.1.6.6.7 Contents of Unit - The Blower Assembly shall consist of the following:

3.4.1.6.6.7.1 Centrifugal Fan - The fan shall be of a continuous operation design and shall deliver sufficient air to comply with performance requirement specified in 3.4.1.6.6.6. The unit shall be 17 by 20 by 25 inches including the air inlet and exhaust unit. Air inlet and exhaust shall be 4-1/2 inches in diameter. Weight shall not exceed 275 pounds.

3.4.1.6.6.7.2 Transition Unit - The Transition Unit shall contain an acoustical baffle and air filter. This unit shall mate the 4-1/2 inch diameter air inlet of the fan through a 12-1/2 inches long duct to an 18 by 18-inch air inlet. The air inlet shall have an oil treated, reusable air filter covering the entry. An acoustical foam baffle shall reduce the fan acoustical noise to the level specified in 3.4.1.6.6.5.

3.4.1.6.6.7.3 Distribution Ducts - The air exhaust of the centrifugal fan shall mate with a series of ducts which shall distribute the air to units requiring cooling air.

3.4.1.6.6.8 Controls - The Blower Assembly shall be activated from the Test and Control Rack by either placing the Radar Set Control (RSC) FUNCTION switch to STDB or OPR or be depressing the AIR COOL push-button switch.

3.4.1.6.6.9 Connections - The electrical connector shall be MS3100R20-4P.

3.4.1.6.7 Remote Safety Unit

3.4.1.6.7.1 Function - This unit shall provide an emergency cutoff for the system and shall be located on the MCS Support Structure at the base of the left member of the rear "A" frame when viewed facing the antenna dish. The unit shall also provide a warning indicator, two 115 VAC single-phase, 60 Hz convenience outlets, and the Klystron Power Amplifier (KPA) relay test connector.

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3.4.1.6.7.2 Form Factor - This unit shall not exceed 4 by 7 by 4 inches. The weight of the unit shall not exceed 2 pounds.

3.4.1.6.7.3 Controls - The unit shall provide a red momentary pushbutton switch which when depressed, shall operate in conjunction with the Power Control Panel (3.4.1.6.2.3.2) to remove all power from the Table-Rack. When the control has been operated, power can be reapplied only by operation of the PWR ON (3.4.1.6.2.3.2.3(a)) switch. A 1000 ohm, 1-1/2 watts potentiometer shall be provided and when used with the Table-Rack Cable Assembly (3.4.4.5.2(m)) shall be capable of testing the body current relays in the KPA.

3.4.1.6.7.4 Indication - The unit shall have a red warning indicator light which shall be continuously illuminated whenever transmitter high voltage is present in the AN/APG-59 Radar Set. The warning light shall commence flashing at any time that the conditions are such as to permit RF radiation from the antenna.

3.4.1.6.8 Test Accessories

3.4.1.6.8.1 Convenience Outlets - These outlets shall supply single-phase, 115 Vac, 60-Hz power and three-phase and single-phase 400-Hz power. Two duplex outlets located both on the left and right sides of the Rack Assembly shall provide 60 Hz power, and 400 Hz power shall be provided by one outlet each on the left and right sides of the Rack Assembly. A three-phase, 400-Hz power outlet shall also be provided on the left side of the Rack Assembly.

3.4.1.6.8.2 Modulator Test Cover Assembly

3.4.1.6.8.2.1 Function - The Modulator Test Cover Assembly shall replace the Modulator Housing cover during testing and maintenance of the RF section of the radar set. The Test Cover shall provide safe access to the units within the Modulator Housing.

3.4.1.6.8.2.2 Form Factor - The Modulator Test Cover Assembly shall be capable of attachment directly to the Modulator Housing. Transparent access covers shall provide visibility of the RF Section and safe access to the section shall be provided by suitable interlocks when the covers are opened. The Cover Assembly shall include a capacitor load which can be used to load the output of the Pulser when the Klystron Power Amplifier is disconnected while checking the Pulser. It shall also contain a capacitor divider for monitoring the output pulse of the Pulser and bring the divided signal to the test points on the front panel. A shelf and tie-down straps shall be provided to mount a Multimeter. An accessory box with

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a hinged cover and latch shall be provided for storage of the High Voltage Probe (not supplied). The overall dimensions of the Cover shall be 31-5/8 by 13-3/8 by 6-3/8 inches. The weight of the unit shall not exceed 15 pounds.

3.4.1.6.8.2.3 High-Voltage Probe - A high-voltage probe shall provided for use in conjunction with the Modulator Test Cover Assembly to make direct diagnostic measurements up to 20,000 Vdc on the Pulser and high-voltage power supplies of the radar. The probe characteristics are:

a. The probe shall provide an attenuation of 100 to 1 of the input voltage.

b. The probe shall be suitable for use with the AN/PSM-4 or other multimeters on the 250 V scale.

c. The probe shall be equipped with banana plug connectors and a closed ground lug.

3.4.1.6.8.2.4 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H-15362.

3.4.1.6.8.3 Kit, Stalo Extension

3.4.1.6.8.3.1 Function - The Stalo Extension Kit shall permit operation of the RF Oscillator at the Table-Rack when it is removed from the system.

3.4.1.6.8.3.2 Form Factor - The Stalo Extension Kit shall consist of an electrical Patch Cable Assembly and an Air Adapter.

3.4.1.6.8.3.2.1 Patch Cable Assembly - The Patch Cable Assembly shall be a cable 36 ± 1-inches long. It shall connect the RF Oscillator to the system through connectors 3J1 and 3P1, 3J2, and 3P2, 3J3 and 3P3, 3J4 and 3P4, and 3J5 and 3P5. These connectors shall be wired pin-to-pin with wire per MIL-W-16878 and RG195/U submininature coaxial cable per MIL-C-17/70 as required.

3.4.1.6.8.3.2.2 Air Adapter - The Air Adapter shall not exceed 65 ± 1 inches in length and shall furnish cooling air from the air connection of the Electrical Equipment Rack to the RF Oscillator.

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3.4.1.6.8.4 Kit, Waveguide Shelf Extension

3.4.1.6.8.4.1 Function - The Waveguide Shelf Extension Kit shall provide the means by which the Receiver Waveguide Assembly and the Transmitter Waveguide Assembly of the AN/APG-59 Radar Set and the components mounted thereon are made accessible for adjustment, servicing, and maintenance. The Waveguide Shelf Extension Kit is replaceable by the Waveguide Extension MX 8491/AWG-10 and in some locations the Radar Maintenance Table-Rack will be so configured. Some cables from the Waveguide Shelf Extension Kit are retained when replacing it with the Waveguide Extension.

3.4.1.6.8.4.2 Form Factor - The Extension Kit shall be in the form of Posts, Microwave Shelf Extensions, Cable Assemblies, and other accessories which shall permit the Radar Set Receiver and Transmitter Waveguide Assemblies to be lowered a nominal 10 inches and operated in the lowered position.

3.4.1.6.8.4.3 Contents - The Waveguide Extension Kit shall consists of the following items:

a. Posts, Extension Tool

b. Extensions, Microwave Shelf

c. Extension Air Duct

d. Cable Assemblies

3.4.1.6.8.4.3.1 Post Extension Tool

3.4.1.6.8.4.3.1.1 Function - Four identical Post Extension Tools shall be furnished. The Posts shall provide the means for supporting the Radar Set Receiving and Transmitting Waveguide Assemblies in the lowered position.

3.4.1.6.8.4.3.1.2 Form Factor - The Post Extension Tool shall be made from hexagonal bar stock and shall have an effective length of 10 inches.

3.4.1.6.8.4.3.1.3 Marking - Each Post shall be marked: USED WITH MICROWAVE SHELF 4 REQ'D SHELF EXTENDER PART OF RADAR MAINTENANCE TABLE RACK.

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3.4.1.6.8.4.3.2 Extension, Microwave Shelf

3.4.1.6.8.4.3.2.1 Function - Three identical Microwave Shelf Extensions shall be provided. These Extensions shall provide waveguide continuity to the Radar Set Receiver and Transmitter Waveguide Assemblies when they are lowered for access.

3.4.1.6.8.4.3.2.2 Form Factor - The Microwave Shelf Extensions shall be constructed from nominal 1/2-inch by 1-inch waveguide. Each end of the extensions shall be fitted with waveguide flanges. Each extension shall be a nominal 10-inches long flange surface-to-flange surface.

3.4.1.6.8.4.3.2.3 Markings - Each extension shall be marked: USED WITH MICROWAVE SHELF 3 REQ'D SHELF EXTENSION PART OF RADAR MAINTENANCE TABLE RACK.

3.4.1.6.8.4.3.3 Extension, Air Duct

3.4.1.6.8.4.3.3.1 Function - The Extension Air Duct shall permit cooling air to be supplied to the Para Amp Pump Housing of the AN/APG-59 Radar Set when the Receiver and Transmitter Waveguide Assemblies of the Radar Set are lowered for access.

3.4.1.6.8.4.3.3.2 Form Factor - The Extension Air Duct shall be fabricated from rectangular tubing a nominal 1 by 1/2-inch with an overall finished length of 11-1/4 inches. The Air Duct shall be fitted with attachment fittings and shall be sealed at the ends. The Air Duct shall be offset and shall contain an air inlet and air outlet, each approximately 5/8 inch in diameter arranged so that they are in the same plane effectively 10-inches apart.

3.4.1.6.8.4.3.3.3 Markings - The Extension Air Duct shall be marked: USED WITH MICROWAVE SHELF AIR DUCT EXTENSION TOOL PART OF RADAR MAINTENANCE TABLE RACK.

3.4.1.6.8.4.3.4 Cable Assemblies - One each of the following Cable Assemblies shall be provided to permit normal operation of the AN/APG-59 Radar Set Receiver and Transmitter Waveguide Assemblies when they are lowered for access.

a. Receiver Waveguide Assembly, Patch Cable Assembly No. 1 - This Assembly shall consist of four individual cables 20 ± 1-inches long. The individual cables shall be formed into one main assembly, leaving the individual cables free at both ends to mate with system connectors 2A8J12

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and 2A8P12; 2A8J13 and 2A8P13; 2A8J14 and 2A8P14; 2A8J27 and 2A8P27. All connectors shall be wired with RG-195/U subminiature coaxial cable per MIL-C-17/70.

b. Receiver Waveguide Assembly, Patch Cable Assembly No.2 - This Assembly shall consist of four individual cables 20 ± 1-inches long. The individual cables shall be formed into one main assembly, leaving the individual cables free at both ends to mate with system connectors 2A8J11 and 2A8P11; 2A8J15 and 2A8P15; 2A8J16 and 2A8P16; 2A8J28 and 2A8P28. All connectors shall be wired with RG-195-U subminiature coaxial cable per MIL-C-17/70.

c. Receiver Waveguide Assembly, Patch Cable Assembly No. 3 - This Assembly shall consist of three individual cables 16 ± 1-inches long. The individual cables shall be formed into one main assembly, leaving the individual cables free at both ends to mate with system connectors 2A8J17 and 2A8P17; 2A8J18 and 2A8P18; 2A8J19 and 2A8P19. All connectors shall be wired with MIL-W-16878 high-voltage wire.

d. Transmitting Waveguide Assembly, Patch Cable Assembly - This assembly shall consist of a cable 20 ± 1-inches long. The Cable Assembly shall mate with system connectors 2A2J2 and 2A2P2. The connectors shall be wired pin to pin with wire per MIL-W-16878. Pins H and B shall be shield tie points at each end of the cable.

3.4.1.6.8.5 Kit, Waveguide Run

3.4.1.6.8.5.1 Function - The Waveguide Run Kit shall provide a means to transfer RF energy from the Radar Set to the Radio Frequency Noise Analyzer, Electronic Units.

3.4.1.6.8.5.2 Form Factor - The Waveguide Run Kit shall be in the form of Waveguide Assemblies, Cable Assemblies, Supports, Bracketry, O-Rings and Hardware which shall make the connection between the Radar Set and the Radio Frequency Noise Analyzer, Electronic Units.

3.4.1.6.8.5.3 Contents - The Waveguide Run Kit shall consist of the following items:

3.4.1.6.8.5.3.1 Rigid and Flexible Waveguide Assemblies - These shall be provided as required to connect the STALO of the MCS to the RFNA in the various modes needed to perform analysis of its proper operation.

3.4.1.6.8.5.3.2 Cable Assemblies These shall be a group of cables, adapters, fittings, and associated hardware to provide the facilities for connecting the Cabinet Electronic Equipment (RFNA) to the MCS in the

Commander
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various ways and combinations as required to perform the analysis of the MCS operation by the RFNA. The cables shall be of sufficient length to reach between connection points in a standard bench installation but within permissable limits of attenuation.

3.4.1.6.8.5.3.3 Mounting and Support Brackets - The necessary mounting and support brackets and hardware shall be provided to attach the waveguide to the Radar Maintenance Table-Rack.

3.4.1.6.8.6 Stalo Terminations - Two electrical dummy loads shall be provided to terminate the outputs (f_o and f_o + 30) of the Radio Frequency Oscillator (Stalo) O-1330C/APG-59, whenever the unit is operated in an extended position or when either or both of the output ports are disconnected from their corresponding flex guide assemblies. The two terminations shall be identical to each other and shall meet the following requirements:

a. Frequency Range: 8.5 to 10.4 GHz

b. Maximum VSWR: 1.20 over the frequency range

c. Average Power to the Load: 250 mW (max)

d. Peak Power to the Load: 250 mW (max)

e. Mechanical Interface: The terminations shall have choke flanges for 1/2-inch W/C with suitable captive hardware. They shall attach to the waveguide bandpass filter 3A1FL1 and/or the power monitor assembly 3A1M1. (Assemblies 3A1FL1 and 3A1M1 have cover flanges similar to UG-135/U, except they shall have 8-32-2B tapped holes.)

3.4.1.6.8.7 DC Voltage Source - A variable dc voltage source shall be provided with the Table Rack, Radar Maintenance as an auxiliary item for use, as required during test and maintenance. It shall be identical to the item provided for Flightline Maintenance in the Cable Assembly Group OA-827A/AWM-22, as specified by MIL-T-81516 paragraph 3.4.5.5.n.

3.4.1.6.8.8 Extender Bracket

3.4.1.6.8.8.1 Function - An Extender Bracket shall be provided for extension and support of the Pitch and Roll Servo Assembly of the Velocity Signal Computer. The assembly shall fasten to the bracket in such a manner that all resolvers are accessible for alignment adjustments. It shall be possible for one person to remove the assembly from the computer and mount it on the alignment bracket unaided. Disconnection or interruption of the resolver circuits at 15A1P2 for alignment purposes shall be possible.

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3.4.1.6.8.8.2 Form Factor - The Extender Bracket shall consist of a bracket 12 inches by 9 inches by 6-3/8 inches, constructed such that the Pitch and Roll Assembly may be mounted on it. It shall be constructed such as to permit its being placed on the bench working surface in at least three positions and remain stable, in order to provide access to the servo alignment screws. Cabling shall be provided to permit extending Pitch and Roll Servo Assembly 4 feet from the Velocity Signal Computer while maintaining proper electrical integrity to perform all alignments.

3.4.1.6.8.9 Relay Tester

3.4.1.6.8.9.1 Function - The relay tester shall provide testing for the standard eight-pin crystal can plug in relays used in the AWG-10 and the current sensitive relays used in the high-voltage power supplies. The relay tester shall be able to test certain relays in their sockets. The relays to be tested shall include the following types:

a. Voltage Relays: 578R040H01 and H02
578R145H01, H02, H03, H07, H08, and H09
578R147H01 and H02

b. Current Relays: 578R040H11
578R147H10

c. Latching Relay: 576R148H03

d. Reed Relay: 579R760H01

3.4.1.6.8.9.2 Power Requirements - The relay tester shall be capable of operating from a 50 to 400 Hz, 115 V source without any of the primary circuit breakers on the bench being closed. The relay tester shall be capable of being operated without being dependent upon the bench and shall have suitable circuit protection.

3.4.1.6.8.9.3 Coil Voltage - Voltage to the relay socket coil terminals shall be applied to binding posts which shall serve as test points for the coil voltage, they shall enable the application of coil voltage to solder hook-type relays such as 478R145H05. Coil voltage of relays whose coil voltage is 26.5 V nominal shall be applied by the actuation of a momentary pushbutton switch. Switching shall be provided to control latching relay testing whereby it shall be possible to latch the relay and reset it.

3.4.1.6.8.9.4 Current Relay Coil Current - Provisions shall be made to test the current sensitive relays in socket 577R480H01. The test circuit shall vary the coil current to determine the specific current at which

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the relay becomes energized. Protection (fusing) shall be provided such that the relay under test or the tester shall not be damaged during testing. A meter shall display coil current.

3.4.1.6.8.9.5 Overload Relays in Circuits - The tester shall provide testing for the following listed relays with their associated circuitry while the relays remain in their sockets. Power shall be supplied to the specified terminals in the following table while the current to the circuit is measured to an accuracy of ±5 percent.

Relay	Test Circuit Connections	Allowable Operating Current	Test Circuit Voltage	Relay Monitor Points
2A5K2	E29 J1/1(gnd)	73 - 95 mA	+6.0 V	J1/4 & J1/8
2A5K2	E29 J1/5	436 - 600 mA	+5.5 V	J1/4 & J1/8
2A5K4	J1/1(gnd) E27	604 - 838 mA	-5.5 V	J1/4 & J1/8
2A5K5	T2/3 J1/1(gnd)	1.6 - 2.0 mA	+24 V	J1/18 & J1/17
2A6K1	J1/P J1/C	50 - 60 mA	+5.5 V	J1/L & J1/8

When the operating current between the points listed in column 2 is within the limits specified in column 3, the circuits are functioning properly. Relay monitor points (for 3.4.1.6.8.9.6) specified in column 5.

3.4.1.6.8.9.5.1 Overload Test Controls - Relay tester controls shall be capable of controlling current over range required for each relay to be tested. Tester circuits shall be protected against damage by large in-rush currents by large capacitors connected across several of the test circuit connections.

3.4.1.6.8.9.5.2 Harnesses - A harness shall be provided for the relays in the High-Voltage Power Supply and one for the relays in the CW Illuminator Power Supply. The harnesses shall be long enough to permit testing while the power supplies remain in the Modulator Housing with the electrical connectors disconnected.

3.4.1.6.8.9.6 Relay Contact Continuity - Proper relay contact continuity shall be determined by visual indication. Test current shall be 10 mA dc +0, -20 percent. Open circuit voltage shall not exceed 6 Vdc. (A lamp may not be used due to initial in-rush current.)

3.4.1.6.8.9.7 Controls - All operating controls shall be suitably marked so that an operator may operate the tester without reference to supporting documents (manuals or check lists). Controls for each of the tests and accept/reject criteria for each test shall be evident.

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3.4.1.6.8.10 Waveguide (W/G) Switch Bypass - Provisions to disable and bypass the Waveguide Switch, (2A2S1), mounted on the Transmitting Waveguide Assembly, in order to make it impossible to radiate from the antenna when the Radar Set is mounted on the bench in the shop.

3.4.1.6.8.10.1 Waveguide Adaptor - The waveguide adaptor shall permanently connect the output of the Transmit Coupler (2A2DC2) to the input of the Dummy Load Coupler (2A2DC1). The waveguide adaptor shall be installed in place of the Waveguide Switch when the Radar Set is mounted on the bench. The Waveguide Switch, upon removal, may be stored in the bench for reinstallation when the Radar Set is removed from the bench.

3.4.1.6.8.10.2 Bracket Assembly - A bracket assembly containing a dummy jack shall be provided. The jack shall mate with the Waveguide Switch connector 2A2S1P1. The dummy jack shall contain jumpers to bypass the +25 V enable signal normally interlocked by the Waveguide Switch. The dummy jack shall also contain provisions to feed the coil voltage for the Waveguide Switch to the RAD ANT indicator on the Aircraft Simulator panel via the bench harness.

3.4.1.6.8.11 Test Set Interface - The Table Rack, Radar Maintenance shall provide any necessary interface with other items of the Test Set described herein. Such interface shall include mechanical mounting provisions, electrical connections for power, and test signals to the extent required for efficient maintenance routines. These provisions shall be made for, but not limited to, the following items:

a. Antenna Maintenance Fixture

b. Waveguide Extension

c. Test Set, Missile Control System

d. Electronic Units, Radio Frequency Noise

3.4.2 Fixture, Antenna Maintenance MX-7340/AWM-21

3.4.2.1 Function - The equipment shall provide a separate facility to service the AN/APG-59 Antenna. It shall permit antenna maintenance without seriously interfering with work on the Table-Rack.

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3.4.2.2 Form Factor - The Antenna Maintenance Fixture shall be deck mounted on a 22-1/4 by 24-5/8-inches rectangular base. The overall dimensions shall be 45-3/4 by 24-5/8 by 75 inches with the guard ring folded and shall be 45-3/4 by 55-1/4 inches with the guard ring open. The total weight of the equipment including cables shall not exceed 450 pounds.

3.4.2.3 Materials and Parts - Materials and parts of the gear reducer, and hand wheel of the Antenna Support Fixture shall be of the best commerical quality, shall meet the performance and environmental requirements imposed by the equipment and shall be protected from corrosion so that its form and function shall not be impaired.

3.4.2.4 MTBF - When maintained in accordance with applicable preventive maintenance procedures, the equipment shall demonstrate at least 6,000 hours MTBF. Accept-reject criteria shall be those defined by paragraph 4.2.8.1 of MIL-STD-781.

3.4.2.5 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H-15362.

3.4.2.6 Primary Input Power Requirements - The Antenna Maintenance Fixture shall provide Outputs (3.4.2.7.2.4) and shall give specified performance when operating from the power sources having limits specified in 3.4.2.6(a) and 3.4.2.6(b). Operation below the specified frequencies shall result in no damage to the unit provided the voltage decreases proportionately (zero voltage at zero frequency). The unit shall resume normal operation when the frequency returns within specified limits. Power required shall have the following characteristics:

a. Three-phase, 109 to 121 Vac (line to neutral), 380 to 420 Hz, not to exceed 865 VA per phase.

b. 27 ± 2 Vdc, 150 watts

3.4.2.7 Contents of Unit - The Antenna Maintenance Fixture shall consist of the following items:

a. Antenna Support Fixture

b. Antenna Control Box

c. Hose Assemblies, Hydraulic

d. Cable Assemblies, Electrical

e. Hydraulic Panel

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3.4.2.7.1 Antenna Support Fixture

3.4.2.7.1.1 Function - The Antenna Support Fixture shall physically support the AN/APG-59 Antenna for servicing. The fixture shall also incorporate provisions for holding the Antenna Control Box.

3.4.2.7.1.2 Form Factor - The Antenna Support Fixture shall be designed for deck mounting either by bolting or welding. Two vertical side members shall extend upwards from a 22-1/4 by 24-5/8 inches rectangular base to position the antenna support which shall be composed of a rotating cradle and a mounting plate. A cradle, 39-3/4 inches above the deck, shall provide the fixture upon which a mounting plate rests. The mounting plate, in the horizontal position, shall hold the AN/APG-59 Antenna Base Assembly which shall be bolted to its surface. The mounting plate shall be capable of 360-degree azimuthal rotation and shall permit locking securely in any desired position. The cradle shall be capable of being tilted-90 degrees from the horizontal plane by use of a hand wheel and gear reducer. An aluminum guard rail, pivoted from an axis 14-1/2 inches above the mounting plate, shall encircle the antenna and shall act as a safety device to prevent accidental contact with the antenna when it is operated in a scanning mode. The guard rail shall be capable of being stowed out of the way in a vertical position when the antenna is not scanning. An extendable oil pan shall be mounted below the cradle and between the vertical support members to retain hydraulic fluid leakage from the antenna connections. An interlock shall prevent antenna electrical operation any time the antenna cradle is rotated from its horizontal plane position.

3.4.2.7.2 Antenna Control Box

3.4.2.7.2.1 Function - The Antenna Control Box shall provide the means to allow the operator to check the electromechanical functions associated with the AN/APG-59 antenna. The unit shall allow the Antenna to be positioned either by manual control or by automatic scan.

3.4.2.7.2.2 Form Factor - The Antenna Control Box shall be an enclosed control panel with dimensions of 12 by 11 by 7-1/2 inches with a 1-1/4 inch mounting flange along one end shall provide switches, controls, test points and connectors as required to perform the function described in paragraph 3.4.2.1. It shall be bracket mounted and accessible from the front of the Antenna Support Fixture.

3.4.2.7.2.3 Inputs - Inputs to the Antenna Control Box shall be as follows:

a. AC power per 3.4.2.6(a)

b. DC power per 3.4.2.6(b)

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c. AN/APG-59 Antenna Azimuth Potentiometer output: 0.346 ± 11 percent Vac/degree off boresight.

d. AN/APG-59 Antenna Elevation Potentiometer output: 0.346 ± 11 percent Vac/degree off boresight.

e. Antenna Position Indicator (API) outputs: Four signal leads with either 0 to 1.2 Vdc or 11 to 13 Vdc.

3.4.2.7.2.4 Outputs - Outputs from the Antenna Control Box shall be as follows:

a. AN/APG-59 Antenna Azimuth Valve Command Signal: 1.63 ± 20 percent milliamperes (mA) per degree of position error.

b. AN/APG-59 Antenna Elevation Valve Command Signal: 1.14 ± 20 percent mA per degree of position error.

c. AN/APG-59 Antenna Azimuth and Elevation Induction Potentiometer Excitation: Phase A, 26.4 Vac ± 8 percent 400 Hz.

d. AN/APG-59 Antenna Feedhorn Spin Motor Excitation: 3-phase 115/200 Vac ± 5 percent, 400 Hz.

e. Simulated Acquisition: 27 ± 2 Vdc

f. Monopulse Mode Select: 27 ± 2 Vdc

g. 12 VDC B+ Supply: 12 ± 0.6 Vdc

h. BIT Control Number 1: 27 ± 2 Vdc

i. BIT Control Number 2: 27 ± 2 Vdc

j. BIT Control Number 3: 27 ± 2 Vdc

k. BIT Control Number 4: 27 ± 2 Vdc

l. BIT Control Number 5: 27 ± 2 Vdc

m. Spoiler Control: 27 ± 2 Vdc

3.4.2.7.2.5 Controls - The following controls shall be provided on the front of the Antenna Control Box. As associated indicator light shall be provided for 3.4.2.7.2.5(d).

a. SCAN MOTOR POWER - A two-position, ON-OFF, toggle switch shall be provided on the panel for the purpose of applying 115/200 Vac, 400 Hz, 3-phase power to the scan drive motor in the Antenna Control Box.

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b. AUTO SCAN DIRECTION (Automatic Scan Direction) - A three-position switch shall be provided for the purpose of programming three separate scan commands into the AN/APG-59 Antenna. Positions shall be marked AZ, EL, and 45 degree. The commanded scan coverage shall be 110 degree + 10 degree -0 degree in the azimuth plane, 110 degrees + 10 degree -0 degree in the elevation plane, and a simulated roll of 45 degrees which shall drive the Antenna in azimuth and elevation simultaneously.

c. SPIN MOTOR POWER - A two-position, ON-OFF toggle switch shall be provided to apply 115/200 Vac, 400 Hz, 3-phase power to the AN/APG-59 Antenna Feedhorn Motor.

d. HYDRAULICS - A two-position, ON-OFF toggle switch shall be provided for control of the Hydraulic Supply to furnish hydraulic power to the AN/APG-59 Antenna. A red lamp shall be located beside the toggle switch to indicate that hydraulic power has been applied to the Antenna.

e. SCAN CMD (Scan Command Select) - A two-position AUTO, MAN (Manual) toggle switch shall be provided. The AUTO position shall provide a scan rate of 60 ± 6 degrees per second. The MAN position shall provide the facility to position the antenna by use of the AZIMUTH MANUAL COMMAND and ELEVATION MANUAL COMMAND controls.

f. AZIMUTH MANUAL COMMAND - A potentiometer shall be provided to position the AN/APG-59 Antenna to any static azimuth position desired. This potentiometer shall provide an input to the azimuth scan command servo amplifier when selected by the SCAN CMD switch.

g. ELEVATION MANUAL COMMAND - A potentiometer shall be provided to position the AN/APG-59 Antenna to any static elevation position desired. This potentiometer shall provide an input to the elevation scan command Servo Amplifier when selected by SCAN CMD switch.

h. WG/SP TEST (Waveguide/Spoiler Test) - A three-position toggle switch shall be provided to apply ±28 Vdc to either the AN/APG-59 Antenna Waveguide Switch or the Spoiler. Positions shall be marked WAVEGUIDE, OFF, and SPOILER.

i. SPIN MOTOR - A two-position toggle switch shall be provided to control the spin motor drive for the AN/APG-59 Antenna Feedhorn. The positions shall be labeled SPIN and STOP.

g. API +28 -VDC BIT (Antenna Position Indicator, BIT Relay) - A five-position switch shall be used to provide the facility for energizing the

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BIT relays on the AN/APG-59 Antenna and to provide the means for B+ excitation to the API photo sensors. Positions shall be marked OFF, 1, 2, 3, and 4.

3.4.2.7.2.6 Protective Devices - The following protective devices shall be provided on the front of the Antenna Control Box:

a. Circuit Breakers - Two circuit breakers shall be provided to protect the primary power source from overloading due to an electrical malfunction on the Antenna Maintenance Fixture.

(1) +28 Vdc - 5A

(2) 400 Hz - 115/200 Vac, 3 phase, 7-1/2A

3.4.2.7.2.7 Electrical Connectors - Three connectors shall be provided on the control panel for interconnection between the AN/APG-59 Antenna, the Table-Rack and the primary power source.

3.4.2.7.2.8 Test Points - Five Test Points (API OUTPUTS) shall be provided on the control panel to monitor the AN/APG-59 Antenna outputs. Test points shall be marked A, B, C, D, and GRD.

3.4.2.7.2.9 Servo Amplifier - The Antenna Control Box shall contain a Servo Amplifier for the azimuth channel and one for the elevation channel. The amplifiers shall accept two inputs:

a. Antenna position information as specified in paragraphs 3.4.2.7.2.3 (c) and (d)

b. SCAN CMD of either AUTO or MANUAL as selected (See paragraph 3.4.2.7.2.5(e)) and provide an output as specified in paragraphs 3.4.2.7.2.4 (a) and (b).

3.4.2.7.3 Hose Assembly Hydraulic - Two hydraulic hose assemblies shall be 480 ± 5 inches in length, consisting of insulated hydraulic hoses to connect the AN/APG-59 Antenna on the Antenna Maintenance Fixture to the Hydraulic Panel. Two additional hose assemblies shall be provided to connect the Hydraulic Pumping Unit to the Hydraulic Panel when this panel is installed on the Table Rack Radar Maintenance. The hoses shall be medium pressure with teflon lining and corrosion-resistant braid reinforcement. The Assembly shall conform to MIL-H-25579, MIL-H-27267 and MIL-F-27272. The hoses shall have end fittings to mate with standard MS flareless tube fittings per MS-33514.

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3.4.2.7.4 Cable Assemblies, Electrical

3.4.2.7.4.1 Function - The Electrical Cable Assemblies shall provide the patch cables necessary to operate the specified electrical functions of the Antenna Maintenance Fixture.

3.4.2.7.4.2 Cable Assembly, Primary Power, 64W2 - The Primary Power Cable shall be nominally 20 feet in length including connectors and pigtails. The cable shall divide into two branches, each nominally 14 feet in length, including pigtails. The cable shall provide 28 Vdc power and 115 V, 400-Hz power connections to the Antenna Control Box.

3.4.2.7.4.3 Cable Assembly, Hydraulic Interlock, 64W3 - The Hydraulic Interlock Cable Assembly shall be 480 ± 2 inches in length including connectors. It shall provide safety interlock with the Hydraulic Panel so that hydraulic power cannot inadvertently be applied to the Antenna Maintenance Fixture when the bench mounted antenna is selected. The cable shall divide into two branches, one for 60 cycle 3-phase and the other for 28 Vdc power.

3.4.2.7.4.4 Cable Assembly, Antenna Interconnecting, 64W1 - The Antenna Interconnecting Cable Assembly shall be nominally 6 feet in length including the connectors and the two-branch division. It shall provide electrical interconnection between the Antenna Control Box and the AN/APG-59 Antenna on the Antenna Maintenance Fixture.

3.4.2.7.4.5 Cable Assembly, System Interconnecting, 64W4 - The System Interconnecting Cable Assembly shall be 38 ± 5/8 feet in length including the connectors. Each end shall be divided into three branches 2 feet long.

3.4.2.7.5 Hydraulics Panel

3.4.2.7.5.1 Function - This unit shall permit the Hydraulic Supply to deliver fluid to either the Antenna mounted on the Table-Rack or the Antenna Maintenance Fixture. An electrical interlock shall also be provided to assure control of the Hydraulic Supply by the facility using it.

3.4.2.7.5.2 Form Factor - The Hydraulic Panel shall consist of the hydraulic transfer valve, fittings, and check valves on a support bracket. This Panel, although a part of the Antenna Maintenance Fixture, shall attach to the MCS Support Structure. The overall dimensions, exclusive of the interconnecting lines, shall be 26-7/8 by 6-1/2 by 11-1/4 inches. The weight of the unit, excluding interconnecting lines, shall not exceed 27 pounds.

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3.4.2.7.5.3 Fittings - The Panel shall have the following quick disconnect fittings:

a. Pressure side of Hydraulic Supply to Panel
b. Return side of Hydraulic Supply to Panel
c. Pressure side of Panel to Table-Rack Antenna
d. Return side of Panel to Table-Rack Antenna
e. Pressure side of Panel to Antenna Maintenance Fixture Antenna
f. Return side of Panel to Antenna Maintenance Fixture Antenna

3.4.2.8 Associated Equipment - The Hydraulic Supply HD413/APM-92 is required to work in conjunction with the Antenna Maintenance Fixture.

3.4.3 Waveguide Extension (WE) MX-8491/AWG-10

3.4.3.1 Function - The WE shall be designed to mount on the bench mounted AN/APG-59 Radar Set, in such a manner that the Receiving Waveguide Assembly is lowered approximately 10 inches. In addition, providing access to the Receiving Waveguide Assembly for maintenance, the WE shall provide capabilities for alignment of the Parametric Amplifier, calibration of the BIT Target Generator, and calibration of the 17-dB Variable Attenuator.

3.4.3.2 Requirements - The detail requirements for the WE are specified in Specification MIL-T-81527.

3.4.4 Cable and Adapter Group OA-8124B/AWM-21

3.4.4.1 Function - The cable and adapter group shall provide the patch cables, test adapters, and certain items of test equipment necessary to operate, or facilitate the operation and test of, the specified components of the AN/APG-59 Radar Set as part of the system, and shall provide necessary test points when components are removed from their normal mounting position for servicing.

3.4.4.2 Form Factor - Four transit cases shall be provided to hold the cables, adapters and test equipment. Each case shall not exceed 35 by 23 by 17 inches. The total weight of the equipment, including transit cases, shall not exceed 440 lb (110 lb each).

3.4.4.3 Cable and Connectors - Cables, cabling and grounding of cables shall be compatible with the requirements of the AN/APG-59 Radar Set.

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3.4.4.4 Service Conditions - The cables, adapters, and certain specified test items shall meet environmental conditions specified in MIL-H-15362. The Transit cases and certain specified test items shall meet environmental conditions of MIL-T-21200.

3.4.4.5 Contents of Unit - The Cable and Adapter Group shall consist of test adapters, patch cables, board extractors, transit cases and certain items of equipment to facilitate or perform tests.

3.4.4.5.1 Test Adapters - The following Test Adapters (one each) shall be provided for readily accessible testing of the AN/APG-59 Radar Set plug-in units specified, except for Test Adapter (aa) where two Test Adapters shall be provided. The adapters shall be constructed to permit measurements of all pin connections for the plug-in units through test points. The adapters shall be marked to indicate the specific plug-in boards used.

a. Test Adapter (55A1A1) - This adapter shall provide means for servicing a general purposes 8-pin, plug-in type relay installed throughout the AN/APG-59 Radar Set.

b. Test Adapter (55A1A2) - This adapter shall be the same as test adapter 55A1A6 except the plug-in board receptacle shall be rotated 180 degrees and test points shall provide for opposite-side accessibility to the pin connections on the board.

c. Test Adapter (55A1A3) - This adapter shall be the same as Test Adapter 55A1A7 except the plug-in board receptacle shall be rotated 180 degrees and test points shall provide for opposite-side accessibility to the pin connections on the board.

d. Test Adapter (55A1A4) - This adapter shall be the same as Test Adapter 55A1A8 except the plug-in board receptacle shall be rotated 180 degrees and test points shall provide for opposite-side accessibility to the pin connections on the board.

e. Test Adapter (55A1A5) - This adapter shall be the same as Test Adapter 55A1A25 except the plug-in board receptacle shall be rotated 180 degrees and test points shall provide for opposite-side accessibility to the pin connections on the board.

f. Test Adapter (55A1A6) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 15A4 through 15A17 installed in the Velocity Signal Computer and 16A15 installed in the Launch Signal Computer.

g. Test Adapter (55A1A7) - This adapter shall have a receptacle to mate with the plug-in board and connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 16A4 through 16A14, 16A16, and 16A17 installed in the Launch Signal Computer.

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h. Test Adapter (55A1A8) - Thi s adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 17A1 through 17A12 installed in the Pilot Command Signal Computer.

i. Test Adapter (55A1A9) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 9AR1, 9AR2, 10AR1, and 10AR2 installed in the Command and Radar Indicators, respectively.

j. Test Adapter (55A1A10) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 5A3A2 and 5A3A3, which are installed in Power Supply Regulators.

k. Test Adapters (55A1A11) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 5A3A7 and 5A3A8 which are installed in the Power Supply Regulators.

l. Test Adapter (55A1A12) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 5A3A5, 5A3A9, and 5A3A10 installed in Power Supply Regulators.

m. Test Adapter (55A1A13) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A3A4 installed in Power Supply regulators.

n. Test Adapter (55A1A14) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A3A6 installed in Power Supply Regulators.

o. Test Adapter (55A1A15) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A3A1 installed in Power Supply, Regulators. A switch shall be provided for opening the closed loop of the regulator circuit. Wire shall be per MIL-W-16878.

p. Test Adapter (55A1A16) - This adapter shall be the same as Test Adapter 55A1A18 except the plug-in board receptacle shall be rotated 180 degrees and test points shall provide for opposite-side accessibility to the pin connections on the plug-in board.

q. Test Adapter (55A1A17) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The

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adapter shall provide means for servicing plug-in boards 14A1 through 14A3 installed in the Selector Test Programmer.

r. Test Adapter (55A1A18) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 11A1 through 11A16 installed in the Indicator Control. This adapter cannot be used for boards 11A12 and 11A14 on SEA modified systems.

s. Test Adapter (55A1A19) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 11A17 and 11A18 installed in the Indicator Control. This adapter shall be used for boards 11A12 and 11A14 on SEA modified systems.

t. Test Adapter (55A1A20) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 5A2A1A1 through 5A2A1A12, 5A2A1A14, and 5A2A1A15 installed in the Timer-Antenna Servo.

u. Test Adapter (55A1A21) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 5A2A3A1 through 5A2A3A4, 5A2A4, 5A2A2A1, 5A2A2A2, 5A2A2A4, 5A2A2A6 through 5A2A2A9, and 5A2A2A11, and 5A2A2A14, 5A2A2A15 for SEA modified systems, all installed in the Timer-Antenna Servo.

v. Test Adapter (55A1A22) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 5A2A2A12 and 5A2A2A13 installed in the Timer-Antenna Servo.

w. Test Adapter (55A1A23) - This adapter shall be the same as Test Adapter 55A1A20 except the plug-in board receptacle shall be rotated 180 degrees and test points shall provide for opposite-side accessibility to the pin connections on the plug-in board.

x. Test Adapter (55A1A24) - This adapter shall be the same as Test adapter 55A1A21 except the plug-in board receptacle shall be rotated 180 degrees and test points shall provide for opposite-side accessibility to the pin connections on the plug-in board.

y. Test Adapter (55A1A25) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 6A1A1 through 6A1A16 installed in the Scan Pattern Generator. Switches shall perform on-off functions of several signal inputs to the plug-in board.

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z. Test Adapter (55A1A26) - This Adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A2A2A10 installed in the Antenna Servo.

aa. Test Adapter (55A1A27)- This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in boards 5A1A1A2 through 5A1A1A4 and 5A1A2A1 through 5A1A2A3 installed in the Tracker, Range-Terrain Clearance.

ab. Test Adapter (55A1A29) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A2A2A5 installed in the Timer-Antenna Servo.

ac. Test Adapter (55A1A30) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A2A2A3 installed in the Timer-Antenna Servo.

ad. Test Adapter (55A1A31) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A1A1A1 installed in the Tracker Range-Terrain Clearance.

ae. Test Adapter (55A1A32) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. The adapter shall provide means for servicing plug-in board 5A2A1A13 installed in the Timer-Antenna Servo.

af. Dummy Mixer (55A1A34) - This unit shall provide means of applying a 30-MHz test signal for the waveguide preamplifiers at 2A8A2 and 2A8A3.

ag. Test Adapter (55A2A1) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in boards 4A1A1 and 4A1A2 installed in Amplifier Assembly, Intermediate Frequency.

ah. Test Adapter (55A2A2) - This Adapter shall have two receptacles to mate with the plug-in board and two connectors to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A1A3 installed in Amplifier Assembly, Intermediate Frequency.

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ai. Test Adapter (55A2A3) - This adapter shall have two receptacles to mate with the plug-in board and two connectors to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A1A4 installed in Amplifier Assembly, Intermediate Frequency.

aj. Test Adapter (55A2A4) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A1A5 installed in Amplifier Assembly, Intermediate Frequency. A three-position switch and three resistors installed on the adapter facilitate testing the "clutter discriminator."

ak. Test Adapter (55A2A5) - This adapter shall have two receptacles to mate with the plug-in board and two connectors to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A1A6 installed in Amplifier Assembly, Intermediate Frequency.

al. Test Adapter (55A2A6) - This adapter shall have two receptacles to mate with the plug-in board and two connectors to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A1A7 installed in Amplifier Assembly, Intermediate Frequency.

am. Test Adapter (55A2A7) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A1A8 installed in amplifier assembly, Intermediate Frequency.

an. Test Adapter (55A2A8) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-R-17/69. This adapter shall provide means for servicing plug-in board 4A1A9 installed in Amplifier Assembly, Intermediate Frequency.

ao. Test Adapter (55A2A9) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A3 installed in the Tracker, Target

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Data. Two switches shall be provided to remove manual input voltage and to apply external voltage during the alignment of the doppler and FM voltage-controlled oscillator.

ap. Test Adapter (55A2A10) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A4 installed in the Tracker, Target Data. Two switches permit alignment of the voltage controlled oscillators on unit 4A3A4.

aq. Test Adapter (55A2A11) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A5 installed in the Tracker, Target Data.

ar. Test Adapter (55A2A12) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A6 installed in the Tracker, Target Data.

as. Test Adapter (55A2A13) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired with wire per MIL-W-16878, and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A7 installed in the Tracker, Target Data. A rotary switch shall be provided to insert any of four voltage dividers in series with the discriminator output to the velocity track integrators.

at. Test Adapter (55A2A14) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A8 installed in the Tracker, Target Data. A switch shall be provided to facilitate checking and aligning the velocity track resweep circuits.

au. Test Adapter (55A2A15) - This adapter shall have two receptacles to mate with the plug-in board and two connectors to mate with the chassis. Connectors shall be wired per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A9 installed in the Tracker, Target Data. Two

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switches shall be provided, one to simultaneously open four pins (pins P2/7, /8, /9, /20) on the extended board and one to permit the grounding of a pin (P1/34).

av. Test Adapter (55A2A16) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired per MIL-W-16878, and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A10 installed in the Tracker, Target Data. A switch shall be provided to permit opening two wires on the adapter(pins P1/8, P1/44), and insert +25 V from P1/5 to J1/8 and J1/44.

aw. Test Adapter (55A2A17) - This adapter shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis. Connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-188/U subminiature coaxial cable per MIL-C-17/69. This adapter shall provide means for servicing plug-in board 4A3A1 installed in the Mixer, Complexer.

ax. Test Adapter (55A4A6) - This adapter shall be capable of extending plug-in unit 5A2A1A1 and shall have a connector to mate with the chassis and a receptacle to mate with the plug-in board. A switch shall be provided to remove the dither modulation from the signal supplied by 5A2A1A1 to the antenna servo valve drivers. The adapter shall use wire per MIL-W-16878.

ay. Test Adapter (55A4A7) - This adapter, with printed wiring, shall be capable of extending plug-in units 5A2A2A1, PRF Generator Board, and 5A2A2A4, Sequencing Board, and shall have a connector to mate with the chassis and a receptacle to mate with the plug-in boards. The adapter shall contain two switches; one to remove the pulse mode PRF from 5A2A2A1. The second switch shall remove the normal input signal from 5A2A2A4 and provide for inserting a test signal.

az. Test Adapter (55A4A10) This printed circuit adapter shall be able to extending plug-in unit 6A2A3A1 and shall have a connector to mate with the chassis and a receptacle to mate with the plug-in unit.

ba. Test Adapter (55A4A11) - This adapter with printed circuit wiring shall be capable of extending plug-in unit 6A2A3A2 and shall have a connector to mate with the chassis and a receptacle to mate with the plug-in board. The receptacle shall be rotated 180 degrees from the normal position of the plug-in board.

bb. Test Adapter (55A4A1) - This adapter shall be capable of extending relays 2A3K5, 2A3K21, and 2A3K22 and shall have a connector to mate with the chassis and a receptacle to mate with the relay.

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bc. Test Adapter (55A4A3) - This adapter with printed circuit wiring shall be capable of extending plug-in unit 4A2A11 and shall have a connector to mate with the chassis and a receptacle to mate with the plug-in unit.

bd. Test Adapter (55A4A5) - This adapter, with printed wiring and a switch to facilitate servicing of plug-in unit 4A2A13, shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis.

be. Test Adapter (55A4A4) - This adapter with an air duct and printed wiring, shall be used to extend plug-in board 4A2A12. It shall have a receptacle to mate with the plug-in board and a connector to mate with the chassis.

bf. Test Adapter (55A4A2) - This adapter shall be used to extend each of the subbanks 4A2A1 through 4A2A10 to facilitate adjustments on the subbanks. The stilt shall have replaceable cables that correspond with the various connector configurations. The five cables required, two of which are permanent to the rigid frame adapter, shall have receptacles at one end to mate with the sub-bank and connectors at the other end to mate with the chassis. The through replaceable cables are 55W44, 55W45, and 55W46. Wire shall be in accordance with MIL-W-16878.

bg. Test Adapter (55A4A8) - This adapter shall be capable of extending plug-in unit 6A2A2A1 and shall have a connector to mate with the chassis and a receptacle to mate with the plug-in board. The receptacle shall be rotated 180 degrees from the normal position of the plug-in board.

bh. Test Adapter (55A4A9) - This adapter shall be capable of extending plug-in unit 6A2A2A2 and shall have a connector to mate with the chassis and a receptacle to mate with the plug-in board.

3.4.4.5.2 Patch Cables - One each unless otherwise noted, of the following Patch Cables shall be provided to permit removal of subassemblies from their operating positions yet permit normal functioning of the specified unit.

a. Power Supply, CW Illuminator Assembly (55W1) - This cable shall supply 115 VAC, 400 Hz, Phase A from the Convenience Outlets to connector 2A6J1 of the Power Supply. The cable shall be 144 ± 6 inches in length with a UP103M (MIL-C-3767/5) connector to mate with the Table-Rack (3.4.1) and a connector to mate with the Power Supply.

b. Test Cable Adapter (55W2) - This adapter shall be used with the General Line Replaceable Unit (LRU) Test Cable Assembly 55W3 to provide means to make electrical measurements through test points for the 25 pin

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Line Replacement Unit (LRU) connectors. The Adapter shall be 8-3/4 inches in length. Connectors shall be wired pin to pin.

c. General Line Replaceable Unit (LRU) Test Cable Assembly (55W3)- This Cable Assembly shall provide means to make electrical measurements through test points for the 50 pin Line Replaceable Unit (LRU) connectors. The cable shall be 96 ± 3 inches in length. Connectors shall be wired pin to pin and shall have 1-to-1 correspondence with test point identity numbers. A 50-pin connector shall be at one end and 50 MS16108-3A Test Jacks in a box at the other end.

d. Fault Isolation Meter Power Cable (55W4) - This cable shall provide power to the Fault Isolation Meter from the Selector Test Programmer through connectors 14J3 and 14J4 and 51P2 and 51P3. The cable shall be wired 132 ±14 inches in length with wire per MIL-W-16878 and RG195/U coaxial cable per MIL-C-17/70.

e. Command Indicator and Radar Indicator Extension Cable (55W5) - When the subassemblies of the Command Indicator or the Radar Indicator are extended for servicing this cable shall be used to electrically connect either unit to the system through connectors 9A2J1 or 10A2J1 and 9A2P1 or 10A2P1. The cable shall be 36 ±1 inch in length. The connectors shall be wired pin to pin with wire per MIL-W-16878.

f. Command Indicator and Radar Indicator Extension Cable (55W6) - When subassemblies of the Command Indicator or Radar Indicator are extended for service this cable shall be used to electrically connect either unit to the system through connectors 9A1J1 and 9A1P1 or 10A1J1 and 10A1P1. The cable shall be 36 ±1 inch in length. The connectors shall be wired pin to pin with wire per MIL-W-16878.

g. Receiving Waveguide Assembly Extension Cable (55W17) - When the Receiving Waveguide Assembly is removed from the system this cable shall be use to electrically connect the Receiving Waveguide Assembly and the system through connectors 2A8A2J5 and 2A8A2P5 or 2A8A3P5. The cable shall be 60 ± 1 inch in length. The connectors shall be wired with coaxial cable per MIL-C-17/70.

h. C.W. Illuminator Power Supply Test Cable Assembly (55W18) - This cable shall provide a 2.54 OHM ±1 percent load for the CW Illuminator Power Supply Test. The cable shall be 24 ±1 inch in length with a connector to mate with the Power Supply Connector 2A6PS2J1. Test Jacks shall be MS 16108-3A.

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i. Receiving Waveguide Assembly Extension Cable (55W19) - When the Receiving Waveguide Assembly is removed from the system this cable shall be used to electrically connect the Receiving Waveguide Assembly and the system through connectors 2A8J1 and 2A8P1 or 2A8J2 and 2A8P2. The cable shall be 60 ±1 inch in length. The connectors shall be wired pin to pin.

j. Control, Radar Set Cable Assembly (55W24) - This cable shall be used when the Radar Set Control is removed from the system to electrically connect the Radar Set Control and the System through connectors 12J1 and 12P2. This cable shall be 36 ±1 inches in length. All connectors shall be wired pin to pin.

k. Control, Radar Set Cable Assembly (55W25) - This cable shall be used when the Radar Set Control is removed from the system to electrically connect the Radar Set Control and the System through connectors 12J2 and 12P2. This cable shall be 36 ±1 inches in length. All connectors shall be wired pin to pin.

l. Control, Antenna Cable Assembly (55W26) - This cable shall be used when the Antenna Control is removed from the system to electrically connect the Antenna Control and the system through connectors 13J1 and 13P1. This cable shall be 36 ±1 inches in length. All connectors shall be wired pin to pin.

m. Table-Rack Cable Assembly (55W29) - This cable shall be used to supply variable currents and voltage from the Remote Safety Unit Connector to service the KPA body current relays. This cable shall be 120 ± 3 inches in length. An on-off switch shall be provided to supply power to the relays under test and test point shall be provided to monitor the variable power being supplied to the relay. All wiring used shall be per MIL-W-16878.

n. Selector-Test Programmer, Cable Assembly (55W32) - This cable shall permit the removal from the Selector-Test Programmer of printed circuit board assemblies through connectors 14A6A and 14A6B for servicing. The cable shall be 18 ± 3 inches in length. All connectors shall be wired pin to pin. The connectors shall be properly keyed to permit connection to the board assemblies.

o. Selector-Test Programmer, Cable Assembly (55W33) - This cable shall permit the removal from the Selector-Test Programmer of printed circuit board assembly 14A5. The cable shall be 18 ±1 inches in length. The connectors shall be wired pin to pin.

p. Parametric Amplifier Power Supply Extension Cable (55W34) - When the Parametric Amplifier Power Supply is removed from the system this cable

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shall be used to electrically connect the Parametric Amplifier Power Supply to the system through connectors 2A7J1 and 2A7P1. This cable shall be 50 ±1 inch in length. The connectors shall be wired pin to pin.

q. Pulse Amplifier Transmitter Extension Cable (55W35) - When the Pulse Amplifier Transmitter is removed from the system this cable shall be used to electrically connect the Pulse Amplifier Transmitter and the system through connectors 2A1J1 and 2A1P1. The cable shall be 72 ±1 inch in length. The connectors shall be wired pin to pin per MIL-W-16878.

r. Pulse Amplifier Transmitter Extension Cable (55W36) - When the Pulse Amplifier Transmitter is removed from the system this cable shall be used to electrically connect the Pulse Amplifier Transmitter and the system through connectors 2A1J2 and 2A1P2. The cable shall be 72 ±1 inches in length. The connectors shall be wired pin to pin per MIL-W-16878.

s. Pulse Amplifier Transmitter Extension Cable (55W37) - When the Pulse Amplifier Transmitter is removed from the system this cable shall be used to electrically connect the Pulse Amplifier Transmitter and the system through connectors 2A2J12 and 2A2P12. The cable shall be 62 ± 1 inch in length. The cable shall be coaxial cable per MIL-C-17/70.

t. Pulse Amplifier Transmitter Extension Cable (55W38) - When the Pulse Amplifier Transmitter is removed from the system this cable shall be used to electrically connect the Pulse Amplifier Transmitter and the system trhough connectors 2A2J13 and 2A2P13. The cable shall be 62 ±1 inch in length. The cable shall be coaxial cable per MIL-C-17/70.

u. Parametric Amplifier Power Supply Extension Cable (55W39) - This cable shall be used to electrically join connector 2A7A1J1 of the Parametric Amplifier Power Supply through connector 2A7A1P1 to a meter. The cable shall be 30 ±1 inch in length. The cable shall be coaxial cable per MIL-C-17/58. The connector shall be UG932A/U and the tip plug shall be black plastic with a 0.080-inch-diameter metal probe.

v. Parametric Amplifier Power Supply Extension (Cable 55W40) - This cable shall be used to electrically join connector 2A7A1J2 of the Parametric Amplifier Power Supply through connector 2A7A1P2 to a meter. The cable shall be 30 ±1 inch in length. The cable shall be coaxial cable per MIL-C-17/58. The connector shall be UG932A/U and the tip plug shall be red plastic with an 0.080-inch-diameter metal probe.

w. Power Supply - Oscillator, CW Offset Cable Assembly (55W47) - This cable shall consist of a coax with BSM connectors, used when extending

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the C-Frequency Filter Unit 6A2A2A1 or the offset-generator unit 6A2A1. The cable shall be 24-inches long.

x. Offset Generator Cable Assembly- This cable, (55W48) shall be used when extending 6A2A1. It shall be wired pin to pin and shall be 24-inches long.

y. Power Supply High Voltage Cable Assembly - Each of these cable assemblies (55W49, 55W50, 55W51) identical to each other, shall be used when extending units from the power supply through 6PS1J1, J2 and J3.

z. FIM Extender Cables - Eight FIM Extender Cables extend various boards from the FIM as indicated below:

Unit Reference Designation	Extends (51A9)	From FIM Connector	To Board Connector
55W52	Address BIT Board	51A1J9	51A9P1
55W53	Address BIT Board	51A1J10	51A9P2
55W54	AC Circuit Board (51A6)	51A1J8	51A6P1
55W55	Relay Board C (51A5)	51A1J6	51A5P1
55W56	Relay Board C (51A4)	51A1J7	51A5P2
55W57	Relay Board B	51A1J17	51A4P1
55W58	Relay Board B	51A1J18	51A4P3
55W59	Relay Board B	51A1J19	51A4P2

The cables shall be 24 inches in length and are terminated with keyed connectors compatible with those of the FIM and extended board. The cables are wired with No. 22 gage white hook-up wire per MIL-W-16878. The connectors are firmly held by threaded or tapped jackscrews into the mating connectors.

aa. RFNA Extender Cables - The RFNA, Extender Cables extend the AM (53A6A3) and FM (53A6A4) preamplifiers. Seven cables can be used as the table shows on the next page.

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Unit Reference Designation	From (RFNA)	To (PREAMP)
55W66	53A6A3/J1 or 53A6A4/J1	53A6A3/P1 or 53A6A4/P1
55W67	53A6A4/J2	53A6A4/P2
55W68	53A6A3/J3 or 53A6A4/J3	53A6A3/P3 or 53A6A4/P3
55W69	53A6A3/J4 or 53A6A4/J4	53A6A3/P4 or 53A6A4/P4
55W70 (3 each)	53A6A3/J5 or 53A6A4/J5, J6, J7	53A6A3/P5 or 53A6A4/P5, P6, P7

The cables are 24 inches in length and are terminated with coax connectors in the case of all but 53A6A3(4) J4, P4 which is a hook-up wire connector. 55W66 and 55W67 are a single R6-233/U coax cable. 55W69 is made up of AWG-22 unshielded MIL-W-16878. The remainder is made of RG-195 of cable.

ab. STALO Extender Cables - The STALO Extender Cable shall be of two types: One shall be a single coax with a coax connector at each end. The other shall be a multiconductor wire cable. There shall also be two TSM (female) to BNC (female) adapters and a 50-ohm coax termination. The extender cables shall be used to extend 3A2 (IF AMP), 3A9 (Filter Box and Shifter) and 3A11 (VFO). The cables shall extend the following connections:

Unit Reference Designation	From	To	Uses
55W62	3A4/J2, J3, or J4	3A4/P2, P3 or P/4	RG-188U-AWG-22 unshielded MIL-W-16878
55W63			
55W64	3A3A3/P1	3A3A3/J1	
55W65			

ac. LRU Probes - Three types of signal probes shall be provided.

(1) (55W71) - This probe shall consist of a female coaxial contact, mounted on a handle, connected to a BNC connector through 95-ohm coaxial cable per MIL-C-17, 6 feet long .

(2) (55W72) - This probe shall consist of a male coaxial contact, mounted on a handle, connected to a BNC connector through 95-ohm coaxial cable per MIL-C-17, 6 feet long.

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(3) (55W73) - This probe shall consist of a standard male contact, mounted on a handle, connected to an 0.080 plug through an AWG No. 18 lead, 6 feet long.

ad. Radar Set Control, Cable Assembly (55W74) - This cable shall be used when the Radar Set control is removed from the system to electrically connect the Radar Set Control and the System through connectors 12J3 and 12P3. The cable assembly shall include a box containing test points connected to each pin on the connectors. The cable from the box to the 12P3 end shall be 48 inches long and the other end shall be 8 inches.

3.4.4.5.3 Precision Attenuator Assembly 55A3A2 of OA-8124B/AWM-21

3.4.4.5.3.1 Function - The Precision Attenuator Assembly (PAA) shall provide the capability for detailed adjustment and troubleshooting of the Receiving Waveguide Assembly (LRU2) of the AN/AWG-10 Missile Control System. The PAA permits both intermediate and organizational level maintenance on both a scheduled and unscheduled basis.

3.4.4.5.3.2 Form Factor - The weight of the unit shall not exceed 10 pounds. The unit shall not exceed 1-cubic foot and shall be able to be rested on a bench during operation.

3.4.4.5.3.3 Service Conditions - The PAA shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.4.5.3.4 Power Requirements - The attenuators shall be capable of withstanding the following input power:

J1	2 watts
J2	10 watts
J3	20 watts
J4	10 watts

3.4.4.5.3.5 Performance - The PAA shall provide a means of:

a. Placing a calibrated 20 dB or 60 dB attenuator in series with the AN/APG-59 Waveguide Switch 2A2S1 and routing the signal to a waveguide choke flange for measurement with a power meter. These capabilities are required for measuring the insertion loss of 2A2DC1 and adjusting variable attenuators 2A2AT1 and 2A2AT9.

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b. Placing a calibrated 0 to 50 dB attenuator in series with either the 20 dB or 60 dB lines, or be routed directly to a coax port. These capabilities are required for inserting an attenuated Main Beam Clutter signal into the Receiver, from the Stalo, for troubleshooting procedures.

3.4.4.5.3.6 Contents of Unit - The PAA shall contain the following components:

a. A four port coaxial switch to route the signals from OSM female input ports, labelled J1, J2, and J3 (20 dB, straight-thru, and 60 dB lines respectively) to a waveguide choke flange. One switch position shall be vacant for nulling of the power meter. The switch shall be labelled CHANNEL SELECT, and positions labelled 1,2,3, and 4 (J1, J2, J3, and open respectively).

b. A single-pole, double throw coaxial switch shall be connected to the switch wiper, and the output connected between the waveguide choke flange and OSM female port labelled J4 (0 to 50 dB line). The switch shall be labelled ATTENUATOR MODE and the switch positions FIXED and VARIABLE (to flange and 0 to 50 dB respectively).

c. The following items external to the unit are supplied:

(1) Two (2) 55W41 flexible coaxial cables with OSM male connectors at each end, matched to ± 0.1 dB at the MCS RF band, nominally 18 in and equipped with captive female coaxial adapters.

(2) Two (2) 55W42 flexible coaxial cables with OSM male connectors at each end and nominally 40 in.

(3) One (1) 55W43 semi-rigid coaxial cable with OSM female connectors and 5.00 ± 0.10 in.

(4) One (1) OSM female-to-waveguide cover flange.

(5) One (1) 90 degree H-bend choke-to-choke waveguide section with gaskets and captive hardware.

(6) One (1) waveguide flange cover plate.

d. A calibration chart containing calibration information shall be supplied on each unit as four (4) factors:

(1) Factor "A" is found by - "A" = 2X + Y - 99.4
where $X = J_1 - J_2$

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and $Y = J_3 - J_2$

and J_i = insertion loss of path from J_i to the waveguide flange, measured in dB. (i = 1, 2, 3, 4)

(2) Factor "B" is the insertion loss of the paths from J1, J2, and J3 to the flange as measured in dB and shall be shown on a schematic diagram.

(3) Factor "C" is the insertion loss of the path from J1 to J4 when the variable attenuator is set at zero.

(4) Factor "D" is the insertion loss of the path from J2 to J4 when the variable attenuator is set at zero. All calibrations shall be to an accuracy of ± 0.1 dB (± 0.2 dB for J3) at center frequency of the AWG-10 RF band.

3.4.4.5.3.7 Connectors - The PAA shall contain four (4) OSM female connectors, mounted on the top panel, marked J1, J2, J3, and J4. A waveguide choke flange shall be mounted on the panel containing the variable attenuator dial.

3.4.4.5.4 Self-Test Adapter, Selector-Test Programmer (55A3A7)

3.4.4.5.4.1 Function - The Self-Test Adapter, Selector-Test Programmer (STASTP) provides the capability for performance verification and detailed maintenance of the Selector-Test Programmer (STP) of the AN/AWG-10 Missile Control System. The STASTP permits intermediate level maintenance and organizational-level maintenance on both a scheduled and unscheduled basis. The STASTP will test the STP to a confidence level of 95 percent.

3.4.4.5.4.2 Form Factor - The weight of the unit shall not exceed 5 pounds. The unit shall not exceed 8 by 7 by 4 inches, except where an extension is required to mate connectors to the Selector-Test Programmer. The unit shall be configured so the appropriate connections can be made to the Selector-Test Programmer while the RIO's seat is in place, not interfering with any other operating controls in the cockpit, and allowing visual access to the STP and the STASTP's and access to controls and test points on the STASTP.

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3.4.4.5.4.3 Service Conditions - The STASTP shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.4.5.4.4 Primary Input Power Requirements - The STASTP shall perform when supplied with power as follows:

Voltage	Power Consumption (max amperes)
+25 Vdc±1%	20 mA
+ 6 Vdc±1%	25 mA
+28 Vdc±1%	500 mA
+12 Vdc±1%	15 mA
+115 Vdc±10%	150 mA

3.4.4.5.4.5 Performance - The STASTP shall be capable of making the following measurements and providing the following test inputs to the STP:

a. Check the ability of the STP voltage comparator board to measure voltages by testing 15-percent max tolerance, 7-percent max tolerance and 1, 2, and 4-percent tolerances. The STASTP voltages are used as references and compared to the "unknown" voltage comparator voltages.

b. Provide precision voltages 10.24, 9.30, 5.12, 4.53, 2.56, 1.28, 0.64, 0.32 and 0.31 to compare to the STP's programming voltages. A relay tree is used to connect the various voltages to VI and V2 of the voltage comparator and compare with the programming voltages 10.24 through 0.32 Vdc, and the sum of the remaining voltages 0.160 + 0.080 + 0.040 + 0.020 + 0.010 which equals 0.31 Vdc. The relays are switched by the sensor commands through relay drivers. A meter reading in the green area will indicate a passing test.

c. Check the relay Digital/Analog outputs and its associated sensors U1, U2, and U3 which produce -0.8, -1.6 and -3.2 Vdc.

d. Check the presence of the 28 Vdc latch signal between frames.

e. Check system command sensor operation in both on ON and OFF states.

f. Check Range Voltage line Voltrip.

g. Check input lines from the MCS to Voltrip (Radar 1, Radar 2, Computer 1 and 2).

h. Provide a 500 millisecond, +26 Vdc, time delay to gate the 900-Hz clock. A specific Digital/Analog output is fed into Voltrip for measurement.

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3.4.4.5.4.6 Contents of Unit - The STASTP shall contain the following components mounted in the unit:

a. Time Delay Module (500 millisecond)

b. "AND" Gate Module

c. "NAND" Gate Module

d. Zener Diode Reference Module (9.300 V)

e. Voltage Divider String

f. Relay Tree

g. Diode Matrix

h. Zero Center Panel Meter

3.4.4.5.4.7 Panel Controls - The STASTP shall have the following controls mounted on the front panel.

a. A momentary contact switch to allow adjustment of 9.30 reference voltage and to place the STASTP in the test mode. The switch shall have the following settings TEST-OFF-CALIBRATE, with momentary contact in the CALIBRATE position.

b. A potentiometer CAL to allow adjustment of the 9.30 reference voltage.

c. A screw adjustment on the panel meter to allow mechanical nulling.

3.4.4.5.4.8 Connectors - The STASTP shall contain connectors which will mate with the STP to provide all inputs and test outputs required for operation. The STASTP shall contain test jacks on the front panel to make the required measurements.

3.4.4.5.5 Alignment Fixture, Launch Signal Computer (55A3A9)

3.4.4.5.5.1 Function - The Alignment Fixture provides the capability for alignment of the three (3) servo potentiometer assemblies of the Launch Signal Computer. The Fixture permits intermediate level maintenance on both a scheduled and unscheduled basis.

3.4.4.5.5.2 Form Factor - The Fixture shall not exceed 70 cubic inches exclusive of cables required. The Fixture shall not weigh more than 5 pounds including cables.

3.4.4.5.5.3 Service Conditions - The Fixture shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 3 equipment.

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3.4.4.5.5.4 Primary Input Power Requirements - The Fixture shall be supplied with 115V ±10 percent, 380 to 420 Hz, single phase crest factor 1.27 to 1.56 with 5V maximum transients.

3.4.4.5.5.5 Performance - The Fixture shall be capable of:

a. Providing the drive voltage for both clockwise and counterclockwise rotation of a series connected two-phase ac servo motor. This voltage shall be no less than 40V nor greater than 65V and its phase relation to the fixed field voltage shall be as close to 90 degrees lead or lag (depending on direction of rotation) as is practical.

b. Providing for slow speed operation (1/3 normal).

c. Providing a 6Vdc ± 10 percent, 0 to 6 mA input and switching required for checking and aligning the cups of the servos.

d. Providing motor reversal and also removing the voltage from the motor.

e. Providing for removal of all voltage from the motor, and applying full voltage and voltage for one-third speed operation.

f. Providing connections for each of the wipers of the servo pots sequentially to a test point for measurement.

3.4.4.5.5.6 Contents of Unit - The Fixture shall contain the following components:

a. Motor Drive Circuit

b. Rectifier Circuit (6 Vdc)

c. Power Cable to mate with convenience outlets (36 inch)

d. Three (3) 26-inch cables to supply power to servos

3.4.4.5.5.7 Panel Controls - The Fixture shall have the following controls mounted on the front panel:

a. Motor Control Switch labeled CW-MAN-CCW

b. Operating Mode Switch labeled OFF-NORM-SLOW

c. Servo Selector Switch with each switch position connected to a Servo wiper.

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3.4.4.5.5.8 Test Point - A red test point is connected to the wiper of the Servo Selector switch.

3.4.4.5.5.9 Connectors - The Fixture shall have connectors to mate with 115V convenience outlets and three (3) to mate with the servos.

3.4.4.5.6 Self-Test Adapter, Fault Isolation Meter (55A3A6)

3.4.4.5.6.1 Function - The Self-Test Adapter, Fault Isolation Meter (STAFIM) shall provide the capability for performance-verification and detailed maintenance of the Fault Isolation Meter, ME373/AWM-21 at intermediate and organizational level maintenance on both a scheduled and unscheduled basis.

3.4.4.5.6.2 Form Factor - The STAFIM shall occupy 1-cubic foot and the length ratio from any one side to another does not exceed four. The weight of the unit shall not exceed 20 pounds. The unit shall be configured to mate with the BIT-FIM Umbilical Cable at one end and J1 and J2 of the FIM at the other. There shall be two connectors which will receive the 50 pin connector on the FIM signal cable.

3.4.4.5.6.3 Service Conditions - The STAFIM shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class II equipment.

3.4.4.5.6.4 Primary Input Power Requirements - The STAFIM shall perform as specified when supplied with power as follows:

a. 115 VAC

Operating voltage limits: 107.5 to 119.5 Vac, three phase
Operating frequency limits: 380 to 420 Hz
Power requirements: 100 Volt-amperes (VA) maximum

b. +26 Vdc Protected

Operating voltage limits: 24 to 29 Vdc
Transients limited to 43 Vdc maximum
Power requirements: 175 watts maximum

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c. 28 Vdc

Operating voltage limits: 24 to 29 Vdc
Power requirements: 100 watts maximum

d. 26 VAC

Operating voltage limits: 24 to 27 Vac single phase
Operating frequency limits: 380 to 420 Hz
Power requirements: 30 VA maximum

e. +6 Vdc

Operating voltage limits: -5.94 to -6.06 Vdc
Power requirements: 300 milliwatts max

f. -6 Vdc

Operating voltage limits: -5.94 to 6.06 Vdc
Power requirements: 6 watts maximum

g. 12 Vdc

Operating voltage limits: 11.78 to 12.12 Vdc
Power requirements: 12 watts maximum

h. +25 Vdc

Operating voltage limits: 24.75 to 25.25 Vdc
Power requirements: 5 watts maximum

i. -25 Vdc

Operating voltage limits: -24.75 to -25.25 Vdc
Power requirements: 5 watts maximum

j. -60 Vdc

Operating voltage limits: -59.4 to -60.6 Vdc
Power requirements: 1 watt maximum

3.4.4.5.6.5 Performance - The STAFIM shall be capable of providing the following test inputs to the FIM in addition to the signal normally available from the Selector Test Programmer (STP):

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a. Precision dc voltages, reference voltages as described by the appropriate FIM Program tape

b. Precision loads as described by the appropriate FIM Program tape

c. AC voltages other than those provided by the STP as described by the appropriate FIM Program tape

d. Ripple voltages as described by the appropriate FIM Program tape

3.4.4.5.6.6 Contents of Unit - The STAFIM shall contain the following components mounted within the unit:

a. A precision 9.3-V reference module

b. AC voltage divider string

c. DC voltage divider string

d. 28-volt latch memory

3.4.4.5.6.7 Panel Controls - The STAFIM shall have the following controls mounted on the front panel:

a. AC voltage adjust

b. DC voltage adjust

c. Negative Peak Detector Switch

d. Test Points (2)

3.4.4.5.7 Power Divider Assembly (55A3A4)

3.4.4.5.7.1 Function - The Power Divider Assembly (PDA) shall provide means of simultaneously monitoring AN/APG-59 pulse Klystron Power Amplifier (KPA) power with a power meter and waveshape with an oscilloscope. These measurements, taken at the radar interface normally occupied by the Power Monitor 2A2A3, are required for minimum performance tests and tuning of the pulse KPA.

3.4.4.5.7.2 Form Factor - The PDA shall be a minimum size and weight consistent with the requirements; but, the input port of the PDA shall be extended a suitable distance to clear the MCS and allow support equipment to be conveniently connected.

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3.4.4.5.7.3 Service Conditions - The PDA shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.4.5.7.4 Primary Input Power Requirements - The PDA shall perform as specified when supplied with average power of 10 dB at the center frequency of the MCS RF band.

3.4.4.5.7.5 Performance - The PDA shall provide the means of taking a signal from the flange normally occupied by the Power Monitor (2A2A3) and splitting the signal to two (2) outputs; one, a waveguide flange upon which a thermistor mount shall be connected, and the other an X-band crystal detector for connection to an oscilloscope. The power from the waveguide output shall be 3.0 ±0.3dB down from the input power.

3.4.4.5.7.6 Contents of Unit - The PDA shall consist of a waveguide power splitter with captive hardware on the input flange. One (1) output port shall terminate in a waveguide flange and incorporate quick-disconnect hardware for mounting the thermistor of the power meter. The second output port shall terminate in an X-band crystal detector with a BNC female connector. The third output port shall be terminated with a matched load.

3.4.4.5.8 KPA By-Pass (55A3A5) - The KPA By-Pass shall consist of two waveguide (cover) flanges to OSM-type coaxial transitions and a flexible coax cable to insert between these two transitions. The cable has a length of 18 ± 1 inch. The cover flanges are equipped with captive finger-operated mounting hardware.

3.4.4.5.9 20 dB Attenuators (2) (55A3A3) - The waveguide attenuators shall be of the fixed type, 20 ±0.3dB (maximum) VSWR of 1.15, 1-watt average power and operate over the range 8 to 12 GHz. The units shall have a cover flange on one end and a choke flange on the other end. The units shall utilize a fiberglass card as the attenuating element and shall withstand the service conditions, or a reasonable combination of them, of MIL-T-21200 for Class 3 equipment.

3.4.4.5.10 Waveguide Crystal Detectors (2) (55A3A1) - The Crystal Detectors' response shall be ±0.3dB over the range 9.0 to 11.0 GHz, maximum VSWR 1.80, 100mW maximum input power and low-level sensitivity greater than 0.4 mV/microwatt. The Crystal detector waveguide flange shall be an insulated cover type and the output termination a BNC female type. The flange shall contain finger-operated captive hardware and guide pins. The units shall withstand the service conditions, or a reasonable combination of them, of MIL-T-21200 for Class 3 equipment.

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3.4.4.5.11 Reed Relay and Time Delay Tester (55A3A8)

3.4.4.5.11.1 Function - The Reed Relay and Time Delay Tester provides the capability for detailed maintenance of the reed relays and their time delay circuits in the Launch Signal Computer and the Pilot Command Signal Computer of the AN/AWG-10 Missile Control System.

3.4.4.5.11.2 Form Factor - The Tester shall not exceed 7 inches by 3 inches by 1 inch; and shall not weigh more than 3 pounds. The Tester shall be configured to extend the relay boards from the chassis and allow access to test jacks for test signal monitoring.

3.4.4.5.11.3 Service Conditions - The Tester shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 3 equipment.

3.4.4.5.11.4 Primary Input Power Requirements - The Tester shall perform when supplied with +28 Vdc ±1 percent, 50 mA maximum.

3.4.4.5.11.5 Performance - The Tester shall be capable of the following:

a. Check the operation of the 30 ±15 millisecond time delay by applying +28 Vdc and providing a point for measurement of the delay.

b. Verify that all contacts on each red relay open and close with application of +28 Vdc to the coils.

c. Check that the contact resistance of all contacts in series does not exceed 5 ohms.

3.4.4.5.11.6 Contents of Unit - The Tester shall contain the following components:

a. Sixteen (16), 1000 ohm ±1 percent resistor arranged so that each resistor will be connected in parallel with a set of relay contacts.

b. A grounding lead to provide power ground for the time delay measuring circuit.

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3.4.4.5.11.7 Connectors - The Tester shall contain:

a. Connectors to mate with the chassis reed relay sockets and the relay boards.

b. A BNC connector connected to the relay coil for measurement by a counter.

c. Two (2) 0.080 tip jacks for measuring resistor string resistance with a voltmeter.

3.4.5 Tool Kit, Maintenance, TK-216/AWM-21

3.4.5.1 Function - The equipment described by this paragraph shall be for organizational and intermediate level use. Typical maintenance uses of the equipment are:

a. Removal and installation of LRU's, requiring manipulation of bolts and screws.

b. Removal, installation and torqueing of bolts and screws of the antenna, feedhorn, and other antenna assembly components.

c. Repair of connectors and wiring of harnesses and assemblies involving contacts and coaxial pins from sizes 2 to 20.

d. Removal, installation and tuning of transmitter waveguide assembly requiring use of non-magnetic tools.

e. Pressurization and leak detection of the waveguide assemblies.

f. Adjustment to the antenna synchros.

3.4.5.2 Form Factor - A single transit case shall store those tools which are normally used at both organizational and intermediate levels of maintenance and shall contain hinged trays to provide access to the tools. The weight of the case, loaded, shall not exceed 65 pounds and the dimensions shall not exceed 18-1/2 by 17 by 13-1/4 inches. The transit case shall not exceed 32 pounds, empty.

3.4.5.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental conditions, or reasonable combination of these conditions, as specified in MIL-T-21200 for Class 2 equipment.

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3.4.5.4 Contents - The Tool Kit, Maintenance, shall consist of the following:

a. Transmitter Waveguide Tools - The following shall be provided, to be used primarily in the area of the transmitter waveguide assembly for installation, removal, waveguide maintenance and work with system KPA's.

(1) Klystron Power Amplifier Maintenance Screwdrivers - There are seven nonmagnetic screwdrivers. Four of the screwdrivers are cross tip types, two are slot types and one is a special CW KPA tuning type. Two cross tip types are 11-1/2 inches long, one is 15-1/2 inches long, and the fourth is a double offset. The two slot type screwdrivers are 11-5/8 and 15-7/8 inches in length; these screwdrivers are for No. 6, No. 8, No. 10 and 1/4-inch screws. The KPA screwdriver has a 0.08-inch diameter tip on a 1/4-inch slot blade.

(2) Klystron Power Amplifier Maintenance Wrenches - There are eight nonmagnetic, hexagonal wrenches. Three with plastic handles are 16 inches in length and are sizes 9/64 (No. 8 screw), 5/32 (No. 10 screw and 3/16 (1/4-inch screw). Four "L" type wrenches are of the same sizes, but one of the 9/64 sizes shall have a special short length (0.250 inches) foot of the "L." One 1/4-inch drive extension hexagonal wrench is 16 inches in length and is a 3/16 size.

(3) Klystron Power Amplifier and STALO Double-Ended Wrenches - There are four pairs of double-ended, nonmagnetic, wrenches. A ninth nonmagnetic type is a special wrench for the STALO. Sizes are:

(a) 8 inches long, 9/16 to 5/8
(b) 9-1/2 inches long, 11/16 to 25/32
(c) 9-1/2 inches long, 3/4 to 13/16
(d) 6-1/2 inches long, 7/16 to 1/2
(3) Special Wrench, approximately 3-1/2 inches long, 1/4-inch box at one end and 1/4-inch open at the other end.

(4) Attenuator Adjusting Tool - A nonmagnetic tool, 14 inches long, used to adjust the waveguide attenuator. It shall have a 3/8-inch aluminum socket for placing over the adjustment nut on the waveguide attenuator.

(5) Drive and Socket Assembly - A nonmagnetic tool used for performing the transmitter KPA cavity adjustments. It has a 3/8-inch socket wrench which slides down over a special screwdriver tip to engage the control locknut. After the locknut is loosened, the screwdriver tip, a 1/8-inch diameter tip on a 1/4-inch slot blade, is used to perform the adjustment.

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b. General Purpose Tools - For general use on the antenna, waveguide assemblies and throughout the MCS where removal and installation is performed, the following shall be provided:

(1) Torque Wrench - The torque wrench shall have a 1/4-inch drive with a maximum length of 15 inches. Maximum capacity is 150 inch-pounds. The wrench is for torqueing the antenna base assembly and shall conform to GGG-W-686, Type III, Class 2, Size 1.

(2) Torque Wrench Adapters - Five hexagonal adapters are used with the 1/4-inch drive and are for No. 6, No. 8, No. 10 and 1/4-inch hexagonal cap-cylindrical socket head screws. These adapters are sizes 7/64, 1/8, 9/64, 5/32 and 3/16 and are approximately 2 inches long.

(3) Torque Wrench Socket - A 7/16-hexagonal socket, deep enough to fit over a bolt 1-21/32 inches long, shall conform to GGG-W-641, Type II, Class 2, Style B.

(4) Adjustable T-Handle - This is a handle, 4-1/2 inches in length, with a sliding 1/4-inch drive which permits use as either a "T" or an "L." It may be used in lieu of the torque wrench with the adapters, socket and extension bar.

(5) Extension Bar - This is a 14-inch extension bar for use with the T-handle, adapters and socket. The extension bar shall conform to GGG-W-641, Type IV, Class 4, Style A.

(6) Flexible Driver - This is a flexible drive shaft, 9-3/16 inches long, with a handle and a 1/4-inch drive.

(7) Wrench, Screwdriver Handle - This is an 8-1/4-inch long hexagonal wrench with plastic handle for securing the 9/64-inch bolts on the STALO filter box.

(8) Wrench - This is a steel L-type 1/8-inch hexagonal wrench for No. 8 cap cylindrical socket screws. It is approximately 2-1/4-inches by 3/4 inch.

c. Special Purpose Tools

(1) Synchro Adjust Tool - This tool is used to rotate and adjust the antenna azimuth and elevation synchros. It shall have the form of a pair of tongs. The handles are approximately ten (10) inches long, with a clasp to permit locking when the semicircular clamps grasp and hold the body of the synchros. The entire tool, except for the pivot, is coated with neoprene.

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(2) Alignment Tool - This tool is used for trimmer adjustments to the crystal oscillator of the STALO 30 MHz oscillator. It shall have tuning slot head on both ends of a plastic shank approximately 10 inches long.

(3) Pressure (Air) Plate Assembly - This item is used to check air pressure and leakage in the input waveguide on the antenna base assembly and the output waveguide on the transmitter assembly. An air check valve shall be provided to permit air pressure to be applied to the waveguide and to maintain pressure when the pressure source is removed. The assembly shall include a pressure gage to measure 0 to 30 psig.

(4) Air Pressure Test Adapter - This adapter permits the nose package to be pressurized when it has been extended from the aircraft. It shall consist of a pressure plate and valve to be used with the Compressor-Dehydrator, HD-755/AWM-21.

(5) API Sensor Alignment Tool - This is an L-type 3/32-inch diameter round rod, approximately 6 inches long used to align the Antenna Position Indicator sensors on the antenna.

(6) Spanner Wrench - A wrench used to remove the antenna feedhorn retaining rings, it consists of a 1/4 arc with lugs to engage slots in the retaining ring. A handle approximately 5 inches long provides a means of applying torque.

(7) Boot Extractor - This consists of two 2-1/2-inch stainless steel hooks used to grasp the plastic boot in the antenna feedhorn, break the pressurization seal and remove the boot. One end has a 1/4-inch hook with which to grasp the boot and the other a 3/8-inch radius hook with which to exert finger pressure.

d. Cable and Connector Repair

(1) Wire Crimping Tool - This tool conforms to MS 3191-1 and consists of a crimp tool frame and three positioners for size 12, 16 and 20 contacts.

(2) Contact Positioners - Three positioners in addition to those furnished with the wire crimping tool are provided for size 12, 16 and 20 contacts and a size 20 Cannon contact in a subminiature connector.

(3) Crimping Gage - The gage provides a GO-NO-GO check on the crimping action of the wire crimp tool and conforms to MS 3196-2.

(4) Coax Removal Tool - This consists of two half-shell, 2-inch long clamps used to grasp and disconnect hard-to-reach coax connectors.

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(5) Coax Extraction Tool - This extractor is used to remove a special 0.190 size coax subminiature pin.

(6) Coax Crimp Tool - This tool provides a 0.160 nominal crimp for coax ground sheath.

(7) Tool, Crimp Type Contact, Insertion and Removal - This tool conforms to MS 18137 and includes an insertion and removal tool handle (MS 18137-20-1 or equivalent), an insertion bit (MS 18137-20-2 or equivalent), a female contact removal bit (MS 18137-20-3), and a male contact removal bit (MS 18137-20-4). These items are used for insertion and removal of No. 20 crimp contacts equivalent to MS 18134-20 and 18136-20.

(8) Insertion Tools - There are four tools for insertion of No. 2 shielded contacts, and for No. 10, No. 12, No. 16 and No. 20 pin and socket contacts for crimp type connectors.

(9) Removal Tools - There are six tools for the removal of No. 2 pin and socket shielded contacts, and No. 10, No. 12, No. 16 and No. 20 pin and socket contacts for crimp type connectors.

(10) Insertion and Removal Tools - These are two insertion/removal tools used to accommodate the special cannon connector contacts. One tool is used for sizes No. 16 and No. 18 contacts, while the second tool is for No. 20 and No. 22 contacts.

(11) Wrench, Connector - A connector wrench is used to facilitate loosening and tightening hard-to-reach line replaceable unit (LRU) Bendix-type connectors.

(12) Cable Splice Crimp Tools - Two crimp tools for three sizes of cable splice crimps.

(13) Hughes Connector Tool - These include for repair of special Hughes connectors, 2 crimp tools, 3 positioners for shielded contacts, one insertion tool and one removal tool.

e. Transit Case - The transit case shall be comprised of an upper and a lower shell, an extendible tray for each shell, and a hinged pad to cover each tray. The transit case shall be of aluminum, painted yellow enamel and equipped with polyurethane compartments in which the tools will be stored. A specific location shall be made and labeled for each tool.

3.4.6 Tool Kit, Test-Set, (Intermediate Level) TK217/AWM-21

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3.4.6.1 Function - This equipment shall be used for the following typical maintenance functions:

a. Removal and replacement of the antenna assembly actuator.

b. Flushing of the antenna hydraulic system.

c. Removal and replacement of the antenna spin motor rotor.

d. Removal of multipin modules from printed circuit assemblies.

e. Repair of cable and chassis connectors on faulty subassemblies and LRU's.

f. Maintenance and repair of the Fault Isolation Meter and the Selector Test Programmer.

g. Splicing and threading program tape of the Fault Isolation Meter and the Selector Test Programmer.

h. Replacement of computer servo motors and attenuators in the microwave assembly area.

i. Installation of inserts into castings where original threads have been damaged or stripped.

3.4.6.2 Form Factor - Two transit cases shall store the tools to be used primarily at intermediate level. One case, Box No. 1, shall contain hinged trays to provide access to the tools. The weight of the case, empty, shall not exceed 32 pounds. The other case, which shall contain the helical coils and facilities for repairing damaged tapped holes, shall be a steel compartmented tool box and weigh not more than 10 pounds empty.

3.4.6.3 Service Conditions - The equipment shall operate satisfactorily under the environmental service conditions for Class 3 equipment, per MIL-T-21200.

3.4.6.4 Contents - The Tool Kit, Test Set, Intermediate Level, shall consist of the following:

a. Box No. 1

(1) Auxiliary Tape Transport Control - This is a small control unit, 4-1/2 by 4-1/8 by 2 inches used to control the operation of the tape drive mechanism in the Fault Isolation Meter and the Selector Test Programmer.

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Control shall be provided for POWER ON-OFF, MOTOR REV-FWD-OFF, and BRAKE OFF-ON operation. Three 24-inch control cables are integral to the control for connection to the FIM and the Selector-Test Programmer.

(2) Tape Splicer - The tape splicer is a tool, 5 by 3 inches, fitted with locating sprockets and thumb screws to prepare the program for splicing.

(3) Splicing Tape - This is 66-foot roll of perforated, transparent, 35 mm tape with adhesive backing.

(4) Knife and Blade - This shall be a handle and blade check assembly conforming to FED STD GGG-K-00450, with five blades, style 4, straight cutting edge types.

(5) Screwdrivers - These are three screwdrivers for maintenance of the Test Set, Missile Control Group, conforming to FED STD GGG-S-121. One screwdriver shall be 6-1/4 inches long with a 3/16-inch flat tip, one with a 0.070-inch tip, 3-3/4 inch nonmetallic handle, and one double offset, cross tip, point size No. 1, 3-3/4 inches long.

(6) Gage, Shoehorn Type - This is a 0.005-inch gage to check roller clearances on the Selector Test Programmer and the FIM.

(7) Bearing Extractors - These are three bearing extractors for removal of the three sizes of high-speed bearings in the FIM. Two are approximately 7 inches long and one is approximately 6 inches.

(8) Flushing Block & Plate - This consists of two flushing blocks and two flushing plates to replace azimuth and elevation servo valves.

(9) Rotor Jacking Screws - These are six special 2-56 size screws for use when removing the Spin Motor Rotor from the Antenna Base Assembly. The screws have a 1-inch shank threaded for 0.7 inch nearest the head. The remainder of the shank is turned down to 0.06 inch diameter. Three are used to remove the rotor and three are spares.

(10) Actuator Removal Screws - These are four screws which have hexagonal cap-cylindrical socket heads, 10-32 by 1-1/2 inches long with full length thread.

(11) Actuator Installation Screws - These are four screws, headless, 10-32 by 1-1/2 inches long with continuous thread. Both ends have screwdriver slots.

(12) Wrenches, Spline - There are three spline wrenches for computer servo motor removal/repair and for removal of certain attenuators from the STALO and Waveguide Assembly. One wrench has four flutes, key size No. 2, for set screws size No. 2 and 3; one wrench is the same key size, for the same set screw sizes but with six flutes; and one wrench is key size No. 4 for set screws No. 4 and socket head cap sizes No. 0 and 1 with 6 flutes.

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(13) Handcrank, Tape Transport - This is a mechanical handcrank to assist in changing the program tape in the Selector Test Programmer and the FIM. It consists of a bracket to mount on the tape transport, a gear to engage the transport gear and a handcrank to rotate the gear.

(14) Lamp Removal Tool - This is a 3-inch long spiral spring, 0.210-inches inner diameter. for use in removing lamps from switch lights on the Test and Control Rack.

b. Box No. 2

(1) Insert Modules - There are six sheet metal modules, each consisting of a quantity of self-locking and nonself-locking helical coil inserts of a single size, an inserting tool, an extracting tool, a tang break-off tool, a bottoming tap, and a gage for that particular size of helical coil insert. Sizes of inserts are: No. 4-40NC, No. 6-32NC; No. 8-32NC; No. 10-32NF, 1/4-28NF and 3/8-24NF.

(2) Twinset Modules - These are six sheet-metal modules, each containing double (outer and inner) inserts, used when the damaged tapped holes are worn to a size beyond the corrective range of the regular inserts. Each module shall consist of a self-locking and nonself-locking inner and outer inserts for a single size hole, an oversize bottoming tap, an outer insert installation tool, an outer insert tang break-off tool, and an outer insert extraction tool. Sizes shall be the same as those for the Insert Module.

(3) Adjustable Tap Wrench - This consists of two T-handles fitted with an adjustable chuck bit used with the bottoming taps. One tool, for use with taps sizes 1/4 to 1/2, shall have a 5-inch long handle and a shank 3-3/8 inches long. The other tool shall be for tap sizes No. 0 to 1/4 and shall have a 3-1/4 inch long handle with a 2-3/8 inch long shank.

3.4.6.5 Transit Cases

a. Box No. 1 - This shall be an aluminum, gray epoxypolymide coated box with two hinged sections. Each section shall be approximately 18-3/8 by 17 by 6-3/8 inches. When closed the dimensions shall be approximately 18-3/8 by 17 by 13-1/8 inches. Molded polyurethane compartments shall provide storage slots for each tool. Each hinged tray shall have a padded lid. Empty weight shall not exceed 34 pounds.

b. Box No. 2 - This shall be a steel gray compartmented tool box, approximately 22 by 8 by 9 inches with a carrying handle on top center. The emply weight shall not exceed 10 pounds. This box shall conform to GGG-T-558, Type 1, Class II with the exception of tote trays and brackets.

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3.4.7 Fixture, Antenna Transporting, MX-7339B/AWM-21

3.4.7.1 Function - The Antenna Transporting Fixture, hereafter referred to as the Fixture, shall be for Organizational and Intermediate Level Maintenance use. The Fixture shall protect the Antenna while it is being transported, and shall be suitable for resting on the deck.

3.4.7.2 Form Factor - The Fixture shall be made of welded tubular, noncorrosive material formed to cradle the Antenna in its normal system-mounted attitude. The Fixture shall provide a protective structure around the lower half of the feedhorn and reflector of the Antenna. A base footing for the Fixture shall be provided, so located that the Fixture shall be stable when placed on the floor or deck. The design of the Fixture shall not permit any part of the Antenna to contact the floor or deck. Four carrying handles, two on each end parallel to the antenna boresight axis, with bicycle type grips and capable of being raised to horizontal attitude and lowered to reduce storage space, shall be provided. The Fixture shall not exceed 34 pounds in weight. Overall dimensions shall not exceed 37 by 32 by 21 inches with the handles down and 64-1/2 by 32 by 21 inches with the handles up.

3.4.7.3 Service Conditions - The Fixture shall function satisfactorily under any of the environmental service conditions specified in MIL-H-15362, or any reasonable combination thereof.

3.4.8 Test Set, Reference Rate Gyro, TS-2512/AWM-21

3.4.8.1 Function - The equipment shall be for Organizational and Intermediate Level use and shall provide the facility to check the zero rate and calibration of the AN/APG-59 Antenna Rate Gyros when afloat and ashore.

3.4.8.2 Form Factor - The Test Set, Reference Rate Gyro shall not exceed 2.4 by 4.0 by 5.7 inches. The total weight of the equipment, including cables, shall not exceed 5 pounds.

3.4.8.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.8.4 Primary Input Power Requirements - The equipment shall obtain all power from the AN/APG-59 Antenna Rate Gyro connector.

3.4.8.5 Contents of Unit

3.4.8.5.1 Gyro - An AN/APG-59 Antenna Rate Gyro with performance characteristics specified in MIL-R-81114 shall be provided.

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3.4.8.5.2 Mounting Fixture - A Fixture shall be provided to mount the Reference Rate Gyro and its connector on the antenna test pad in either the same azimuth or same elevation attitude as the respective system gyros. For test of the azimuth rate gyro, the mounting fixture shall support the Reference Rate Gyro so that the gyro mounting ring surface is parallel to the machined surface of the test pad within ±0.5 degree. For test of the elevation rate gyro the mounting fixture shall support the Reference Rate Gyro so that the plane of the gyro mounting ring surface is perpendicular to the machined surface of the test pad within ±0.5 degree, and the edge of the vertical slot in the pad is perpendicular to said plane within ±0.5 degree. The rotational alignment of the test gyro shall be held to ±0.25 degree of the reference position.

3.4.8.5.3 Adapter Cable Assembly - An Adapter Cable shall be provided approximately 5 feet in length. Adapter Cable shall be spliced to an AN/APG-59 Antenna Rate Gyro cable, pin to pin and with identical wire color code.

3.4.8.6 Transit Case - A Transit Case 14 by 12 by 5 inches shall be provided to hold and secure all items. The weight of the case, excluding equipment and cables, shall not exceed 7 pounds.

3.4.9 Generator, Signal, SG-696B/AWM-21.

3.4.9.1 Function - The Signal Generator shall provide the means to inject test signals into the AN/APG-59 Radar Set.

3.4.9.2 Form Factor - This unit shall not exceed 11 by 10 by 15 inches. The total weight of the equipment, including cables, shall not exceed 25 pounds.

3.4.9.3 Service Conditions - The equipment shall meet the temperature, shock and humidity requirements of MIL-T-21200 for Class 2 equipment, both operating and nonoperating. Other operating conditions shall conform to MIL-H-15362.

3.4.9.4 Primary Input Power Requirements - The test equipment shall be capable of satisfactory operation when supplied with -25 Vdc ±1 percent, 60 mA, 1.5 VA.

3.4.9.5 Performance

3.4.9.5.1 General - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced

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performance requirements for service conditions are acceptable, such variation will be specified.

3.4.9.5.2 Signal Output - The equipment shall generate a sine wave in the following band:

3.4.9.5.2.1 Frequency Band

a.	Frequency Range:	Continuously variable at a nominal frequency of 30 MHz over range specified in paragraph 3.4.4.4.1.3.1 (1) (a) of MIL-R-81114 for the cutoff frequencies.
b.	Frequency Indication:	Uncalibrated dial
c.	Frequency Resolution:	±100 Hz
d.	Distortion:	10 percent maximum
e.	Output:	-13 to -145 decibels reference to 1 milliwatt (dBm) in 1 dB steps, into a 50 ohm load.
f.	Drift:	Less than 100 Hz in 5 minutes
g.	Output Impedance:	50 ohms nominal
h.	FM Noise:	-107 dBm maximum in a 1 Hz bandwidth offset from the carrier by 1000 Hz.

3.4.9.6 Contents of Unit

3.4.9.6.1 Panel Controls

a. ATTENUATION/DB - There shall be two frequency output attenuation controls. One control shall be calibrated in 10 dB increments and the other control shall be calibrated in 1 dB increments.

b. FREQ/MHz (Frequency in Megahertz) - Means shall be provided for manual control of the frequency of the Signal Generator. A coarse frequency dial indication shall be contained within the knobs of this potentiometer.

3.4.9.6.2 Connectors - The following connectors shall be loaded on the front panel of the Signal Generator.

a. J1 OUTPUT - J1 shall provide signal output for the Frequency Band (3.4.9.5.2) and shall be a modified MS35175-291C.

b. J2 INPUT POWER - J2 shall provide input power connections and shall be MS3120E-10-6P.

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c. J3 TEST - J3 shall provide a test output for the High Frequency Band and shall be a modified MS35175-291C.

3.4.9.6.3 Cable Assembly, Power - A cable assembly, 72 ±1 inches shall be provided to connect the equipment to the 25-pin LRU connector of Regulators Power Supply which shall provide power. The cable shall be terminated at one end with a connector MS3126E-10-6S and terminated at the other end with a 25-pin connector.

3.4.9.6.4 Cable Assembly, Signal - Two cable assemblies, 72 ±1 inches, shall be provided to connect the Signal Generator output to the unit under test. Each cable shall be coax, RG-58/U per MIL-C-17/28, and shall be terminated in a connector MS35168-88E at each end.

3.4.9.6.5 Termination - A 50-ohm termination shall be provided consisting of a MS35173-274B connector and a 50-ohm ±1 percent, 1/8 watt resistor.

3.4.9.6.6 Dummy Crystal - A dummy crystal shall be provided to be used for signal injection in the AN/APG-59 receiver-preamplifier from the Signal Generator.

3.4.9.6.7 Combination Case - The equipment shall be provided with a Combination Case conforming to MIL-T-21200 and having maximum dimensions of 11 by 10 by 15 inches. The weight of the case, without equipment installed, shall not exceed 4 pounds.

3.4.10 Compressor-Dehydrator, Electronic Equipment, HD-755/AWM-21.

3.4.10.1 Function - The equipment shall be for Organizational and Intermediate Level use and shall provide the facility to pressurize the AN/APG-59 waveguide with dry air to check for leakage and to charge the Filler and Bleeder, Coolant.

3.4.10.2 Form Factor - The Compressor-Dehydrator shall not exceed 12 by 8 by 10 inches and total weight shall not exceed 20 pounds. The unit shall be gray in color.

3.4.10.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.10.4 Primary Input Power Requirements - The equipment power requirements shall be 103.5 to 126.5 V, 3-phase 380 to 420 Hz, 80 VA max per phase.

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3.4.10.5 Contents of Unit

3.4.10.5.1 Compressor - The Compressor at sea level shall deliver 650 cubic inches of air per minute minimum at 35 pounds per square inch gauge (psig), when supplied with the input power of 3.4.10.4. The Compressor outlet shall be quick-disconnect nipple, part number ECC206D4 or equivalent.

3.4.10.5.2 Dehydrator - The Dehydrator shall contain 26 ounces (oz) minimum of silical gel per MIL-D-3716, Grade H, Type IV. The desiccant life shall be 1 hour with outlet air dew point from -37.2°C (-35°F), to 1.7° C (35°F) with 32.2°C (90°F) saturated inlet air. The Dehydrator shall have an indicator to show the condition of the desiccant.

3.4.10.5.3 Pressure Gage - A gage indicating 0 to 50 psi shall be located on the front of the unit. Marking shall indicate 2 psi intervals.

3.4.10.5.4 Pressure Regulator - The unit shall contain a hand controlled pressure regulator capable of regulating the outlet pressure from 5 to 35 psig. Resolution shall be less than ±1 psig.

3.4.10.5.5 Hose Assembly - An air delivery hose 20 ±0.5 feet long shall be supplied with the equipment. The hose shall have a quick disconnect at one end to mate with Compressor (3.4.10.5.1) outlet and an air check at the other end to mate with air valve MS28889-1.

3.4.10.6 Control - A two-position toggle switch shall be located on the unit and shall control application of input power. Positions shall be marked ON and OFF. One type UP106M connector, per MIL-C-3767, shall be provided for the power connection.

3.4.11 Test Set, Missile Control Group - A Test Set, Missile Control System TS-2511C/AWG-10 or equal shall be provided.

3.4.12 Radar Set AN/APG-59 - A Radar Set AN/APG-59 shall be provided in accordance with MIL-R-81114. This equipment shall be GFE.

3.4.13 Counter, Electronic, Digital - An Electronic Digital Counter CP-870/AWM-21 or equal shall be provided in accordance with MIL-C-24165.

3.4.14 Multimeter - A Multimeter Electronic TS-505/U as specified in MIL-M-55088 or equal is required. This Multimeter is general and standard equipment and is not supplied as part of the AN/AWM-21 Test Set.

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3.4.15 Multimeter - A Multimeter AN/PSM-4 as specified in MIL-M-15134 or equal is required. This Multimeter is general and standard equipment and is not supplied as part of the AN/AWM-21 Test Set.

3.4.16 Electronic Units, Analyzer, Radio Frequency Noise - A Radio Frequency Noise Analyzer TS-2573C/AWG-10 shall be provided.

3.4.17 Cabinet, Electrical Equipment - An Electrical Equipment Cabinet CY-6667A/AWM-21 or equal shall be provided as specified in MIL-C-81583. This equipment shall house the equipment of 3.4.16 and shall be compatible with it.

3.4.18 Oscilloscope, OS-180/AWM-21

3.4.18.1 Function - The equipment shall provide the capability for the observation and measurement of time, phase, amplitude, repetitive characteristics and signal abnormalities of waveforms within the Missile Control System being tested. The Oscilloscope is used for intermediate level and organizational level maintenance.

3.4.18.2 Form Factor - The unit shall not exceed 11-3/4 by 14 by 25-1/4 inches. The weight shall not exceed 55 pounds.

3.4.18.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.4.18.4 Radio Interference - The instrument shall meet applicable radio interference susceptibility requirements of MIL-I-6181.

3.4.18.5 Power Requirements - The oscilloscope shall operate from either a 115 or 230 Vac, 10 percent, single phase, 50-to 1000-Hz power source. Operation at 115 to 230 Vac shall be selectable by an internal switch. The total power to operate the instrument shall not exceed 125 watts.

3.4.18.6 Performance Requirements - The Oscilloscope shall be capable of making the following measurements:

a. Voltage - 1 V to 250 V ac ±3%; 5 mV to 100 mV ac ±5% within the Intermediate frequency range.

b. Waveshape - Amplitude of -100 V peak ±5% to + 150 V peak

c. Frequency - DC to 50 MHz

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d. Electrical Phase difference of two simultaneous signals ±10 deg to ± 180 deg.

e. Time Interval - 20 nanoseconds ±10% to 1.0 second ±5%.

3.4.18.6.1 Warmup - The oscilloscope shall be fully operable within 3 minutes after power is turned on.

3.4.18.7 Contents of Unit - The Oscilloscope shall consist of the following items:

a. Oscilloscope Housing

b. Dual Channel Vertical Amplifier

c. Time Base and Display Generator.

3.4.18.8 Front Panel Controls - The following controls shall be mounted on the front panel of the Oscilloscope:

a. Oscilloscope Housing

(1) INTENSITY - This control shall vary the visible brightness of the oscilloscope trace.

(2) ASTIGMATISM - This shall be a screwdriver slotted control to adjust roundness of the writing beam.

(3) FOCUS - This control shall vary sharpness of the writing beam.

(4) TRACE ALIGN - The control shall be a screwdriver slotted control to adjust trace position.

(5) SCALE - The control shall adjust overall brightness on the face of the CRT.

(6) POWER - The push button POWER control shall turn the oscilloscope on and off.

(7) POSITION - This control shall provide coarse and fine adjustment of the display's horizontal position.

(8) MAGNIFIER - This control shall provide for magnification of the horizontal presentation.

(9) DISPLAY - The DISPLAY switch shall select the source of the horizontal input signal.

(10) AC-DC - The AC-DC switch shall provide for coupling for horizontal input signals.

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b. Dual Channel Vertical Amplifier

(1) DISPLAY - The display mode selector shall provide for selection of the following operating modes

A - Display of channel A
B - Display of channel B
A+B - Display algebraic sum of both channels
ALT - Presents a separate display of each channel input on alternate sweeps
ALT B TRIGGER - Presents separate displays of each input
CHOP B TRIGGER - Allows each input to be displayed separately on alternate sweeps with B channel input used as a trigger.

(2) DC BAL A, DC BAL B - An adjustment to minimize vertical shift of trace when polarity is switched.

(3) DC BAL A+B - An adjustment to balance trace when in A+B operation.

(4) POSITION (Channel A and B) - This control varies the vertical position of the display.

(5) POLARITY (Channel A and B) - This switch selects between a normal +UP or inverted -UP display.

(6) CAL (Channel A and B) - An adjustment to align amplifier with setting of the VOLTS/CM switch.

(7) VOLTS /DIV (Channel A and B) - This control selects the input amplitude necessary to give 1 cm of deflection.

(8) GND-AC-DC (Channel A and B) - This switch permits selection of the coupling desired.

c. Time Base and Delay Generator

(1) TIME/DIV - A time base control with 22 calibrated sweep ranges from 0.1 μsec/cm to 1.0 sec/cm for main sweep and 18 calibrated sweep ranges from 0.1 μsec/cm to 50 msec/cm for delayed sweep.

(2) DELAY (DIV) - Calibrated delay sweep from 1 μsec to 10 sec.

(3) SWEEP MODE - A switch to allow selection of NORM, AUTO, or SINGLE sweep.

(4) RESET - A pushbutton for resetting after a triggered display.

(5) INTENS RATIO - A potentiometer screwdriver adjustment for intensity.

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(6) VERNIER

(a) MAIN - The main vernier provides continuous adjustment between time base ranges and extends slowest sweep to at least 2.5 sec/cm.

(b) DELAYED - The delayed vernier provides continuous adjustment between ranges and extends slowest sweep to at least 125 msec/cm.

(7) TRIGGER (MAIN or DELAYED)

(a) LEVEL - The control shall vary the trigger level over the range of ±5 volts and ±50 volts in the mode switch EXT ÷10 position.

(b) MODE - The mode switch shall permit selection of (INT) internal vertical deflection signal, (LINE) internal source of line frequency; (EXT) ac or dc external signal and (EXT÷10) an external signal.

(c) SLOPE - The SLOPE switch provides selection of (-) or (+) slope.

(d) COUPLING - The coupling switch provides coupling in accordance with the following:

ACS - Attenuates signals above 30 kHz.
ACF - Attenuates signals between 15 and 30 kHz.
ac - To prevent erratic triggering at low frequency below 20 Hz.
dc - To preserve low frequencies and maintain reference level with random repetition rates.

3.4.18.9 Front Panel Connectors - The following connectors shall be mounted on the front panel of the Oscilloscope.

a. Oscilloscope Housing

(1) CALIBRATOR - GND - 250 mV - 10 V: This connector shall provide a square wave (1 kHz) at two amplitudes.

(2) EXT INPUT - A BNC connector for coupling an external horizontal input signal to the Oscilloscope.

b. Dual Channel Vertical Amplifier

(1) INPUT (A) - BNC

(2) INPUT (B) - BNC

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c. Time Base and Delay Generator

(1) EXT INPUT (MAIN) - This connector provides for a main sweep input (BNC).

(2) EXT INPUT (DELAYED) - This connector provides for a delayed sweep input (BNC).

(3) DEL'D TRIG OUTPUT - This connector provides a 1.5-V signal with < 50 nsec risetime at the end of delay time (BNC).

d. Lower Portion of the Oscilloscope

(1) Z-AXIS - A BNC connector for coupling an external horizontal input signal to the Oscilloscope.

(2) Input Power - A receptacle for input power to the Oscilloscope.

3.4.18.10 Oscilloscope Housing - The oscilloscope housing shall consist of an enclosure for containing the cathode ray tube, power supply, and intermediary circuitry. The enclosure shall have two cavities to accept the time base and delay generator and dual channel vertical amplifier.

3.4.18.10.1 Cathode-Ray Tube - The waveforms shall be displayed on a 4-inch rectangular cathode-ray tube, operating with a 12 kV accelerating potential. The screen shall be P31 phosphor and be aluminized for maximum light output.

a. Display Area - The oscilloscope shall have a display area of 8 by 10 cm.

b. Graticule - Display area of the graticule shall be marked in 5 minor/major divisions.

3.4.18.10.2 Horizontal Amplifier - External Input - The horizontal amplifier shall meet the following requirements.

a. Bandwidth - The bandwidth shall be dc to 5 MHz, 3 dB, dc coupled and 5 Hz to 5 MHz, ac coupled.

b. Sensitivity - Sensitivity shall be 1 V/cm in X1 position, 0.2 V/cm in X5 position and 0.1 V/cm in X10 position. A vernier control shall provide continuous adjustment between ranges. Dynamic range shall be ±5V.

c. Input Impedance - The signal input impedance shall be one megohm ±2 percent shunted by 30 pf ±3 percent.

d. Sweep Magnifier - The oscilloscope shall provide for 5X and 10X expansion of any position of the displayed sweep. Magnified sweep accuracy shall be within ±5 percent.

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3.4.18.10.3 Amplitude Calibration - A voltage calibrator meeting the following requirements shall be provided.

a. Output - The voltage calibrator shall provide a 250 mV to 10V peak-to-peak output.

b. Accuracy - Amplitude accuracy shall be within ±1 percent from -28 to 55°C.

c. Waveshape - The calibrated voltage waveform shall be a square wave.

d. Rise and Fall Time - The rise time and fall time of the calibrated wave shall be less than 3 µsec.

e. Frequency - The frequency of the calibrated output shall be 1000 Hz.

3.4.18.10.4 Intensity Modulation - Means for internal and external intensity modulation shall be provided.

a. Trace Blanking - A signal of approximately 2 volts positive dc to 15 MHz shall blank a trace of normal intensity.

b. Input Resistance - Input resistance for external dc coupling shall be approximately 5.1 k ohms.

3.4.18.10.5 Power Supply - The oscilloscope housing shall incorporate the necessary power supplies required to operate the plug-in units.

3.4.18.10.6 Internal Control - An internal control shall be provided for phase/bandwidth selection of normal XY-operation.

3.4.18.11 Dual Channel Vertical Amplifier - Each of the vertical channels shall provide input signals to the display circuit in accordance with the following requirements:

a. Bandwidth - The frequency response of the instrument when measured at -3 dB from an 8 cm, 50 kHz reference level shall be dc to 50 MHz dc coupled and 2 Hz to 50 MHz ac coupled.

b. Deflection Factor - The attenuator controls and circuitry shall provide 12 calibrated sensitivity ranges in 1, 2, 5 sequence from 5 mV/cm to 20V/cm, and continuous uncalibrated vernier adjustment from 5 mV/cm to 50V/cm. A sensitivity calibration adjustment shall be provided for each channel. When calibrated at the maximum sensitivity, calibrated range accuracy shall be ±3 percent or better.

c. Rise Time - The overall response of the vertical circuit shall be < 7 nsec with an 8 cm input step.

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d. Input Impedance - The signal input impedance shall be 1 megohm shunted by 25 pf ±3 percent.

e. Input Voltage - An input signal having a total of 600 volts, combined dc and peak ac, shall not cause damage to the instrument.

f. Isolation - There shall be a minimum of 50-dB isolation between the two channels under any possible combination of attenuator control and input signals within the range of this specification.

g. Mode Selection Circuit - The mode selection circuit shall provide for the selection of the following operating modes.

(1) Normal display of either channel A or B

(2) Inverted display of channel A or B

(3) Alternate Display - One sweep causes the display of the input to one channel and the next causes the display of the input to the other channel.

(4) Chopped Display - Sweep is alternately switched between the outputs of the two channels at a 400 KHz rate

(5) Added Display - Common mode rejection 40 dB minimum at 5 mV/cm, 20 dB on other ranges for frequencies up to 1 MHz.

h. Vertical Linearity - A two cm signal positioned to the extreme left and right of the display area shall show less than 0.15 cm compression.

i. Triggering - The circuits shall permit triggering in any of the modes specified in g. Triggering shall be possible over the range of dc to 50 MHz with 0.5 cm p-p signal or more displayed on the CRT.

3.4.18.12 Time Base and Display Generator - The horizontal circuitry of this unit shall provide recurrent or triggered linear sweeps at the choice of the operator.

3.4.18.12.1 Time Base Calibration - A single control (TIME/DIV) shall provide 22 calibrated sweep time ranges in 1, 2, 5 sequence from 0.1 μsec/cm to 1 sec/cm. An uncalibrated vernier control shall provide for continuous adjustment of the sweep time between the calibrated ranges to a minimum of 2.5 sec/cm.

3.4.18.12.2 Sweep Expansion - The instrument shall provide for 10X expansion of any portion of the displayed sweep.

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3.4.18.12.3 Sweep Accuracy - The accuracy of the time base sweep shall be ±3 percent or better

3.4.18.12.4 Sweep Modes - The following sweep modes shall be selected by the SWEEP MODE control

a. AUTO - provides a baseline in the absence of a trigger; allows main sweep to free run.

b. NORM (Normal Trigger)

c. SINGLE (Single Sweep)

3.4.18.12.5 Trigger Requirements

3.4.18.12.5.1 Trigger Mode. The sweep circuit shall permit triggering by any of the following signals.

a. Internal, from display channel (vertical deflection signal)

b. Internal source of line frequency

c. External signal, dc or ac

d. External signal - 10

3.4.18.12.5.2 Trigger Input Impedance - The input impedance of the external trigger shall be 1 megohm ±2 percent centered by less than 35 picofarads.

3.4.18.12.5.3 Trigger Level - The trigger level on external signal shall be continuously adjustable over the range of ±5 volts and ±50 volts in - 10 position.

3.4.18.12.5.4 Trigger Slope - The polarity of the trigger signal shall be selected by a trigger slope control.

3.4.18.12.5.5 Trigger Sensitivity - Trigger sensitivity shall be 0.5V peak-to-peak from dc to 50 MHz increasing to 1 volt at 90 MHz.

3.4.18.12.5.6 Maximum Input Voltage - An input signal having a total of 600 volts combined dc and ac peak shall not cause damage to the instrument.

3.4.18.12.6 Variable Time Delay - The instrument shall provide for continuously variable sweep delay times described as follows.

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3.4.18.12.6.1 Delay Ranges - A single control (TIME/DIV) shall provide 18 calibrated time delays in 1, 2, 5 sequency from 0.1 µsec/cm to 50 msec/cm. A calibrated 10 turn vernier multiplier shall provide for continuous adjustment of the delay time between the calibrated ranges and extend the slowest sweep to a minimum of 125 msec/cm.

3.4.18.12.6.2 Delay Time - The delay time shall be from 0.1 µsec to 10 µsec.

3.4.18.12.6.3 Delay Accuracy - The accuracy of the time delays over the temperature range - 15 to 55°C shall be ±3 percent or better.

3.4.18.12.6.4 Multiplier Linearity - Multiplier accuracy shall be ±1 percent, ±2 percent over the temperature range of 15 to 55°C.

3.4.18.12.6.5 Jitter - Time jitter of the input signal shall not exceed 1 part in 20,000 of maximum delay.

3.4.18.13 Transit - Instrument Case and Accessories - The Oscilloscope shall be contained in a transit case, the front cover of which shall contain the following accessories:

2 Voltage probes, 10X, 10 ft ±5 in. long, with a 0.080 male tip plug

1 Power cord

2 Attenuators, 10X, 93 ohm, BNC termination

2 Cables, 72 ±3 inches long, BNC termination

2 Termination, 93 ohm, BNC

2 Tee's, BSM

2 Adapters, BSM to BNC

1 Adapter, right angle, TSM

1 Adapter, TSM to BNC

2 Tee's, TSM

1 Adapter, BNC female to two 0.080 male tip plugs, blue and red.

1 Screen and RFI filter

2 Hook tips

2 Spanner tips

2 Ground leads

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2 Tip plugs (male) 0.080

2 Probe insulators

3.4.19 Voltmeter Digital ME371/AWM-21

3.4.19.1 Function - The Digital Voltmeter (DVM) shall provide the means to measure ac and dc voltage and resistance in subassemblies, printed circuit boards, components and wiring of the AN/AWG-10/11/12 missile control systems.

3.4.19.2 Form Factor - The DVM shall be capable of rack mounting in a standard 19-inch rack. The dimensions for rack mounting shall be approximately 18 by 21 by 6 inches. The front panel shall be approximately 20 by 6 inches. The unit shall also be made portable by the addition of transport adapter blocks which, when attached to the sides of the DVM, shall change the dimensions to approximately 21 by 21 by 6 inches. Addition of the front cover shall change the dimensions to approximately 21 by 5 by 6 inches. The unit with side blocks installed shall be capable of being installed on the maintenance cart V-387A/AWM-22. The weight of the unit shall not exceed 65 pounds in the rack-mounted configuration, nor shall it exceed 80 pounds in the portable configuration. Both limits do not include the remote readout unit.

3.4.19.3 Service Conditions

3.4.19.3.1 Environmental - The DVM shall meet the environmental requirements (when tested in accordance with the procedure specified by MIL-T-5422) of MIL-T-21200 for Class 2 equipment. Limits of the tests will be as specified in MIL-T-21200.

3.4.19.3.2 Interference - The DVM shall meet the interference requirements specified in MIL-T-6181.

3.4.19.4 Primary Input Power - The DVM shall perform as specified in paragraph 3.4.19.5 when supplied with power as follows.

Voltage: 115V ±10 percent or 230V ±10 percent
Frequency: 50 to 60 Hz ±10 percent or 400 Hz ±15 percent
Power Consumption: 55 watts above 0°C ambient.
455 watts below 0°C ambient.*

3.4.19.5 Performance - Measurements shall be presented in five (5) digital visual readings with a sixth digit overrange extending each

* At ambient temperatures below 0°C the internal heats shall operate.

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range by 10 percent. Front Panel Controls shall be provided for selection of measurement range, measurement function and reading mode, millivolt scale zeroing, power and remote control. There shall be five (5) dc voltage ranges from 100 millivolts to 1000 volts, three (3) ac voltage ranges from 10 volts to 500 volts (1000 Vrms full scale), five (5) resistance ranges from 1 ohm to 10,000 kohms and three (3) dc to dc ratio ranges from 1 to 100. External commands may be used to program the Function, the Range and the Reading Mode from a remote position. Measurement readings for dc voltages shall be made in either the filtered or nonfiltered mode. Non-filtered measurements shall be obtained in 45 milliseconds while filtered measurements shall be obtained in 515 milliseconds. In the filtered mode, 60 dB of rejection shall be obtained against all normal-mode noise greater than 58 Hz. Measurement accuracies shall be based on an ambient temperature of +25°C.

3.4.19.5.1 DC Voltage - Filtered - The DVM shall measure filtered dc voltages as follows:

a. Full-Scale Ranges - ±10.9999, ±109.999, ±1099.99 (1000 V. Max)

b. Accuracy, 24 hours - With variation in ambient temperature of ±1°C, ±0,005 percent of the reading or ±0.001 percent of full scale

c. Temperature Coefficient - ±0.0005 percent of reading and ±0.0002 percent of full scale per °C

d. Measure Time (max) - 515 milliseconds (ms)

e. Normal Mode Rejection - 60 dB min. 59 Hz and up

f. Common Mode Rejection

dc	-140 dB with up to 100 ohms in line
ac (≤ 61 Hz)	-120 dB with up to 100 ohms in line

g. Input Resistance -

10 range	-10,000 megohms
all others	-10 megohms

3.4.19.5.2 DC Voltage - Fast - The DVM shall measure fast dc voltages as follows:

a. Ranges	Same as 3.4.19.5.1
b. Accuracies	Same as 3.4.19.5.1
c. Measure time	50 milliseconds max (AVG) 100 milliseconds max (worst case)

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d. Common mode rejection:

dc	Same as 3.4.19.5.1
ac (≤ 61 Hz)	94 dB with up to 100 ohms in line

e. Input resistance — Same as 3.4.19.5.1

3.4.19.5.3 DC/DC Ratio - The DVM shall measure DC/DC ratios (+2 to +11 volts external reference) as follows:

a. Full-scale ranges - ±1.09999:1, ±10.9999:1, ±109.999:1

b. Accuracy (24 hours ±1°C) - All ±0.004 percent of reading, ±0.001 (with 10V external reference) percent of full scale.

c. Input resistance - 20 megohms

d. Measure time - Same as 3.4.19.5.1

e. Mode rejection - Same as 3.4.19.5.1

3.4.19.5.4 Millivolt D.C. - The DVM shall measure millivolts dc as follows:

a. Full-scale ranges - ±109.999, ±1099.99 millivolts

b. Accuracy 3 months

100-mV range - ±0.01 percent of reading ±0.020 percent of full scale
1000-mV range - ±0.01 percent of reading ±0.005 percent of full scale

c. Temperature coefficient - ±0.001 percent of reading ±2 microvolts per °C

d. Measure Time (max)	1 second
e. Input resistance	100 megohms

3.4.19.5.5 AC Volts - The DVM shall measure ac voltages as follows:

a. Full-scale Ranges - 10.9999, 109.999, 500 V rms (1099.99 Vrms full scale)

b. Accuracy (3 Months)

50 Hz to kHz - ±0.1 percent of reading or ±0.01 percent of full scale
10 kHz to 50 kHz - ±0.25 percent of reading or ±0.05 percent of full scale
50 kHz to 100 kHz - ±1 percent of reading or ±0.1 percent of full scale

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c. Temperature Coefficient - ±0.01 percent of reading and ±0.0002 percent of full scale per °C

d. Measure time (max) - 0.3 seconds

e. Input resistance - 1 megohm in series with 0.22 microfarads.

3.4.19.5.6 OHMS - The DVM shall measure resistance as follows:

a. Full-scale ranges — 1.09999, 10.9999, 109.999, 1099.99 10999.9 kohms.

b. Accuracy (3 months)

1, 10, 100 kohms ranges - ±0.01 percent of reading ±0.001 percent of full scale

100 kohms range - ±0.03 percent of reading ±0.001 percent of full scale

10,000 kohms range - ±0.1 percent of reading ±0.001 percent of full scale

c. Temperature coefficient -

1, 10, 100 kohms ranges - ±0.001 percent of reading and ±0.0002 percent of full scale per °C

1000 kohms and 10,000 kohms ranges - ±0.0025 percent of reading and ±0.0002 percent of full scale per °C

d. Measure time - 2 seconds

e. Max voltage across unknown - 11 volts

f. Max current through unknown - 10 milliamperes

3.4.19.6 Contents of Unit - The DVM shall consist of the following items:

a. Meter

b. Transport Adapter

c. Input Signal Cable

d. Power Cable

e. Test Extender Board

f. Remote Readout at Control Unit (Optional)

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3.4.19.6.1 Meter - The Meter shall consist of a single enclosed unit, adaptable to both rack mounting in a standard 19 inch equipment rack as well as being man-transportable. The meter shall be wired to operate with the remote readout and control unit described in paragraph 3.4.19.6.6 but the necessary board components to accomplish this function shall require separate provisioning action.

3.4.19.6.1.1 Controls - The meter will have five (5) controls on the front panel as follows:

a. Power ON-OFF - A two(2)-position switch to electrically connect the power supply. The DVM display shall be illuminated when POWER is on.

b. READ/HOLD - A two (2)-position switch allowing a choice of READ mode in which new readings are commanded at a fixed rate of 4 per second, or a HOLD mode which freezes the last previous reading indefinitely.

c. FUNCTION SWITCH - A seven (7)-position switch which provide the capability of selecting one of the seven (7) operating modes. These modes are:

(1) PROGRAM - Selection of Function to be measured shall be made externally through the remote programming input when the appropriate boards are plugged into the chassis in conjunction with the remote readout control unit described in paragraph 3.4.19.6.6.

(2) FAST DC - To select the high speed, nonfiltered dc measurements, from 100 microvolts to 1000 volts.

(3) DC - To select medium speed dc measurements, filtered, from 100 microvolts to 1000 volts.

(4) 10X RATIO - To select measurement of the ratio of the input voltage to a reference voltage. The reference voltage will not exceed 11 volts.

(5) AC - To select measurement of ac voltages from 50 Hz to 100 kHz.

(6) MV - To select measurement of low level dc voltages from 1 microvolt to 1000 millivolts.

(7) KOHM - To select measurement of resistance from 0.10 HM to 10 megohms.

d. Range Switch - A seven (7) position switch to provide the capability of selecting one (1) of the seven (7) ranges (7) range modes.

(1) 1.0999 fs Range - Fixed, for resistance measurements only (referred to as 1 range)

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(2) 10.9999 fs Range - Fixed; for resistance, ac, dc, and fast dc measurements (referred to as 10 range)

(3) 109.999 fs Range - Fixed; for resistance, ac, dc, fast dc and millivolts measurements (referred to as 100 range)

(4) 1099.99 fs Range - Fixed; for resistance, ac, dc, fast dc and millivolts measurements. (referred to as 1000 range)

(5) 10999.9 fs Range - Fixed; for resistance measurements (referred to as 10,000 range).

(6) AUTO - Range shall be automatically selected by internal sensing circuitry

(7) REMOTE - Range selection shall be made externally through the remote programming input.

e. M V Zero - used to zero adjust to readout when the Function Switch is on MV and Range Switch is on 100 and the input is shorted.

3.4.19.6.1.2 Connectors - All connectors except the signal input connector shall be on the rear panel.

a. Signal Input Connector - Glass Seal P/N-BE1H14-19P (var B) or equivalent - shall be mounted on the front panel and provide input connections for +Volts, -Volts, +Current, -Current and external reference voltage.

b. Rear Signal Input Connector - Glass Seal P/N-BE7H14-19P (var-A) or equivalent shall be mounted on the rear panel and provide input connections for +Volts, -Volts, +Current, -Current and external reference voltage.

c. Remote Programming Connector - Glass Seal P/NBE7H16-26P or Equivalent - shall provide input connections for use with the optional remote readout and control unit as described in 3.4.19.6.6.

d. Power Connector - Glass Seal P/N-BE7H12-3P or Equivalent shall provide input connections for 3 phase, 115 volt, 400 Hz power.

3.4.19.6.1.3 Indicators

a. Power Fuse Indicator - Located on the front panel - it shall illuminate when the power fuse is blown

b. Heater Fuse Indicator - Located on the Front Panel - It shall illuminate when the Heater Fuse is blown

c. Fan Fuse Indicator - Located on the front panel - it shall illuminate when the Fan Fuse is blown.

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3.4.19.6.1.4 Fuses - Meter power supply, heater and fan input lines shall be fused. The Fuses and Fuse Holders will be located on the rear panel.

3.4.19.6.2 Transport Adapter - To convert a DVM as described in 3.4.19.6.1 to a cart mounted or portable unit, a Transport Adapter shall be provided which consists of two (2) side blocks or covers, each approximately 2 by 21 by 6 inches, and one (1) front cover, approximately 5 by 6 by 20 inches. The two (2) blocks shall fasten to the left and right sides of the DVM and each block shall have a recessed handle. The blocks shall fasten to the front cover. The front cover shall protect the front panel of the DVM while the unit is in transit and in storage and fasten to the side blocks. The blocks and cover need not effect a watertight seal. The front cover shall provide storage space for the signal cable, power cable and test extender board.

3.4.19.6.3 Input Signal Cable - The Input Signal Cable shall be a minimum of 8 feet long. It shall be a four (4) conductor cable with the conductors designated as +Volts (+V), -Volts (-V), +Current (+I) and -Current (-I). The +V and +I conductors will be tied together at the end furthest from the meter. Also, the -V and -I conductors will be tied together at the end furthest from the meter. The probe tip for the positive (+V, +I) lead will be one (1), 0.080 inch-nominal, red point type. The probe tip for the negative (-V, -I) lead will be one (1), 0.080 inch-nominal, black type. An RFI shield, grounded at the end nearest the meter will be provided. The signal input connector will be provided for connecting the cable to the signal input on the meter front panel (cannon P/N KSP06F14-19S or equivalent). The cable will be capable of being used to measure potentials of 1000 volts dc. A 12-inch long grounding lead terminated in an alligator clip will be attached to the negative (black) probe tip lead.

3.4.19.6.4 Power Cable - A Power Cable extendible to minimum of 8 feet from a 2-1/2-foot coiled length shall be provided. It shall be of coiled, molded construction comprised of three (3) conductor AWG 18 wire with a female plug (Cannon KSP06F12-3S or equivalent). At one end and a MIL-type, male, three (3) prong with movable third prong at the other.

3.4.19.6.5 A Test Extender Board - A Test Extender Board shall be provided and shall serve as a patch cable between the plug-in board and its mating connector. The purpose of this board is to allow for troubleshooting while maintaining normal electrical contact between the plug-in board and the DVM. The board shall be as small and as lightweight as practical and capable of extending any plug-in board far enough so that all of its components are beyond the outer surface of the DVM chassis.

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3.4.19.6.6 Remote Readout and Control Unit - A Remote Readout and Control Unit shall be provided as an optional accessory. It will be honored in a separate portable case and shall be compatible with the basic DVM. It shall provide a means of selecting the function range, and mode; and of displaying the DVM readout. A cable shall be provided for connecting the unit to the DVM. The unit, cable and internal DVM circuit board components shall be an optional feature and require separate provisioning.

3.4.19.7 Construction - MIL-STD-108 will be used as a guide for the enclosure of the unit. Internal circuitry will be of the solid state variety to provide a highly reliable easily serviced unit.

3.4.19.8 Interchangeability and Replaceability - All plug-in units and replaceable subassemblies of the same part number shall be physically and functionally interchangeable one for the other without selection for fit or performance.

3.4.19.9 Reliability - The mean time between failure (MTBF) of the DVM shall be, as a design goal, a minimum of 300 hours. The appropriate reliability discipline in design, test, manufacturing and inspection to assure achievement of high reliability in the areas of piece part or component fabrication shall be exercised by the supplier.

3.4.19.10 - Storage Life - The minimum storage life of the DVM, as a design goal, shall be 30 months under any combination of the environmental conditions specified herein, after which the equipment upon being recalibrated will be capable of operation in accordance with the requirements of the specification for no less than the total specified service life.

3.4.19.11 Service Life - The DVM shall have as a design goal a minimum service life of 10,000 hours when operated in accordance with the performance requirements of this specification and under any combination of the environmental conditions specified herein, when subject to reasonable maintenance and parts replacement.

3.4.20 Generator, Signal, SG-844/AWM-21

3.4.20.1 Function - The equipment shall provide means of adjusting and calibrating X-band portions of an AN/AWG-10 Missile Control System by the generation of CW (Continuous Wave) and Swept RF Signals, and Frequency markers in the 8-to 12.4-GHz range. It shall be used at intermediate maintenance level.

examination and tests will be [illegible]

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3.4.20.2 Form Factor - The SG-844/AWM-21 (herein referred to as RF Sig. Gen) will be enclosed in a single unit, with dimensions approximately 17 inches wide x 9 inches high x 19 inches deep. The front panel will be compatible with a standard 19-rack and exceeds the front dimensions of the unit. The weight will not exceed 70 pounds.

3.4.20.3 Performance

3.4.20.3.1 General - The sweep shall be capable of start-stop operation where end points may be manually set and the SWEEP may be either UP or DOWN. Frequency markers (a minimum of two) be amplitude adjustable and capable of being set to any point within the band. Markers shall be easily detectable regardless of the width of the frequency sweep, when the spectrum is visually displayed. Controls shall be minimized consistent with the functional operation, nonambiguous and simple to operate. Frequency settings shall be visually identified and adjustable to the accuracy required herein.

3.4.20.3.2 Frequency Ranges (8 to 12.4 GHz)

3.4.20.3.2.1 Frequency Accuracy

a. Maximum Power - The frequency accuracy of the RF unit sweep end points on broad band sweep, center frequency on narrow band sweep, and CW operations in ±2 percent of the frequency setting except ±2 percent of full scale for the frequency markers.

b. At 50-milliwatts Power Output - Same as (a) above except that the tolerance becomes ±1.5 percent instead of ±2.0 percent.

3.4.20.3.3 Sweep Modes - The equipment shall provide for automatic, manual or triggered modes.

a. START-STOP SWEEP OPERATION - Settings adjustable over the entire band for START and STOP of sweep.

b. SINGLE FREQUENCY OPERATION

c. NARROW BAND SWEEP - Centered on single frequency with width up to 10-percent full band.

d. MARKER TO MARKER SWEEP OPERATION - Settings adjustable over entire band using MARKED controls.

3.4.20.3.4 Sweep Time - Sweep Time shall be continuously adjustable in four decade ranges 0.01 to 100 seconds and synchronized with the power frequency.

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3.4.20.3.5 Frequency Stability - Frequency stability shall be as follows:

a. With temperature: ±0.01 percent per °C

b. With 10-percent change in line voltage: ±1 MHz

c. Residual FM; (1) 50 kHz peak at power line frequency of 60 Hz as measured with equipment having a 5 kHz noise bandwidth

(2) 150 kHz peak at power line frequency of 380 to 420 Hz as measured with equipment having a 5 kHz noise bandwidth.

3.4.20.3.6 Amplitude Modulations - The RF Signal Generator shall be amplitude modulated as follows:

a. Internal - Square wave continuously adjustable from 950 to 1050 Hz on all sweep times.

b. External

(1) Frequency Response: DC to 350 kHz unleveled

(2) Sensitivity: -10V reduces RF output level at least 30 dB below noted CW output

3.4.20.3.7 Power Input - 115V or 230V ±10 percent 50 to 60 Hz and 380 to 420 Hz, power input shall be approximately 350 watts.

3.4.20.3.8 Power Output - The CW Power Output shall not be less than 50 milliwatts over the frequency range of 8 to 12.4 GHz.

3.4.20.4 Construction - MIL-STD-108 shall be used as a guide for construction of the enclosure for the unit.

3.4.20.5 Interchangeability and Replaceability - All plug-in units and replaceable subassemblies of the same part number shall be physically and functionally interchangeable. However, calibration may be required when a part is replaced.

3.4.20.6 Workmanship - Workmanship shall excel or equal the standards of paragraph 3.5, MIL-T-21200f.

3.4.20.7 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination thereof as specified in MIL-T-21200 for Class 3 equipment.

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3.4.20.8 Reliability - The appropriate reliability dicipline in design, test, manufacturing and inspection shall be exercised to assure achievements of high reliability in the areas of piece part or component fabrication. As a design goal the MTBF (mean time between failure) of the unit shall be 300 hours.

3.4.20.9 Storage Life - Except for the backwards wave oscillator, the unit shall have, as a design goal, a minimum storage life of 30 months under any combination of the environmental conditions specified, after which the equipment upon being recalibrated, shall be capable of operation in accordance with the requirements of this specification for no less than the total specified service life.

3.4.20.10 Service Life - The unit shall have, as a design goal, a minimum service life of 10,000 hours when operated in accordance with the performance requirements of this specification and under any combination of the environmental conditions specified herein, when subject to reasonable maintenance and parts replacement.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

a. Preproduction tests

b. Acceptance tests

c. Life tests

NOTE

Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

a. Contractor's demonstration tests

b. Service approval tests

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4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be conducted in accordance with the approved test procedure of 4.5. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration test and when requested by the procuring activity the equipment shall be delivered to the Government for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests necessary to determine that the equipment meets all requirements of this specification and the contract. Preproduction tests include environmental testing in accordance with procedures of MIL-T-5422 with test limits modified to correspond with the limits specified herein or as specified in detail equipment specification. Interference tests shall be conducted in accordance with MIL-T-6181 for those items designated in accordance with MIL-T-21200.

4.2.4 Preproduction Approval - Approval of the preproduction models shall be by the Naval Air Systems Command upon satisfactory completion of all tests. All preproduction models, except those consumed in testing or those rendered irrepairable for their intended use, shall be returned to the Contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction models shall not be considered as equipments under this contract; however, such models may be reworked and submitted for acceptance as production equipments. Incorporation of design changes, in delivered production equipments, found to be necessary as a result of the preproduction test program, shall be the subject of separate and independent negotiations on an individual basis.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling tests

c. Special tests

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4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of product

b. Operational test

c. Reliability test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material, workmanship, and interchangeability requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. Only those adjustments as are accessible to the operator during the normal use of the equipment.

Temperature	-Room Ambient
Humidity	-Room Ambient
Vibration (10 minutes)	-Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum double amplitude of 0.036 inch.

The equipment shall be vibrated (without vibration isolators between the equipment and test fixtures) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be cycled periodically through its phases of operation. Should a failure occur, it should be repaired and tests started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.3.2 Sample Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected for Sampling Tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2 additional
101 to 200	1 additional
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests after successful completion of the tests of 4.3.1.

a. Complete operational tests at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Repeat the test of (a) above using the highest and lowest specified input voltages and frequencies.

c. Perform operational test at environmental conditions when the results from the preproduction test indicate that these environmental conditions cause degradation(s) which result in marginal or unsatisfactory performance.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.3.1 Special Test Schedule - Selection of equipments for special test shall be made as follows:

a. On an early production equipment.

b. On an equipment midway through the production run.

c. On an early equipment after an extensive engineering change.

d. When failure reports or other information indicate additional equipment should be tested. (This will be determined by the procuring acitivity.)

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Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control. The maximum number of special tests shall be defined by the procuring agency at the time for request for bids.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as are selected by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted:

a. Temperature - Altitude tests

b. Vibration test (90-minutes cycling in each of 3 planes)

c. Humidity tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

a. Determine the cause of failure

b. Determine if the failure is an isolated case or design defect.

c. Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.1. The life tests sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling test or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1 additional
201 to 500	1 additional
Over 500	1 for each additional 500 or fraction thereof

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4.4.1 Test Conditions - The life test shall be conducted under the following conditions:

Temperature	-Room Ambient
Altitude	-Normal Ground
Humidity	-Room Ambient

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.4 Reconditioning of Life Test Samples - An equipment which has been subject to the life test shall be reconditioned as follows:

a. On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

b. After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedure - Preproduction tests, acceptance tests, and life tests shall be performed in accordance with a test plan and procedures mutually agreeable to the procuring activity and the contractor. The test plan shall be prepared by the contractor. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, the procuring activity will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

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4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability, or interchangeability of maintenance parts of assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with MIL-T-23336.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of equipment shall be preserved, packed, packaged, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

a. Title, Number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1).

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specifications.

b. Specifications - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Notation of Revisions - A Bar in the margin denotes a change made in that paragraph from the previous version of the document. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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6.4 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 Connectors

6.5.1 ECP SM-79 - Wiring at connectors noted includes changes authorized by ECP SM-79.

6.5.2 ECP SM-46 - Wiring at connectors noted includes changes authorized by ECP SM-46.

6.5.3 ECP SM-83/99 - Wiring at connector noted includes changes authorized by ECP SM(A)-83/99.

6.5.4 ECP SM-124 - Wiring at connectors noted includes changes authorized by ECP SM(A)-124.

6.5.5 ECP SM-128/69R/123 - Wiring at connectors noted includes changes authorized by ECP SM-128/69R/123.

6.6 Fixture, Antenna Maintenance - Antenna Maintenance Fixture MX-7340C/AWM-21 is designed to operate with Table Rack, Radar Maintenance (sidebench). Its requirements are defined in MIL-T-81527A.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81517B(AS) TEST SET, MISSILE CONTROL SYSTEM AN/AWM-21

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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